

An Introduction to the Analysis of Slender Structures



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What it's all about

Slender structures are defined as structures in which the cross-sectional dimensions are much smaller than their axial length. Many structures encountered in civil or industrial engineering can be classified as slender structures. To study systems such as those shown in the figure below one may wish to develop a one dimensional continuum theory.



Taney bridge in Dublin



pen



stacks

Some examples of slender structures.

Here we focus on the analysis of one-dimensional linear elastic systems in static equilibrium and under the hypothesis of small displacements. All forces are applied gradually, without shock or impact. The analysis of the governing differential equations of these slender systems will serve as basis for the introduction of some basic concepts of matrix structural analysis.

These lecture notes have been written with the aim of giving a self-contained introduction to the analysis of slender structures. Needless to say, I make no claim of originality. There are many excellent books on the market and some of them have been heavily used/abused in compiling these lecture notes.

A.S.

Delft, the Netherlands
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Chapter 1

Axial deformation

Axial deformation is one of the simplest deformation mechanisms. Nevertheless, it can describe many important engineering problems. In this chapter, we shall introduce a general procedure which will be employed to derive the governing equations of a bar undergoing axial deformation. We will make use of this procedure in the following chapters.

1.1 Kinematic assumptions

A straight homogeneous bar is under the action of a distributed load q acting along its axis as shown in Figure 1.1(a). We make the hypothesis that cross sections can only translate and remain orthogonal to the longitudinal axis of the bar which coincides with the centroidal axis. Under the action of the axially applied load, the bar in Figure 1.1(a) undergoes an elongation which is described by means of the axial degree of freedom $u(x)$. A cross section at x will displace by $u(x)$ and the displacement of a cross-section at $x + dx$ will be $u(x + dx) = u + du$. Due to its deformability, the infinitesimal element dx undergoes a change in length equal to du . This deformation is measured by the axial strain

$$\varepsilon = \frac{du}{dx}. \quad (1.1)$$

1.2 Constitutive relation and stress resultant

Stresses and deformation in a an elastic body can be related by means of Hooke's law. In the case of axial deformation, the ratio of stress to strain is equal to the modulus of elasticity

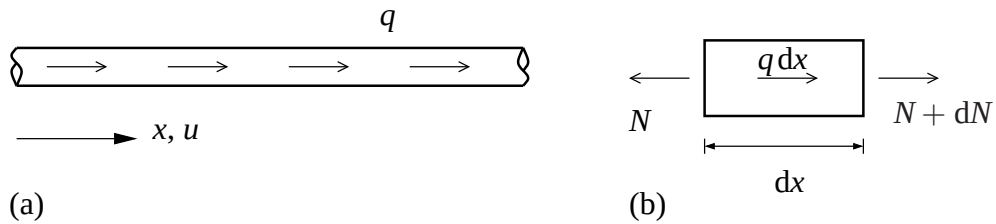


Figure 1.1

E or, equivalently,

$$\sigma = E\varepsilon. \quad (1.2)$$

The stress distribution over the cross section gives rise to the stress resultant or the net internal force

$$N = \int \sigma \, dA = \int E\varepsilon \, dA = \frac{du}{dx} \int E \, dA, \quad (1.3)$$

where in the last equation we have moved ε out of the integral as it is a function of x and the integral is on the cross section.

For prismatic bars with homogeneous cross-sections with E independent of y and z , the normal force becomes

$$N = EA \frac{du}{dx}. \quad (1.4)$$

1.3 Equilibrium equation

Figure 1.1(b) shows a free body diagram of a bar segment isolating the internal forces. For the purpose of applying the condition of equilibrium, the applied distributed load q with the dimension of a force/length is replaced by its resultant qdx . From the equilibrium in the horizontal direction we derive the equilibrium equation

$$-\frac{dN}{dx} = q \quad \text{or} \quad -\frac{d}{dx} \left(EA \frac{du}{dx} \right) = q \quad (1.5)$$

valid for the case in which the extensional stiffness EA , also known as axial stiffness, is a function of x , as it would be for a tapered bar. Obviously, the extensional stiffness may be brought out of the derivative in case of a homogeneous prismatic bar to obtain

$$-EA \frac{d^2u}{dx^2} = q. \quad (1.6)$$

1.4 Boundary conditions

Boundary conditions are imposed on a differential equation to fit the solutions to the actual problem. With reference to Figure 1.2, and with the displacement field as unknown, a boundary condition at a given point is drawn from conditions on displacements or stresses (either one of the two).

– A Dirichlet (or essential) boundary condition specifies the value of a solution on the boundary of the domain:

$$u(0) = 0.$$

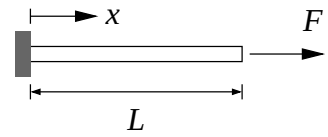


Figure 1.2

– A Neumann (or natural) boundary condition specifies the value of the normal derivative of a solution on the boundary of the domain:

$$\frac{\partial u}{\partial \mathbf{n}} = \frac{\partial u}{\partial x} \cdot \mathbf{n} \rightarrow \text{1D bar} \rightarrow EA \frac{du}{dx} \Big|_{x=L} = F \rightarrow N(L) = F.$$

Example 1.1

Consider a bar with a uniformly axial distributed load q_0 (the bar resembles the one depicted in Figure 1.2). The axial displacement u obeys the differential equation

$$EA \frac{d^2 u}{dx^2} = -q_0. \quad (1.7)$$

This equation can be integrated once to obtain the axial force

$$N = EA \frac{du}{dx} = -q_0 x + C_1. \quad (1.8)$$

A second integration yields the general solution

$$EA u(x) = -\frac{1}{2} q_0 x^2 + C_1 x + C_2, \quad (1.9)$$

where C_1 and C_2 are integration constants to be determined by the boundary conditions $u(0) = 0$ and $N(L) = 0$. This implies that $C_1 = q_0 L$ and $C_2 = 0$ from which

$$N(x) = q_0 (L - x) \quad \text{and} \quad u(x) = \frac{q_0 x}{2EA} (2L - x). \quad (1.10)$$

1.5 Equilibrium equation for continuously distributed elastic reaction forces

Consider a bar embedded in a medium as shown in Figure 1.3(a). This situation can be thought of as being representative of a reinforcement bar in concrete, as a pole embedded into soil etc. An approximation to this problem consists in replacing the action of the surrounding medium with a set of spring distributed along the surface of the bar as shown in Figure 1.3(b). Another assumption is to consider the surrounding material, and the springs, as a linear elastic medium so that its action on the bar can be characterized by a uniformly distributed load of the type $p = ku$, where k [F/L²] is the stiffness of the surrounding medium.

The bond between the bar and the surrounding material varies from situation to situation and is far from linear. In any case, the medium exerts a resistance against the extraction of the embedded bar. This resistance is modelled by considering the distributed force p as acting in the direction opposite to the displacement, as shown in Figure 1.3(c).

The governing equation is derived considering the contribution coming from the surrounding medium in the equilibrium of an infinitesimal element dx as shown in Fig-

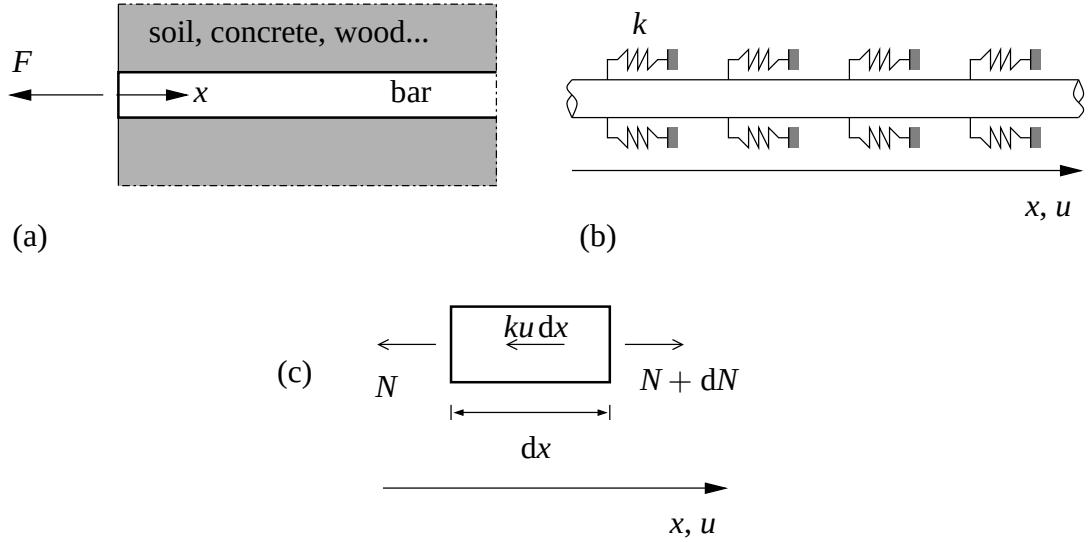


Figure 1.3

Figure 1.3(c). The equilibrium in the horizontal direction yields

$$\frac{dN}{dx} = ku \quad \text{which is expanded to} \quad \frac{d}{dx} \left(EA \frac{du}{dx} \right) - ku = 0, \quad (1.11)$$

or

$$EA \frac{d^2 u}{dx^2} - ku = 0, \quad (1.12)$$

in case of a homogeneous prismatic bar.

Example 1.2

A preliminary analysis of any pull-out problem can be pursued by considering the approach described in Section 1.3. The differential equation is written as

$$EA \frac{d^2 u}{dx^2} - ku = 0 \quad \text{or} \quad \frac{d^2 u}{dx^2} - \alpha^2 u = 0 \quad (1.13)$$

with $\alpha^2 = k/EA$.

This is a second order homogeneous differential equation with a constant coefficient α whose solution can be expressed by the homogeneous solution as:

$$u(x) = C_1 e^{\alpha x} + C_2 e^{-\alpha x}. \quad (1.14)$$

With reference to the bar depicted in Figure 1.3(a), the two integration constants follow after application of the boundary conditions at $x = 0$ and $x \rightarrow \infty$. The boundary condition at

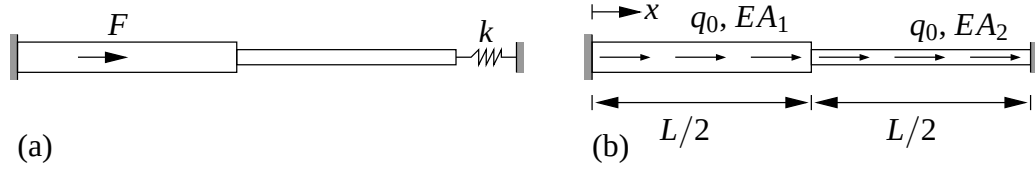


Figure 1.4

$x \rightarrow \infty$ is written as $u(\infty) \rightarrow 0$, since at infinite distance from the point of application of the concentrated load the axial displacement must be 0. This implies

$$C_1 e^{\alpha\infty} + C_2 e^{-\alpha\infty} \rightarrow 0 \quad (1.15)$$

from which $C_1 = 0$. The second boundary condition $N(0) = EA \frac{du}{dx}(0) = F$ yields $C_1 - C_2 = F/EA\alpha$ from which, making use of $C_1 = 0$, $C_2 = -F/EA\alpha$.

We can now express the displacement field as

$$u(x) = -\frac{F}{EA\alpha} e^{-\alpha x} \quad (1.16)$$

and the axial force as

$$N(x) = EA \frac{du}{dx} = F e^{-\alpha x}. \quad (1.17)$$

1.6 Matching conditions at discontinuities

Situations similar to those depicted in Figure 1.4 cannot be dealt with by a simple integration of the differential governing equation. The infinitesimal element which was used in the derivation of the above derivations excluded the presence of discontinuities. Discontinuities in extensional stiffness EA , distributed load q or the presence of concentrated forces and support reaction forces may be dealt with by using special boundary conditions known as matching conditions. The differential equations are considered in each domain, separately, and the matching conditions are employed to solve for the unknown integration constants. This procedure, illustrated in the example below, is general and can be employed in the solution of many other differential equations.

Example 1.3

Consider the case shown in Figure 1.4(b). The differential equation (1.6) with $q = q_0$ will be solved in two separate domains by successive integrations. Considering the first domain

($0 < x < L/2$) we have:

$$EA_1 \frac{d^2 u_1}{dx^2} = -q_0, \quad EA_1 \frac{du_1}{dx} = -q_0 x + C_1, \quad EA_1 u_1 = -\frac{1}{2} q_0 x^2 + C_1 x + C_2.$$

Proceeding similarly for the second domain ($L/2 < x < L$) we obtain:

$$EA_2 \frac{d^2 u_2}{dx^2} = -q_0, \quad EA_2 \frac{du_2}{dx} = -q_0 x + C_3, \quad EA_2 u_2 = -\frac{1}{2} q_0 x^2 + C_3 x + C_4.$$

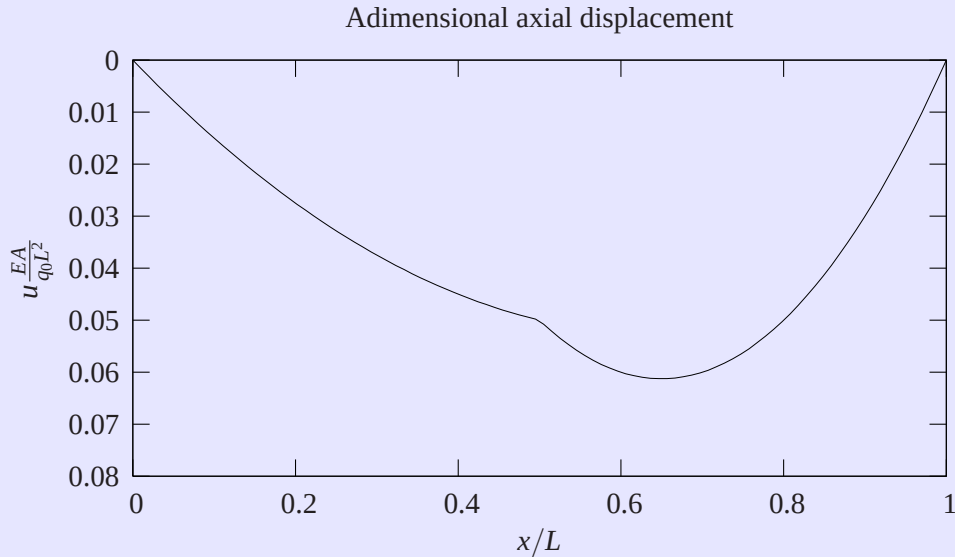
The boundary conditions $u = 0$ at $x = 0$ and $x = L$ yield $C_2 = 0$ and $C_4 = q_0 \frac{L^2}{2} - C_3 L$. The two unresolved integration constants C_1 and C_3 can be determined by enforcing two matching conditions at the interface. These matching conditions are an equilibrium condition ($N_1(L/2) = N_2(L/2)$) and a kinematic condition ($u_1(L/2) = u_2(L/2)$). These two conditions yield, letting $A_1 = 4A_2 = 4A$, $C_1 = C_3$ and $C_1 = 13/20 q_0 L$.

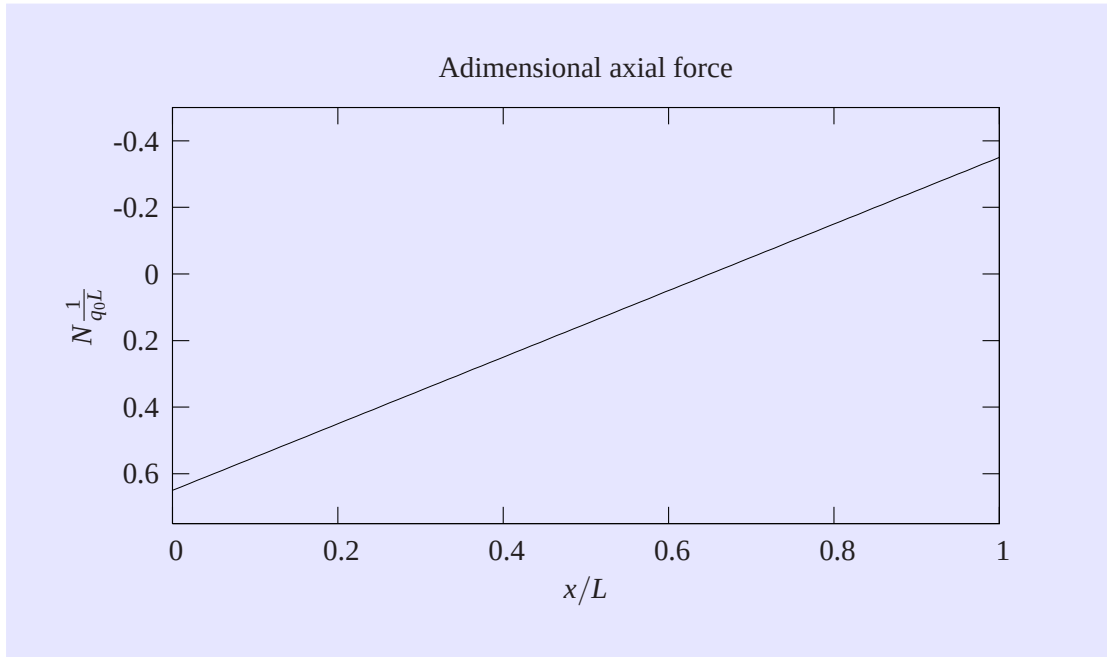
In summary, we have $C_1 = 13/20 q_0 L$, $C_2 = 0$, $C_3 = 13/20 q_0 L$ and $C_4 = -3/20 q_0 L^2$. With these integration constants the displacement field and the axial force read as, respectively,

$$u = \begin{cases} \frac{q_0 L^2}{EA} \left(-\frac{1}{8} \left(\frac{x}{L} \right)^2 + \frac{13}{80} \frac{x}{L} \right) & \text{if } 0 < x < L/2 \\ \frac{q_0 L^2}{EA} \left(-\frac{1}{2} \left(\frac{x}{L} \right)^2 + \frac{13}{20} \frac{x}{L} - \frac{3}{20} \right) & \text{if } L/2 < x < L, \end{cases} \quad (1.18)$$

$$N = EA \frac{du}{dx} = \begin{cases} q_0 \left(\frac{13}{20} L - x \right) & \text{if } 0 < x < L/2 \\ q_0 \left(\frac{13}{20} L - x \right) & \text{if } L/2 < x < L. \end{cases} \quad (1.19)$$

A plot of the displacement field and of the axial force is shown below.





Exercises

1.1 A uniform rod of length L is hung vertically under the action of gravity. Show that the loading per unit length is $q(x) = \rho Ag$, where ρ is the mass density and g is the gravitational constant. Consequently, show that the displacement distribution is

$$u(x) = \frac{\rho AL}{EA} gx \left(1 - \frac{x}{2L}\right).$$

[Problem 2.1 from Reference [1]]

1.2 Consider a rod of length L that has a varying area of the form

$$A(x) = A_1 + A_{21} \frac{x}{L}$$

with $A_{21} = A_2 - A_1$. If this is fixed at one end and a load P is applied at the other, show that the displacement distribution is

$$u(x) = \frac{PL}{EA_{21}} \ln \left[1 + \left(\frac{A_2}{A_1} - 1 \right) \frac{x}{L} \right].$$

[Problem 2.3 from Reference [1]]

References

- [1] J. F. Doyle. *Static and Dynamic Analysis of Structures with an Emphasis on Mechanics and Computer Matrix Methods*. Kluwer Academic Publishers, 1991.

Chapter 2

Euler-Bernoulli beam bending

The beam theory presented in this chapter is the results of many years of work by some of the most influential individuals in the mechanics community.

This chapter is based on [1, 2, 3, 4].

2.1 Limitations of the theory

The sign conventions employed in this chapter are shown in Figure 2.1. Beam problems will be solved with the assumption that the shear strains are approximately zero. This model is known as technical theory of bending. The technical theory of bending is valid only under the assumption of small displacements. We also assume that the beam has a straight longitudinal axis with cross section of any shape provided it is symmetric about the y axis. As a consequence of their geometrical proportions, all beams are stable under the action of the applied load: a thin sheet of paper makes a bad beam as it will buckle sidewise and collapse.

2.2 Kinematic assumptions

Kinematics describes how the deflection of the beam is tracked. Here, the deflection v of a beam is defined as the transverse displacement of the center line of the beam in the plane xoy . This is the only unknown. The deflection is accompanied by a rotation of the beam neutral plane and by a rotation of the beam cross section.

The key assumption in Euler-Bernoulli beam theory is known as Bernoulli-Navier hypothesis: *plane cross-sections remain planar and normal to the beam axis in a beam subjected to bending*. This hypothesis is valid when deformations due to shear and torsion result small compared to those deriving from normal stress and flexural deformation. This hypothesis will be somehow relaxed in the Timoshenko beam theory as we shall see in Chapter 4.

2.2.1 Relationship between deflection and curvature

Consider Figure 2.2(a). From simple geometric considerations

$$ds = \rho d\theta \quad \text{and} \quad \kappa = \frac{1}{\rho} = \frac{d\theta}{ds}, \quad (2.1)$$

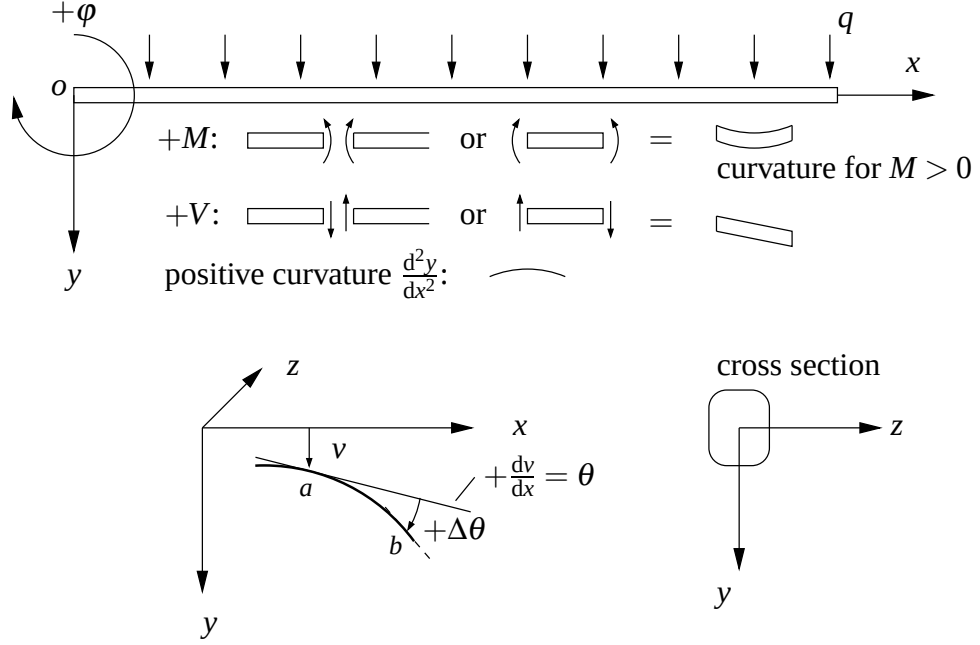


Figure 2.1 Sign convention in beam problems.

with κ the curvature and ρ the radius of curvature. The slope of the deflection curve is evaluated as $\frac{dv}{dx}$ and can be related to the angle of rotation of the axis of the beam θ (see Figure 2.2(b)) by means of

$$\frac{dv}{dx} = \tan \theta. \quad (2.2)$$

This expression has been obtained by approximating ds with a straight line since dx is infinitesimal. Note that it is also $dx = ds \cos \theta$.

Under the assumption of very small rotations, we may set $\sin \theta \approx \tan \theta \approx \theta$ and $\cos \theta \approx 1$. Hence $ds = dx$, $\kappa = \frac{1}{\rho} = \frac{d\theta}{dx}$ and $\frac{dv}{dx} = \theta$. Taking the first derivative of θ and using the expression for the curvature we obtain

$$\frac{d^2v}{dx^2} = \kappa = \frac{1}{\rho}. \quad (2.3)$$

2.2.2 Relationship between curvature and longitudinal strain

In order to seek the relation between curvature and the associated deformation, we consider a portion of beam in pure bending produced by two couples M_0 as shown in Figure 2.3. The couples M_0 generate positive curvature and negative bending moment M (negative according to our convention stated in Figure 2.1). The beam axis is bent into a circular curve. Indeed, the symmetry of the beam and its loading requires that all elements of the beam deform in an identical manner which is possible only if the deflection curve is circular and if the cross

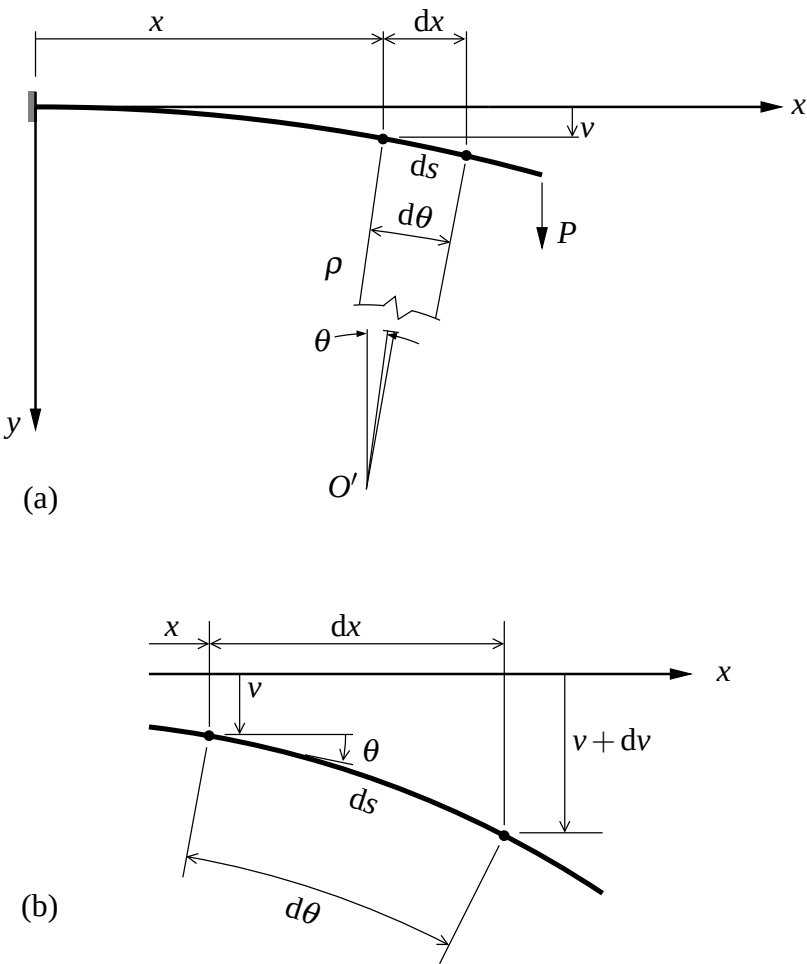


Figure 2.2 [1, Figure 7.1]

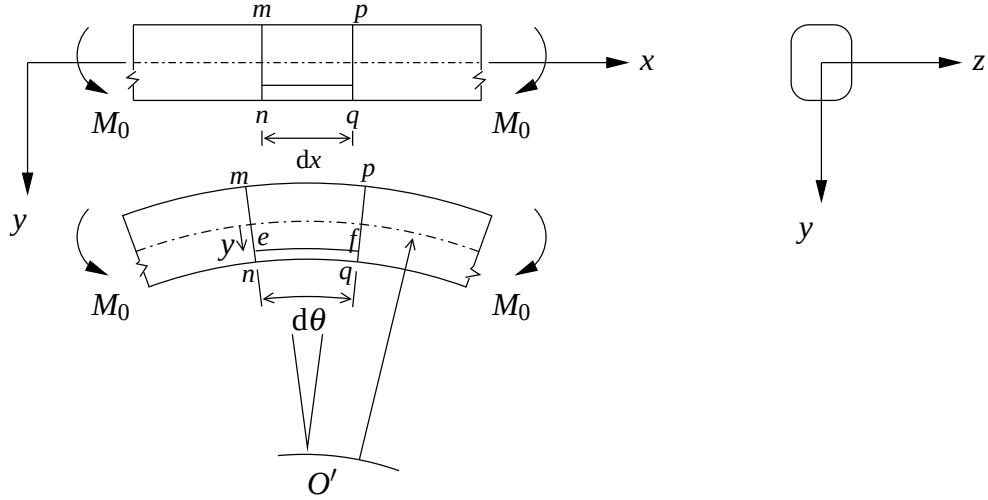


Figure 2.3 [1, Figure 5.6]

section remain plane during loading. Due to the deformation, the fibers in the upper part of the beam are in tension whereas those in the lower part are in compression. The fibers on the neutral surface of the beam do not change in length. The intersection of the neutral surface with a cross section is called neutral axis of the cross section. Here, the z axis is the neutral axis for the cross section (this surface is indicated by the dashed line in Figure 2.3). The planes of cross sections mn and pq of the deformed beam intersect in a line through the center of curvature O' . The angle between the two planes is denoted by $d\theta$ and the distance between O' and the neutral axis is the radius of curvature ρ .

The initial distance dx between the two planes remains unchanged at the neutral surface. The length dx of a segment on the neutral axis can be related to the radius of curvature ρ by means of $dx = \rho d\theta$. On the other hand, a segment ef at distance y from the neutral axis is strained and its length is now equal to $ds_{ef} = (\rho - y) d\theta$. Hence, the strain in the segment ef after bending is equal to

$$\epsilon_{xx}^{ef} = \frac{ds_{ef} - dx}{dx} = -\kappa y. \quad (2.4)$$

2.3 Relationships between load, shear force and bending moment

The relationships between load, shear force and bending moment is derived by expressing the equilibrium of an infinitesimal element dx of a beam in bending loaded with a distributed load of intensity q as shown in Figure 2.4. The relation between shearing forces and distributed load is obtained from equilibrium of forces in the vertical direction and reads as

$$\frac{dV}{dx} = -q. \quad (2.5)$$

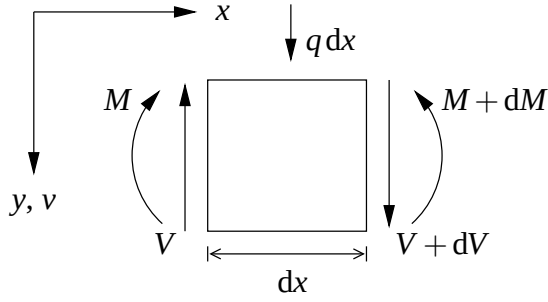


Figure 2.4 Element dx of a beam.

The equilibrium equation obtained by summing moments about an axis through the left hand face of the element and orthogonal to the plane of the figure yields

$$\frac{dM}{dx} = V, \quad (2.6)$$

where we have discarded product of differentials. This equation is valid in regions where there is a distributed load.

It does not hold where there is a concentrated load. A similar set of equation can be derived in case of distributed couples.

2.4 Relationship between internal bending moment and curvature

The longitudinal strain (2.4) is related to the stress by means of Hooke's law through the Young's modulus so that $\sigma_x = E \varepsilon_x = -E \kappa y$.

Consider Figure 2.5 representing a portion of a beam in bending where we have replaced the internal moment at the right-hand cross section with the corresponding stress distribution. For equilibrium, the internal couple resulting from the sum of $\sigma_x dA y$ over the whole section must equal the internal moment M . The element of force $\sigma_x dA$ on the element dA acts in the positive direction of the x axis when σ_x is positive and in the negative direction when σ_x is negative. Hence, its moment about the z axis is $dM = -\sigma_x y dA$. The equilibrium equation obtained by summing moments about the z axis yields

$$M = \int dM = \int \sigma_x y dA, \quad (2.7)$$

from which, considering the expression of the stress σ_x

$$M = \int \sigma_x y dA = - \int E \kappa y^2 dA = -EI \kappa \quad (2.8)$$

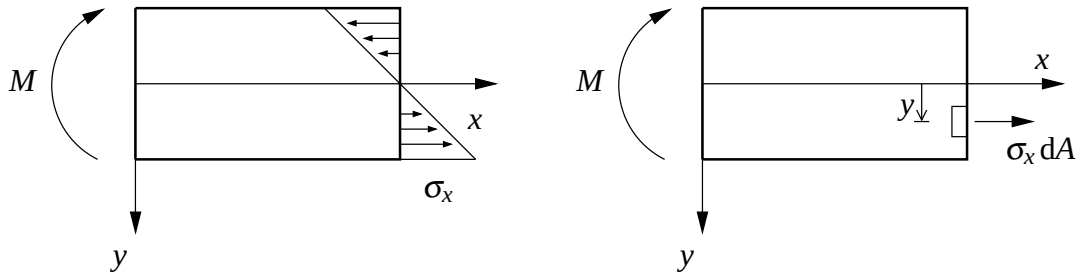


Figure 2.5

where

$$I = \int y^2 dA \quad (2.9)$$

is the moment of inertia around the neutral axis z ($I = I_{zz}$ and $M = M_{zz}$).

2.5 The differential equation of the transverse deflection

The differential equation of the deflection of a beam is obtained by eliminating the curvature κ from (2.8) and (2.3) to obtain

$$\frac{d^2 v}{dx^2} = -\frac{M}{EI}. \quad (2.10)$$

By making use of the relation (2.5) between shearing force and distributed load and (2.6) between shearing force and bending moment, (2.10) can be expressed as

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 v}{dx^2} \right) = q \quad (2.11)$$

or as

$$EI \frac{d^4 v}{dx^4} = q \quad (2.12)$$

if the flexural stiffness EI does not vary with x along the length of the beam.

It is worth noticing that the curvature related to a positive bending moment M is opposite to that related to a positive curvature $\frac{d^2 v}{dx^2}$ of the deflection line. Hence the minus in (2.10).

There is no general consensus on sign convention (cf e.g. [2, 3, 5]). In a system like the one depicted in Figure 2.6(a), the sense of the curvature of the elastic line v and that induced by a positive bending moment M is the same and the governing differential equation reads as

$$\frac{d^2 v}{dx^2} = \frac{M}{EI} \quad (2.13)$$

where $M = M_{zz}$ and $I = I_{zz}$. On the other hand, the system depicted in Figure 2.6(b) is similar to that reported in Figure 2.1 and the governing differential equation reads as

$$\frac{d^2 w}{dx^2} = -\frac{M}{EI} \quad (2.14)$$

where $M = M_{yy}$ and $I = I_{yy}$. The introduction of other sign conventions might seem confusing at first. Nevertheless, it is very important to realize that there is more than one way of looking at things. After all, quoting Den Hartog [3], the “sign convention used for shear force diagrams and bending moments is only important in that it should be used consistently throughout a project.”

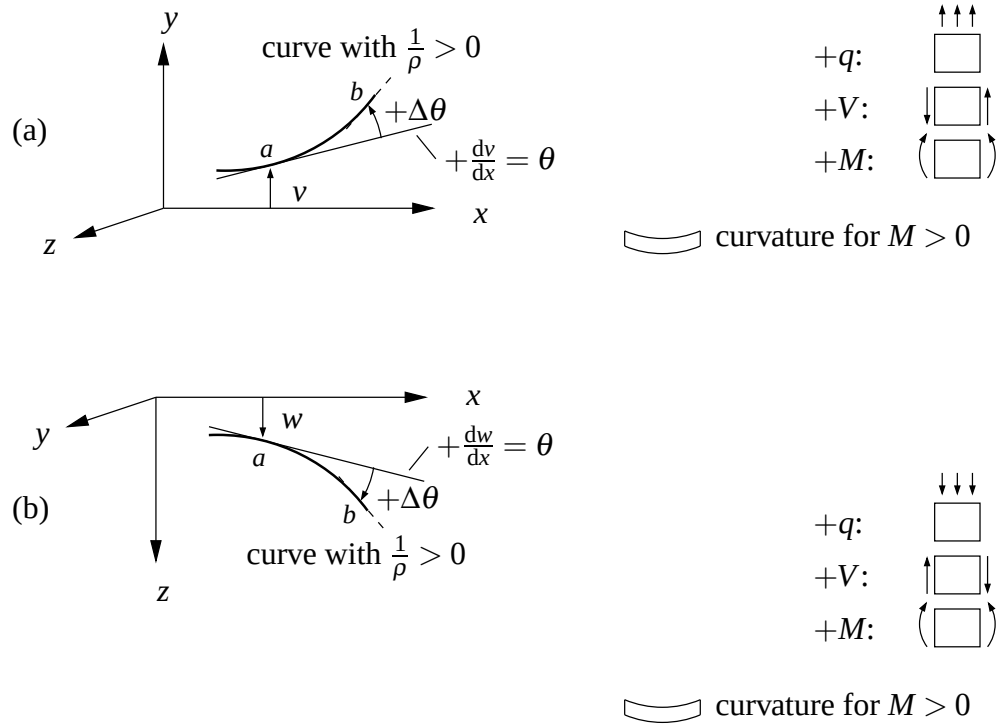
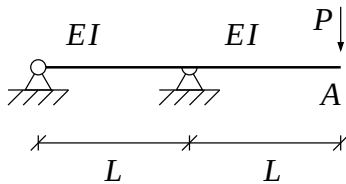


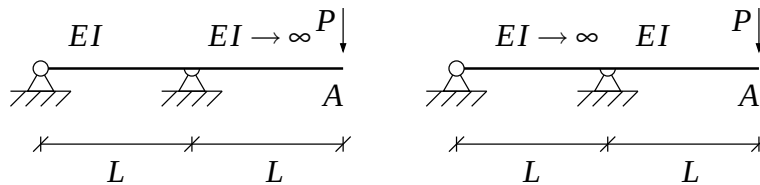
Figure 2.6 Other sign conventions in beam problems (cf Figure 2.1).

Exercises

2.1



Determine the deflection at A, and show that it can be obtained as the sum of the deflection at A of the two beams below. Can this result be generalized to other boundary conditions and/or other quantities such as rotations and reaction forces? Under what conditions it is valid? The deflection at A can be obtained using the second order differential equation function of the bending moment or the fourth order differential equation function of the distributed load. Use both differential equations.



References

- [1] J. M. Gere and S. P. Timoshenko. *Mechanics of Materials*. Wadsworth, Inc., Belmont, California, second edition, 1984.
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- [3] J. P. Den Hartog. *Strength of Materials*. Dover Publications, Inc., New York, 1961.
- [4] S. P. Timoshenko and D. H. Young. *Theory of Structures*. McGraw-Hill Book Company, New York, second edition, 1965.
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Chapter 3

Deflection of shear beams and frames

According to the Euler-Bernoulli beam theory, cross sections carry a resultant shearing force V but the deformation associated to the corresponding shear stress is not taken into account. This anomaly has been resolved by Timoshenko in 1921-22 [1, 2] by approximating the effect of shear as an average over the cross section – in reality, the shear stress and strain vary over the cross section. An extension of the Euler-Bernoulli beam theory that includes transverse shear deformation is discussed in Chapter 4.

In this chapter, we shall concentrate on the ideal shear beam employing a constant shear stress over a cross section. This kind of beam exhibits no flexural deformation but deforms in shear only. However, the beam is subjected to bending moments even if these moments do not contribute to the deformation.

3.1 The governing equation

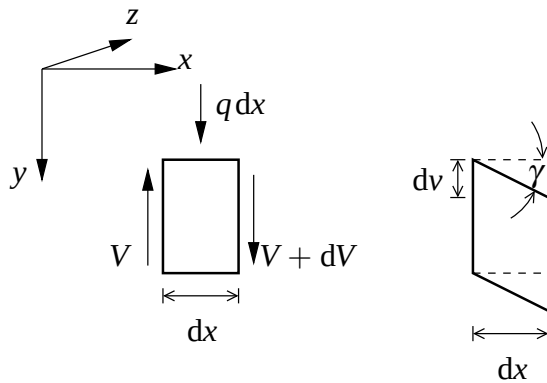


Figure 3.1

The kinematic quantity describing shear deformation is the shear distortion γ caused by the shear force V as shown in Figure 3.1. The shear distortion is related, in small deformation, to the deflection v by means of the kinematic relationship

$$\gamma \approx \frac{dv}{dx} \quad (3.1)$$

which can be derived with simple geometrical considerations analyzing Figure 3.1. Assuming a linear elastic material, the constitutive equation is formulated in terms of Hooke's law by specifying the relationship

$$\tau = G\gamma \quad (3.2)$$

between the deformation, i.e. the shear strain γ , and the stress, i.e. the shear stress τ . Considering an average expression of the shear force τ acting on a section,

$$\tau = \frac{V}{A_s} \quad (3.3)$$



Jules Arthur Vierendeel

★ Leuven, Belgium, April 10, 1852

† Ukkel, Belgium, November 8, 1940

Belgian engineer and writer.

In 1896 he developed a girder with upper and lower beams and rigidly connected vertical members, not braced by diagonal members. The Vierendeel girder was successfully applied in bridge construction.

The first Vierendeel bridge was built of steel, over the River Leie between Ruien en Avelgem in 1902/1904.

where A_s is the effective area in shear, we can express the shear deformation as

$$\gamma = \frac{dv}{dx} = \frac{V}{GA_s}, \quad (3.4)$$

where the quantity GA_s is known as the shear stiffness of the beam [3, Sections 7.12 and 12.9]. We have defined the shear stress τ in terms of the effective area in shear A_s because the shear strain is not constant across the cross section.

With reference to the differential element in Figure 3.1, the equilibrium equation in the vertical direction yields

$$q = -\frac{dV}{dx} \quad (3.5)$$

which can be combined with (3.4) to give the second order differential equation

$$-GA_s \frac{d^2v}{dx^2} = q. \quad (3.6)$$

This second order differential equation is very similar to the one derived for the axial deformation problem (cf (1.6)). Although the flexural deformation is not included in the formulation, the shear beam is subjected to bending moments. These bending moments are related to the shearing forces through relation (2.6).

3.2 The Vierendeel frame

Frame structures with rigid floor diaphragms may be analyzed by means of the shear beam analogy as their deflection is likely to be dominated by a shear mode type of deformation due to bending of the columns. Several building configurations show the presence of rigid concrete floor diaphragms as a lateral-load resisting system. This building configuration is known as the Vierendeel frame and is named after the Belgian engineer Arthur Vierendeel who developed the design in 1896.

The *Vierendeel* frame, sometimes referred to as *Vierendeel* truss or *Vierendeel* beam, was initially employed in some bridges like the ones shown in Figures 3.2 and 3.3. This frame is nowadays rarely used in bridges owing to a lesser economy of materials when compared to other solutions. Although the *Vierendeel* frame is not an efficient means of transmitting transverse load, it is however used to resist lateral load in buildings. It is also popular in the design of unconventional buildings for architectural reasons—the headquarters of the Commerzbank in Frankfurt, shown in Figure 3.4, and the Beinecke Rare Books & Manuscripts Library, Yale, shown in Figure 3.5, are typical examples.

This system is characterized by rigid joints, upper and lower beams and is a statically indeterminate truss in which all members are subject to bending moments.

3.2.1 Commerzbank headquarters

The building of the Commerzbank headquarters was designed by Norman Foster and engineered by Ove Arup. With a structural height of 259 m, the Commerzbank Tower, built in 1997, was the tallest building in Europe until the completion in December 20, 2003, of the Triumph Palace, an apartment building in Moscow.

Floors between sky gardens are supported by eight-story high *Vierendeel* frames which also resist lateral load. Pairs of vertical masts, enclosing the corner cores, support eight-story *Vierendeel* trusses, which in turn support clear-span office floors. There are no columns within the offices and the *Vierendeel* frames enable the gardens to be totally free of structure.

3.2.2 Beinecke Rare Books & Manuscripts Library

The Beinecke Rare Books & Manuscripts Library is located on the campus of Yale University in New Haven, Connecticut. The library opened in 1963 and is a big box of translucent marble on little feet. It was designed by Gordon Bunshaft, of the famous New York City architectural firm of Skidmore, Owings and Merrill.

The library features five-story *Vierendeel* frames supported by four concrete corner columns. Façades are assembled from prefabricated steel crosses welded together at inflection points.

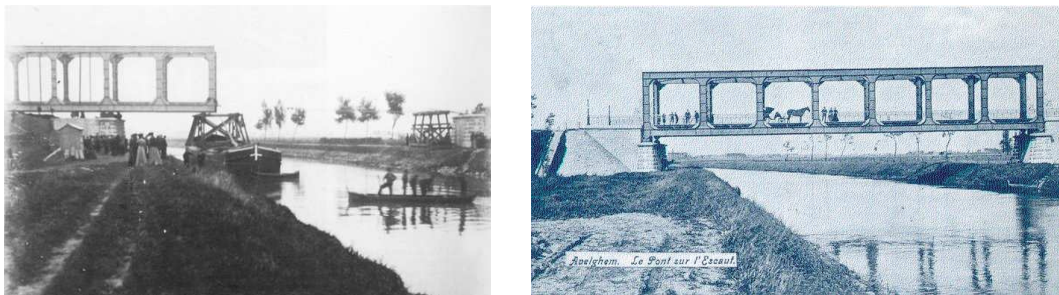


Figure 3.2 The first *Vierendeel* bridge was built in steel over the River Leie (1902/1904). The photos show different stages of construction. The span of the bridge was 42 m.



Figure 3.3 The Lanaye bridge in Belgium (1932) was blown up by the Belgian Army as a precaution measure to obstruct the German troops on May 11, 1940, one day after the German invasion of Belgium. The span of the bridge was 88 m.

Example 3.1 A multi-story building with rigid floor diaphragms as a shear beam

Consider a two-dimensional schematic of a multi-story building with rigid floor diaphragms depicted in the left part of the figure below (this system is known as “rigid jointed unbraced frame” or Vierendeel frame). Since the load is transmitted unaltered from floor to floor, it is possible to study the whole building by analyzing the single bay equivalent (shown in the right part of the figure).

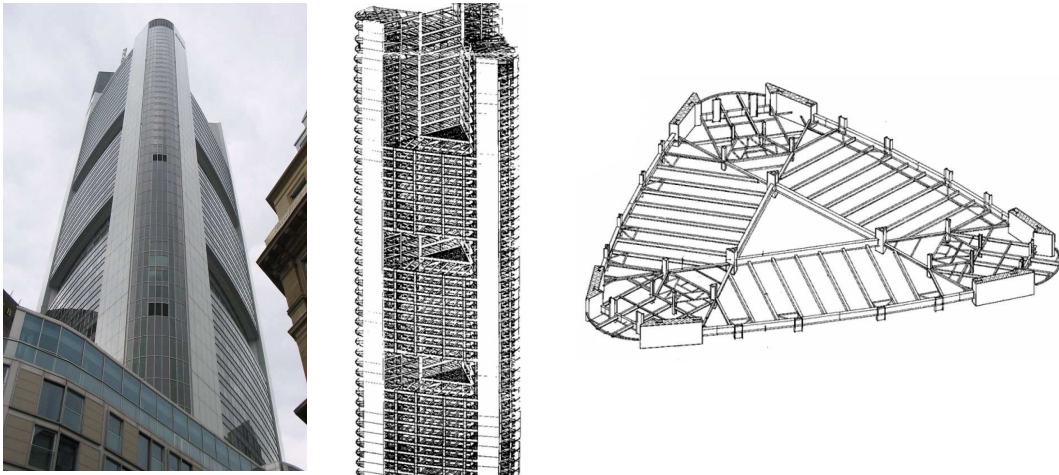


Figure 3.4 Commerzbank headquarters in Frankfurt Am Main.

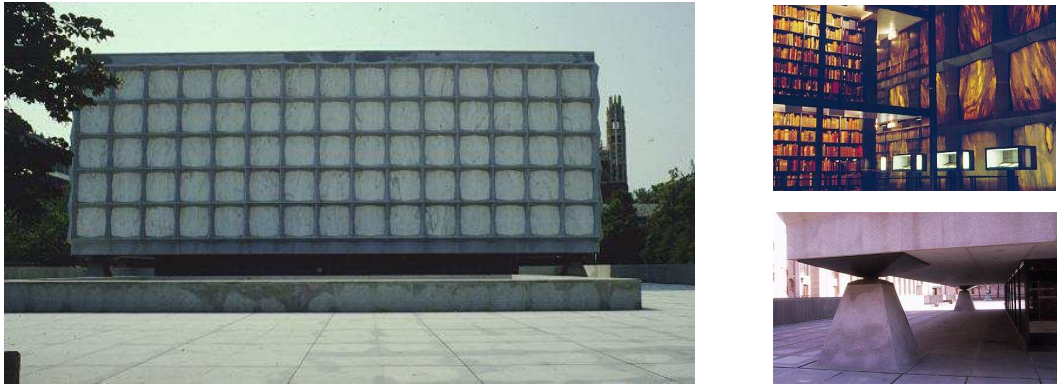
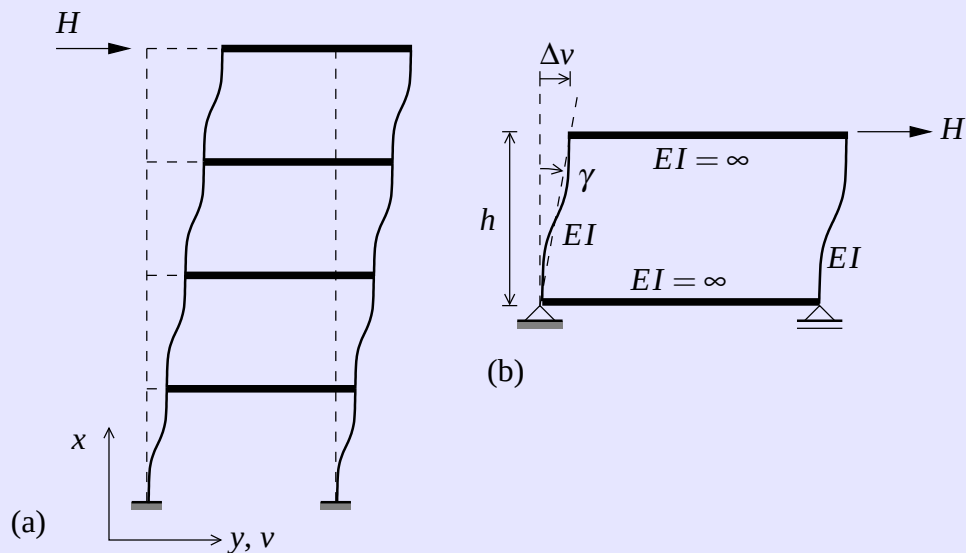


Figure 3.5 Beinecke Rare Books & Manuscripts Library. Length direction span: 131 feet (≈ 40 m) – Width direction span: 80 feet (≈ 25 m).



From simple considerations, the deflection Δv of the single story due to a horizontal force H is equal to

$$\Delta v = \frac{Hh^3}{24EI}, \quad (3.7)$$

where we have considered that the force H is equally distributed between the two columns. Under the assumption of small deflections, (3.7) can be re-written as

$$H = \frac{24EI}{h^2} \frac{\Delta v}{h} = \frac{24EI}{h^2} \gamma = k\gamma \quad (3.8)$$

which is similar to the constitutive equation of a shear beam where k is the shear stiffness of the portal (cf (3.2)).

Since the force transmitted to each floor is the same, this result is valid for all floors. As a consequence, all joints of the columns will remain on a straight line after shear deformation, similar to the deformation of a shear beam. Indeed, with top point load, the internal shearing force and $\frac{dv}{dx}$ are constant and the deflected shape is a straight line. Thus the deflection of a Vierendeel frame can be estimated using the shear beam deflection formula with the equivalent shear stiffness k from (3.8). The situation is however different in the case of a uniformly distributed load as shown in the next example.

Example 3.2 A shear frame with a distributed lateral load

When a distributed lateral load is applied to a shear frame, the deflection is parabolic. Consider the frame depicted below. Equation (3.5) is integrated once to obtain

$$V = -q_0x + C_1 \left(= GA_s \frac{dv}{dx} \right). \quad (3.9)$$

A second integration (or integration of (3.6)) yields

$$GA_s v = -\frac{1}{2}q_0x^2 + C_1x + C_2, \quad (3.10)$$

where C_1 and C_2 are integration constants that can be defined by means of the boundary conditions $v = 0$ at $x = 0$ and $V = 0$ at $x = L$. With these boundary conditions $C_1 = q_0L$ and $C_2 = 0$. Hence

$$V = q_0(L - x) \quad (3.11)$$

and

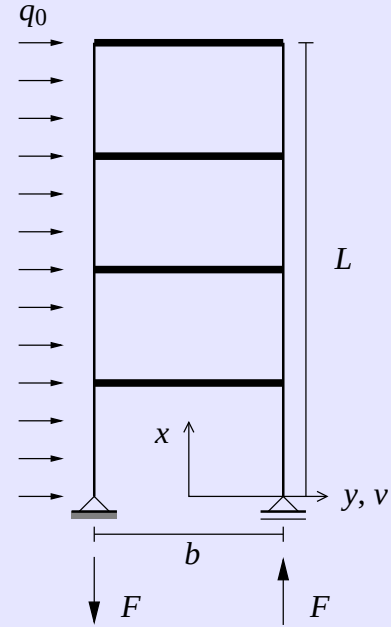
$$GA_s v = \frac{1}{2}q_0x(2L - x). \quad (3.12)$$

Extreme values are $v = 0$ and $V = q_0L$ at $x = 0$ and $v = q_0L^2/2GA_s$ and $V = 0$ at $x = L$.

In addition to the horizontal reaction force $V = q_0L$, the supports provide vertical reaction forces F . These reaction forces acting at distance b can be found by rotational equilibrium at one of the supports ($F = q_0L^2/2b$).

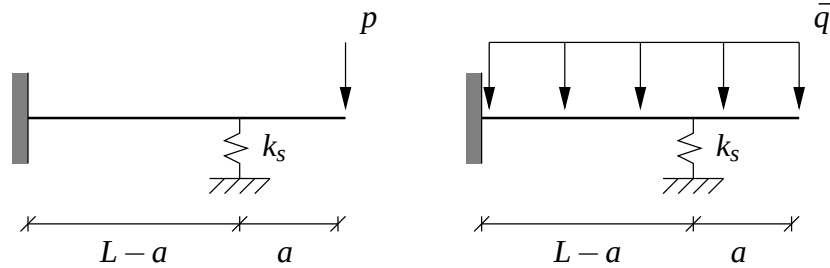
The internal bending moment M can be derived through integration of (2.6) with $M = 0$ at $x = L$ as boundary condition. A simple calculation yields

$$M = q_0x \left(L - \frac{x}{2} \right) - \frac{1}{2}q_0L^2. \quad (3.13)$$

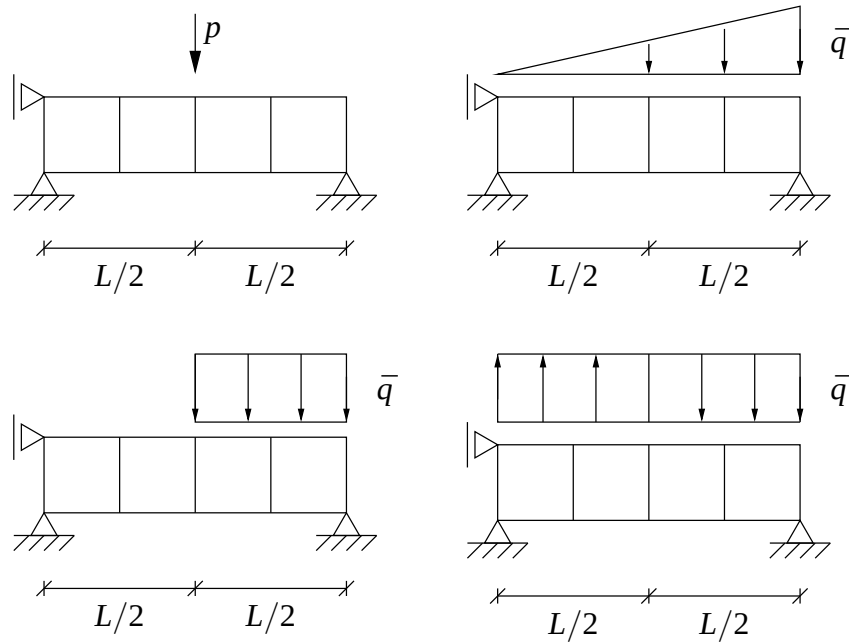


Exercises

3.1 Compare the deflected shape and the shear force diagram of the two beams below considering the shear beam theory and the Euler-Bernoulli beam theory. Discuss the influence of the linear spring and its position.



3.2 Determine the expression of the deflected shape and the shear force diagram of the beams below and sketch them. All the beams have shear stiffness k .



References

- [1] S. P. Timoshenko. On the correction for shear of the differential equation for transverse vibrations of prismatic bars. *Philosophical Magazine*, 41:744–746, 1921.
- [2] S. P. Timoshenko. On the transverse vibrations of bars of uniform cross-section. *Philosophical Magazine*, 43:125–131, 1922.
- [3] J. M. Gere and S. P. Timoshenko. *Mechanics of Materials*. Wadsworth, Inc., Belmont, California, second edition, 1984.

Chapter 4

Timoshenko beam theory

The Timoshenko beam theory [1, 2] is an extension of the Euler-Bernoulli beam theory that includes first-order transverse shear effect. The core assumption of the Euler-Bernoulli beam theory, i.e. plane cross sections perpendicular to the beam axis remain plane and perpendicular to the neutral axis during bending, is relaxed. This relaxation is introduced through an additional degree of freedom which describes the additional rotation to the bending slope. This extra rotation generates a shear strain. This beam model was presented in 1922 in the context of vibration and dynamics. Similar to the shear beam described in Section 3, a constant shear over the beam height is assumed.

4.1 Kinematic assumptions

The relaxation of the normality assumption of plane sections that remain plane and normal to the deformed centerline is what distinguishes the Timoshenko beam theory from the Euler-Bernoulli beam theory. In the Timoshenko beam theory there are two independent kinematic quantities: the transverse deflection $v(x)$ and the cross sectional rotation $\varphi(x)$ – φ is the rotation of the cross section with respect to the vertical axis or, equivalently, the rotation angle of the generic cross section with respect to his tangent.

Consider Figure 4.1. The displacement field for a point p at distance y from the center of



Stepan Prokofyevich Timoshenko

★ Shpotivka in Poltava Gubernia, Russia (now in Chernihiv Oblast, Ukraine), December 23, 1878

† Wuppertal, Germany, May 29, 1972

He is reputed to be the father of modern engineering mechanics. He wrote many of the seminal works in the areas of engineering mechanics, elasticity and strength of materials, many of which are still widely used today.

In 1957 the American Society of Mechanical Engineers established the Timoshenko Medal in his honor, and he was the first recipient of this annual award because “by his invaluable contributions and personal example, he guided a new era in applied mechanics.”

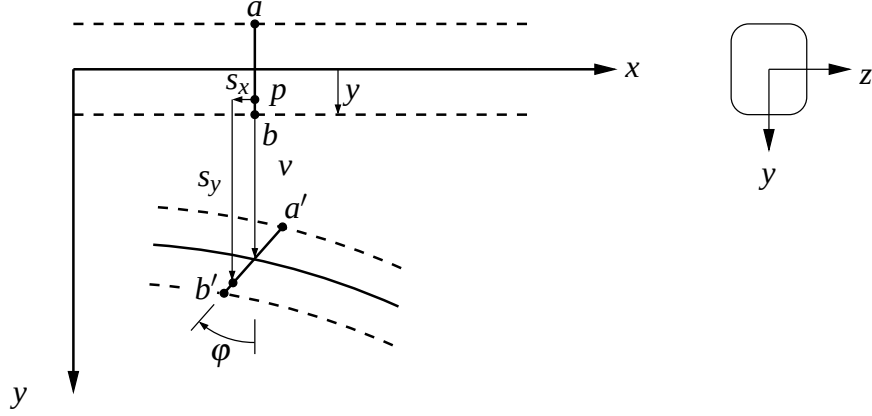


Figure 4.1

the beam on the cross section ab is described by

$$s_x(x, y) = -y\varphi(x) \quad \text{and} \quad s_y(x, y) = v(x). \quad (4.1)$$

From the displacement field we derive the non-zero components of the strain field as

$$\varepsilon_x = \frac{ds_x}{dx} = -y \frac{d\varphi}{dx} \quad \text{and} \quad \gamma_{xy} = \frac{ds_x}{dy} + \frac{ds_y}{dx} = -\varphi + \frac{dv}{dx}. \quad (4.2)$$

4.2 Relationships between deformations and internal forces

The shear deformation γ is related to the shear force through (3.4). By making use of (4.2)₂ we obtain the following expression for the shear force:

$$V = GA_s \gamma = GA_s \left(\frac{dv}{dx} - \varphi \right). \quad (4.3)$$

From Hooke's law and making use of (4.2)₁:

$$\sigma_x = E \varepsilon_x = -Ey \frac{d\varphi}{dx}. \quad (4.4)$$

By making use of the above expression for the longitudinal stress and following considerations similar to those reported in Section 2.4, the bending moment M is expressed as a function of the cross sectional rotation φ according to

$$M = -EI \frac{d\varphi}{dx}. \quad (4.5)$$

4.3 Limit cases

The shear beam and the classical beam theories are recovered by an appropriate choice of the flexural stiffness EI and the shear stiffness GA_s . The Euler-Bernoulli beam is recovered when $GA_s \rightarrow \infty$. In this case there is no shear and $\gamma \rightarrow 0$. This implies that $\frac{dv}{dx} \rightarrow \phi$ and $M \rightarrow -EI \frac{d^2v}{dx^2}$. When $EI \rightarrow \infty$ there is no bending ($\phi = 0$) and the shear beam is recovered. Only the shear deformation is present. In this case $\gamma \rightarrow \frac{dv}{dx}$ and $q = -\frac{dV}{dx} \rightarrow -GA_s \frac{d^2v}{dx^2}$.

4.4 The differential equations governing the transverse deflection and cross sectional rotation

The equilibrium of an infinitesimal beam segment is not affected by the added shear deformation. As a consequence, the relationships derived in Section 2.3 are still valid. In particular, we derive the differential equations governing the transverse deflection and cross sectional rotation of the Timoshenko beam by eliminating the shear force V and the bending moment M from (2.5) and (2.6).

Let us use the equilibrium relation (2.6). We consider V from (4.3) and express the first derivative of M from (4.5). Substituting these expressions of V and M into (2.6) yields

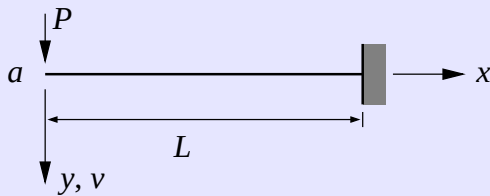
$$EI \frac{d^2\phi}{dx^2} + GA_s \left(\frac{dv}{dx} - \phi \right) = 0. \quad (4.6)$$

We now make use of the second equilibrium relation (2.5). As before, we consider V from (4.3) and replace its derivative in (2.5) to obtain

$$GA_s \left(\frac{d^2v}{dx^2} - \frac{d\phi}{dx} \right) = -q. \quad (4.7)$$

Equations (4.6) and (4.7) are two coupled second order differential equations governing the deflection v and the cross sectional rotation ϕ of the Timoshenko beam. The problem is fully determined with the definition of four boundary conditions.

Example 4.1



Consider a cantilever beam with a concentrated load P at the free end. In this case there is no distributed load and the governing equation (4.7) simplifies to

$$\frac{d^2v}{dx^2} = \frac{d\phi}{dx}. \quad (4.8)$$

Differentiating (4.6) once and combining it

with (4.8) yields

$$EI \frac{d^4v}{dx^4} = 0. \quad (4.9)$$

The differential equations (4.8) and (4.9) are the governing equations for the cantilever beam without distributed load. It is interesting to notice that (4.9) is the ordinary beam theory equation.

Direct integration of (4.9) yields:

$$EIv = C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4. \quad (4.10)$$

The problem is completely specified by the following boundary conditions: 1) $M = 0$ at $x = 0$; 2) $V = -P$ at $x = 0$; 3) $\phi = 0$ at $x = L$; 4) $v = 0$ at $x = L$. Application of the boundary conditions results in the following:

bc 1) $M = 0$ at $x = 0$ implies $\frac{d\phi}{dx} = 0$. By making use of (4.8) we have $C_2 = 0$;

bc 2) $V = -P$ at $x = 0$: we make use of (2.6) from which $V = -EI \frac{d^3v}{dx^3}$. The boundary condition implies $C_1 = P$;

bc 3) $\phi = 0$ at $x = L$: using (4.3) with $V = -P$ at $x = L$ implies $C_3 = -\frac{PEI}{GA_s} - \frac{PL^2}{2}$;

bc 4) $v = 0$ at $x = L$ implies $C_4 = \frac{PLEI}{GA_s} + \frac{PL^3}{3}$.

With these integration constants, the deflection reads

$$v = \underbrace{\frac{Px^3}{6EI} - \frac{PL^2x}{2EI} + \frac{PL^3}{3EI}}_{\text{bending}} + \underbrace{\frac{P}{GA_s}(L-x)}_{\text{shear}} \quad (4.11)$$

while the deflection at the free end is

$$v_a = \frac{PL^3}{3EI} + \frac{PL}{GA_s} = \frac{PL^3}{3EI} \left(1 + \frac{3EI}{GA_s L^2} \right). \quad (4.12)$$

The relative importance of the shear contribution can be appreciated by analyzing the last term in this equation. Usually, for very slender beams, the shear component can be neglected. However, for span/cross section height L/h such that the beam can be considered thick, the shear contribution becomes important.

Exercises

4.1 A Timoshenko beam is clamped at the two ends. A prescribed transverse deflection $\bar{u}$ is applied at one of the two ends. Compute the expression of the deflected shape and of the shearing force and bending moment diagrams and sketch them. Express the value of the reaction forces and moments at the two ends as a function of the parameter $\Phi = \frac{12EI}{GA_s L^2}$. Show that the shear beam and the Euler-Bernoulli beam results are recovered as limit cases (express the limit cases as function of Φ).

4.2 Compare the influence of the shear contribution in the above exercise and in Example 4.1. What are the factors that contribute the most? What is the role of the boundary conditions?

References

- [1] S. P. Timoshenko. On the correction for shear of the differential equation for transverse vibrations of prismatic bars. *Philosophical Magazine*, 41:744–746, 1921.
- [2] S. P. Timoshenko. On the transverse vibrations of bars of uniform cross-section. *Philosophical Magazine*, 43:125–131, 1922.

Chapter 5

Beams and frames on elastic foundation

In this chapter we shall study the behavior of beams and shear frames on elastic foundation. The foundation or soil can be replaced by a set of distributed linear elastic springs. Similar to the pull-out problem described in Section 1.5, we consider the force related to these spring to be proportional and opposite to the displacement. Sign convention follows those used in Chapters 2 and 3.

This chapter is based on [1]. A review of possible approaches to the study of beams on elastic foundation can be found in [2].

5.1 The differential equation of the elastic line

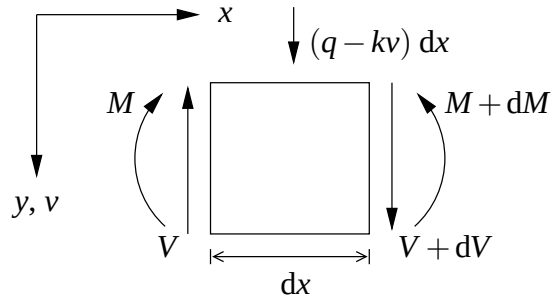


Figure 5.1

The only difference with the Euler-Bernoulli beam and the shear beam is the presence of a distributed load proportional to the displacement. Hence, we can start the derivation of the differential equation of the elastic line by considering the equilibrium of a differential element. With reference to Figure 5.1, equilibrium of forces in the vertical direction yields

$$\frac{dV}{dx} - kv = -q, \quad (5.1)$$

where k is the soil stiffness. The equilibrium equation obtained by summing moments about an axis through the left hand face of the element and orthogonal to the plane of the figure yields the same equation as (2.6).

5.1.1 The shear beam

With trivial manipulations, the differential equation of the deflection of a shear beam (cf (3.6)) can be expressed as

$$-GA_s \frac{d^2 v_s}{dx^2} + kv = q. \quad (5.2)$$

It is convenient to consider the following form of the homogeneous equation

$$\frac{d^2 v_s}{dx^2} - \alpha^2 v = 0 \quad (5.3)$$

with $\alpha^2 = \frac{k}{GA_s}$. This equation is identical to (1.13)₂ and we refer to Section 1.5 for its solution.

Particular solutions

Particular solutions account for the non-homogeneous term in (5.1). For the shear beam, some particular solutions $v(x)$ are as follows.

$$\begin{aligned} \text{Uniform distributed load } q(x) = q_0 : \quad v(x) &= \frac{q_0}{k} \\ \text{Load } q(x) = q_0 x : \quad v(x) &= \frac{q_0}{k} x \\ \text{Sinusoidal load } q(x) = q_0 \sin \frac{\pi x}{L} : \quad v(x) &= \frac{q_0}{GA_s \left(\frac{\pi}{L}\right)^2 + k} \sin \frac{\pi x}{L} \end{aligned}$$

5.1.2 The Euler-Bernoulli beam

The differential equation of the deflection curve of a Euler-Bernoulli beam supported on an elastic foundation can be derived using (5.1) and following the line of reasoning reported in Chapter 2. With some simple manipulation we obtain

$$EI \frac{d^4 v}{dx^4} + kv = q. \quad (5.4)$$

The homogeneous counterpart of (5.4) can be written as

$$\frac{d^4 v}{dx^4} + \frac{k}{EI} v = 0. \quad (5.5)$$

Substituting $v = e^{mx}$ in (5.5) we obtain the characteristic equation

$$m^4 + \frac{k}{EI} = 0 \quad (5.6)$$

which has the roots

$$m_1 = -m_3 = \lambda (1 + i), \quad m_2 = -m_4 = \lambda (-1 + i), \quad (5.7)$$

with $\lambda^4 = k/4EI$. The general solution of (5.5) is then

$$v = \sum_i^{1,4} A_i e^{m_i x} \quad (5.8)$$

which can be written in a more convenient form using

$$e^{i\lambda x} = \cos \lambda x + i \sin \lambda x, \quad e^{-i\lambda x} = \cos \lambda x - i \sin \lambda x \quad (5.9)$$

as

$$v = e^{\lambda x} (C_1 \cos \lambda x + C_2 \sin \lambda x) + e^{-\lambda x} (C_3 \cos \lambda x + C_4 \sin \lambda x). \quad (5.10)$$

The new integration constants C_1, C_2, C_3 and C_4 are related to the old ones through

$$C_1 = A_1 + A_4, \quad C_2 = i(A_1 - A_4), \quad C_3 = A_2 + A_3, \quad C_4 = i(A_2 - A_3). \quad (5.11)$$

The factor λ is called the *characteristics* of the system and has dimensions of length^{-1} . The term $1/\lambda$ is referred to as the *characteristic length*.

Equation (5.10) is the general solution for the deflection line of a straight prismatic Euler-Bernoulli beam supported on an elastic foundation under the action of transverse bending forces. An additional term is necessary if a distributed load q is present. The slope, the bending moment and the shearing force can be obtained by the relationships derived in Chapter 2.

The four integration constants C_1, C_2, C_3 and C_4 can be determined from conditions on the deflection v , the slope θ , the bending moment M or the shearing force V existing at the two ends of the beam.

Particular solutions

Particular solutions account for the non-homogeneous term in (5.4). For the Euler-Bernoulli beam, some particular solutions $v(x)$ are as follows.

$$\begin{aligned} \text{Uniform distributed load } q(x) = q_0 : \quad v(x) &= \frac{q_0}{k} \\ \text{Load } q(x) = \sum_{i=0}^{\max 3} a_i x^i : \quad v(x) &= \frac{q(x)}{k} \\ \text{Sinusoidal load } q(x) = q_0 \sin \frac{\pi x}{L} : \quad v(x) &= \frac{q_0}{\pi^4 \frac{EI}{L^4} + k} \sin \frac{\pi x}{L} \end{aligned}$$

5.2 Examples

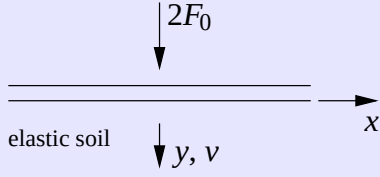
The solutions derived above, with the proper boundary conditions, can be used to solve a wide variety of problems. Below we illustrate some typical examples.

An interesting property of the solution of this class of problem lies in the fact that the principle of superposition can be applied without restrictions to all quantities of interest. Indeed, the deflection, the slope, the bending moment and the shearing force are directly proportional to the load.

Example 5.1 (Beam of infinite length subjected to a concentrated force)

Consider a beam of infinite length subjected to a concentrated force. We can assume that the transverse deflection is zero at infinite distance from the application of the load. This condition implies that the term connected to $e^{\lambda x}$ in (5.10) vanishes and

$$v = e^{-\lambda x} (C_3 \cos \lambda x + C_4 \sin \lambda x). \quad (5.12)$$



The problem is completely identified by imposing $V = -F_0$ at $x = 0$ and, because of symmetry due to the flexural stiffness of the beam, $\frac{dv}{dx} = 0$ at $x = 0$ (the beam has a horizontal tangent where the load is applied). With these boundary condition $C_3 = C_4 = F_0\lambda/k$ and

$$v = \frac{F_0\lambda}{k} e^{-\lambda x} (\cos \lambda x + \sin \lambda x) = \frac{F_0\lambda}{k} A_{\lambda x}, \quad (5.13a)$$

$$\theta = \frac{dv}{dx} = -\frac{2F_0\lambda^2}{k} e^{-\lambda x} \sin \lambda x = -\frac{2F_0\lambda^2}{k} B_{\lambda x}, \quad (5.13b)$$

$$M = -EI \frac{d^2v}{dx^2} = \frac{F_0}{2\lambda} e^{-\lambda x} (\cos \lambda x - \sin \lambda x) = \frac{F_0}{2\lambda} C_{\lambda x}, \quad (5.13c)$$

$$V = -EI \frac{d^3v}{dx^3} = -F_0 e^{-\lambda x} \cos \lambda x = -F_0 D_{\lambda x}, \quad (5.13d)$$

where we have made use of the following quantities:

$$A_{\lambda x} = e^{-\lambda x} (\cos \lambda x + \sin \lambda x), \quad B_{\lambda x} = e^{-\lambda x} \sin \lambda x,$$

$$C_{\lambda x} = e^{-\lambda x} (\cos \lambda x - \sin \lambda x), \quad D_{\lambda x} = e^{-\lambda x} \cos \lambda x,$$

which are related through the following relations:

$$\frac{dA_{\lambda x}}{dx} = -2\lambda B_{\lambda x}, \quad \frac{dB_{\lambda x}}{dx} = \lambda C_{\lambda x}, \quad \frac{dC_{\lambda x}}{dx} = -2\lambda D_{\lambda x}, \quad \frac{dD_{\lambda x}}{dx} = -\lambda A_{\lambda x}.$$

It is worth noticing that when $\lambda x > 1.5\pi$, the value of the functions $A_{\lambda x}$, $B_{\lambda x}$, $C_{\lambda x}$ and $D_{\lambda x}$ is under 0.01. This means that the support conditions of the beam at any point $x > 1.5\pi/\lambda$ from the application of the load does not influence the shape of the deflection line. In other words, a beam of length $l = 3\pi/\lambda$ loaded with a concentrated force P at the middle will exhibit approximately the same deflection curve as an infinitely long beam with the same applied load [1].

Example 5.2 (Beam of infinite length subjected to a concentrated moment)

Following a procedure similar to that used in Example 5.1, we can solve the case of an infinite beam on elastic soil with a clockwise concentrate moment $2M_0$. The following solution

is valid for $x > 0$:

$$v = \frac{2M_0\lambda^2}{k} B_{\lambda x}, \quad (5.14a)$$

$$\theta = \frac{dv}{dx} = \frac{2M_0\lambda^3}{k} C_{\lambda x}, \quad (5.14b)$$

$$M = -EI \frac{d^2v}{dx^2} = M_0 D_{\lambda x}, \quad (5.14c)$$

$$V = -EI \frac{d^3v}{dx^3} = -M_0 \lambda A_{\lambda x}. \quad (5.14d)$$

For those points on the left of the point of application of the moment, the sign of v and M must be reversed. Note that the arguments of the functions $A_{\lambda x}$, $B_{\lambda x}$, $C_{\lambda x}$ and $D_{\lambda x}$ are always taken as positive, irrespective of the location of x with respect to the point of application of the moment.

Example 5.3 (How to speed-up derivations)

Considering again the beam of infinite length subjected to a concentrated force in Example 5.1, we can express (5.13) in a format that allows a quick computation of the various derivatives involved. To this end, we set the integration constants in (5.12) as

$$C_3 = A \sin \omega \quad \text{and} \quad C_4 = A \cos \omega \quad (5.15)$$

with which we have

$$v = e^{-\lambda x} (A \sin \omega \cos \lambda x + A \cos \omega \sin \lambda x) = A e^{-\lambda x} \sin(\lambda x + \omega). \quad (5.16)$$

This is the expression of a sinusoidal curve with decreasing amplitude $A e^{-\lambda x}$, angular frequency λ and phase angle ω . Its derivative is equal to

$$\frac{dv}{dx} = -\lambda A e^{-\lambda x} \sin(\lambda x + \omega) + \lambda A e^{-\lambda x} \cos(\lambda x + \omega), \quad (5.17)$$

which can be expressed as

$$\frac{dv}{dx} = -\lambda \sqrt{2} A e^{-\lambda x} \sin\left(\lambda x + \omega - \frac{\pi}{4}\right) \quad (5.18)$$

if we multiply the first terms by $\sqrt{2} \cos \pi/4 (= 1)$ and the second by $\sqrt{2} \sin \pi/4 (= 1)$. It is worth noticing that the differentiation implied the multiplication of the amplitude by the factor $-\lambda \sqrt{2}$ and a phase decrease of $\pi/4$. Hence

$$M = -EI \frac{d^2v}{dx^2} = -2\lambda^2 E I A e^{-\lambda x} \sin\left(\lambda x + \omega - \frac{\pi}{2}\right), \quad (5.19)$$

and

$$V = -EI \frac{d^3v}{dx^3} = 2\sqrt{2}\lambda^3 EIA e^{-\lambda x} \sin\left(\lambda x + \omega - \frac{3\pi}{4}\right). \quad (5.20)$$

The constants A and ω are found with the boundary conditions at $x = 0$. From the condition on the slope we find that $\omega = \pi/4$ and from the condition on the shearing force $A = F_0 \lambda \sqrt{2}/k$. Hence

$$v = \frac{F_0 \lambda \sqrt{2}}{k} e^{-\lambda x} \sin\left(\lambda x + \frac{\pi}{4}\right), \quad (5.21a)$$

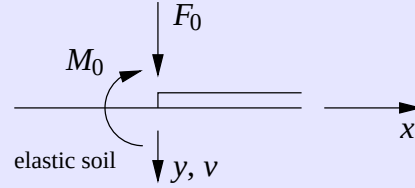
$$\theta = \frac{dv}{dx} = -\frac{2F_0 \lambda^2}{k} e^{-\lambda x} \sin \lambda x, \quad (5.21b)$$

$$M = -EI \frac{d^2v}{dx^2} = -\frac{F_0}{\sqrt{2}\lambda} e^{-\lambda x} \sin\left(\lambda x - \frac{\pi}{4}\right), \quad (5.21c)$$

$$V = -EI \frac{d^3v}{dx^3} = F_0 e^{-\lambda x} \sin\left(\lambda x - \frac{\pi}{2}\right). \quad (5.21d)$$

Example 5.4 (Beam of semi-infinite length)

Consider a semi-infinite beam on elastic foundation under the action of two concentrated loads, force and bending moment. Equation (5.10) is the general solution for the deflection of an Euler-Bernoulli beam on elastic foundation. Since $w(x) \rightarrow 0$ for $x \rightarrow \infty$, we must have $C_1 = C_2 = 0$. The boundary conditions at $x = 0$ determine C_3 and C_4 :



$$M(0) = -EI \frac{d^2v}{dx^2}(0) = M_0 \rightarrow C_4 = \frac{2\lambda^2 M_0}{k},$$

$$V(0) = -EI \frac{d^3v}{dx^3}(0) = -F_0 \rightarrow C_3 = \frac{2\lambda F_0}{k} - \frac{2\lambda^2 M_0}{k}.$$

Armed with these expressions we find

$$w(x) = \frac{2\lambda F_0}{k} D_{\lambda x} - \frac{2\lambda^2 M_0}{k} C_{\lambda x},$$

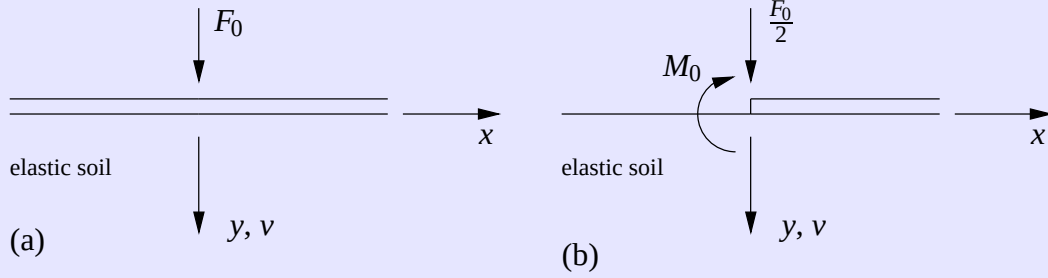
$$\frac{dv}{dx}(x) = -\frac{2\lambda^2 F_0}{k} A_{\lambda x} + \frac{4\lambda^3 M_0}{k} D_{\lambda x},$$

$$M(x) = -EI \frac{d^2v}{dx^2}(x) = -\frac{F_0}{\lambda} B_{\lambda x} + M_0 A_{\lambda x},$$

and

$$V(x) = -EI \frac{d^3 v}{dx^3}(x) = -F_0 C_{\lambda x} - 2M_0 \lambda B_{\lambda x}.$$

Example 5.5 (An application of the principle of superposition)



By using the expressions derived in Example 5.4, we can determine the solution for a beam of infinite length on elastic foundation under the action of a concentrated force.

Indeed, the beam on the left-hand side of the figure can be equivalent to the beam depicted in the lower part. The difference between these two cases lies in the slope $\frac{dv}{dx}$ at the point of application of the force which, for the beam on the left-hand side of the figure is zero. We can make use of this fact to derive a boundary condition at $x = 0$ for the beam in the upper part. At $x = 0$ the slope derived in the previous example is

$$\frac{dv}{dx} = -\frac{2\lambda^2 F_0}{k} + \frac{4\lambda^3 M_0}{k},$$

where we have used a load of intensity $F_0/2$. By setting the slope to zero, we can derive the value of the bending moment that neutralize the slope created by the concentrated force. Proceeding along this line, we obtain

$$M_0 = \frac{F_0}{4\lambda}.$$

Finally, using the expressions from the previous example with $F_0/2$ and $M_0 = F_0/4\lambda$, we obtain the solution for the beam of infinite length:

$$w(x) = \frac{\lambda F_0}{2k} A_{\lambda x},$$

$$\frac{dv}{dx}(x) = -\frac{\lambda^2 F_0}{k} B_{\lambda x},$$

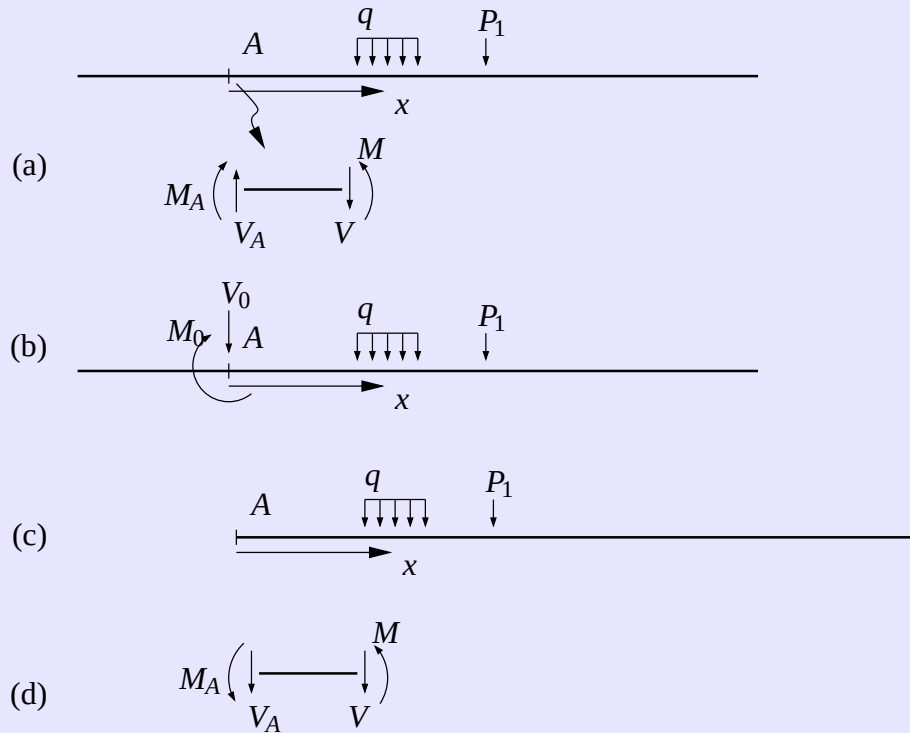
$$M(x) = -EI \frac{d^2 v}{dx^2}(x) = \frac{F_0}{4\lambda} C_{\lambda x},$$

$$V(x) = -EI \frac{d^3 v}{dx^3}(x) = -\frac{F_0}{2} D_{\lambda x},$$

which can be compared to the expressions reported in (5.13). These expressions are valid for $x > 0$. The expressions for $x < 0$ are obtained from the symmetry and antisymmetry conditions: $w(x) = w(-x)$, $\frac{dv}{dx}(x) = -\frac{dv}{dx}(-x)$, $M(x) = M(-x)$, $V(x) = -V(-x)$.

Example 5.6 (Another application of the principle of superposition)

A semi-infinite beam can be solved by using the expressions derived for the beam of infinite length. The free end can be free, fixed or hinged [1, page 10]. To make the idea clear, consider a beam of infinite length (Figure (a)) under an arbitrary loading condition. At a point A there exists an internal moment M_A and an internal shearing force V_A . These two forces maintain the continuity of the beam. If they were both equal to zero we could simply consider the semi-infinite beams on the right and on the left of A as two separate beams. Hence we have to define a strategy to make these internal forces zero on the beam of infinite length. This can be easily accomplished by applying at A an external bending moment M_0 and a shearing force V_0 like depicted in Figure (b) such that the internal forces M_A and V_A are equal to zero at that point.



To determine the expression of the forces that we have to apply we make use of (5.14) for a concentrated moment of intensity $2M_0$ and (5.13) for a concentrated force of intensity

$2V_0$. We then know that a moment of intensity M_0 will produce $M = M_0 D_{\lambda x}/2$ and $V = -M_0 \lambda A_{\lambda x}/2$, and a force V_0 will produce $M = V_0 C_{\lambda x}/4\lambda$ and $V = -V_0 D_{\lambda x}/2$. Our objective is to let these external force and moment generate internal forces opposite and equal to M_A and V_A . Therefore, the equilibrium condition that needs to be fulfilled is like that depicted in Figure (d) which, in combination with the above expressions for internal shearing force and bending moment, gives

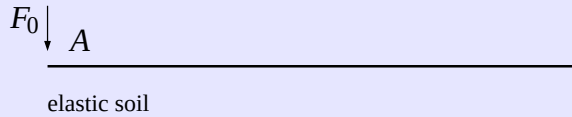
$$\frac{M_0}{2} + \frac{V_0}{4\lambda} + M_A = 0 \quad \text{and} \quad V_A - \frac{V_0}{2} - \frac{M_0 \lambda}{2} = 0, \quad (5.22)$$

from which

$$M_0 = -4 \left(M_A + \frac{V_A}{2\lambda} \right) \quad \text{and} \quad V_0 = 4(\lambda M_A + V_A). \quad (5.23)$$

We may say that V_0 and M_0 in (5.23) are such that the beam in Figure (b) and Figure (c) are identical for $x > 0$. We call V_0 and M_0 the end-conditioning forces.

Example 5.7 (Semi-infinite beam with a concentrated load)



Consider the semi-infinite beam with a concentrated load at the free end shown beside. The boundary conditions at point A are $M = 0$ and $V = -F_0$. By making use of the scheme depicted in Figure (d) of Example 5.6, we found

that $M_A = 0$ and $V_A = F_0$. Substituting these values into (5.23) we obtain the corresponding end-conditioning forces $V_0 = 4F_0$ and $M_0 = -2F_0/\lambda$. If we apply these forces on the infinite beam, using (5.14) for the concentrated moment of intensity M_0 and (5.13) for a concentrated force of intensity V_0 , we obtain the solution for $x > 0$:

$$v = \frac{2F_0 \lambda}{k} D_{\lambda x}, \quad (5.24a)$$

$$\theta = -\frac{2F_0 \lambda^2}{k} A_{\lambda x}, \quad (5.24b)$$

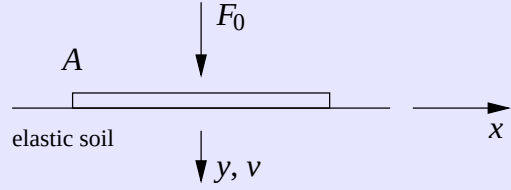
$$M = -EI \frac{d^2 v}{dx^2} = -\frac{F_0}{\lambda} B_{\lambda x}, \quad (5.24c)$$

$$V = -EI \frac{d^3 v}{dx^3} = -F_0 C_{\lambda x}. \quad (5.24d)$$

The same results could have been obtained by directly integrating (5.10) with the boundary conditions $v = 0$ for $x \rightarrow \infty$ ($C_1 = C_2 = 0$) and $M = 0$, $V = -F_0$ at $x = 0$ ($C_4 = 0$, $C_3 = 2F_0 \lambda/k$). Compare these results with those obtained in Example 5.4 by letting $M_0 = 0$.

Example 5.8 (Beam of finite length)

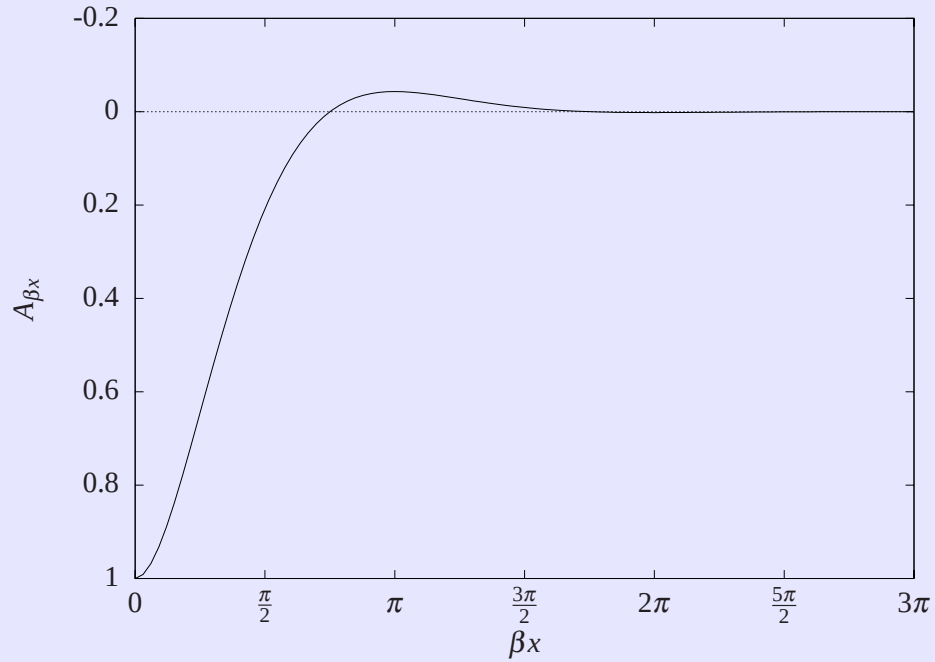
Consider a beam of finite length l on elastic foundation under the action of a concentrated load F_0 . We would like to determine the length of the beam so that the deflection $v(x)$ derived for a beam of infinite length can be used with confidence in this case.



The deflection $v(x)$ can be expressed by

$$v = \frac{F_0 \lambda}{2k} e^{-\lambda x} (\cos \lambda x + \sin \lambda x) = \frac{F_0 \lambda}{2k} A_{\lambda x},$$

with the function $A_{\lambda x}$ shown in the figure below.



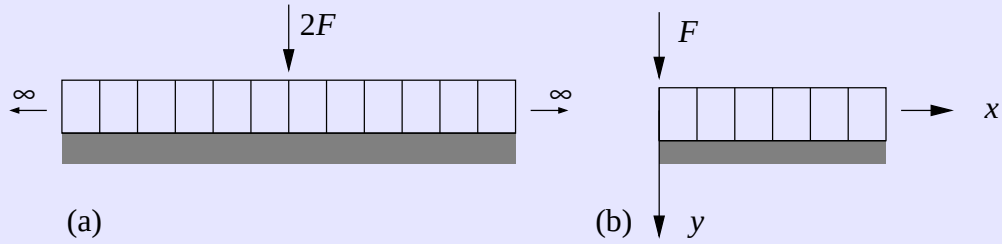
The length of the beam can be determined by evaluating the function $A_{\lambda x}$ at a few points as shown below.

λx	$A_{\lambda x}$	$ A_{\lambda x} $ [%]
0	+0.10000E + 01	100.000
$\frac{\pi}{2}$	+0.20788E + 00	20.788
π	-0.43214E - 01	4.321
$\frac{3\pi}{2}$	-0.89833E - 02	0.898
2π	+0.18674E - 02	0.187
$\frac{5\pi}{2}$	+0.38820E - 03	0.039
3π	-0.80700E - 04	0.008

When $\lambda x > \frac{3\pi}{2}$, $|A_{\lambda x}| < 1\%$. This means that for points at a distance larger than $\frac{3}{2} \frac{\pi}{\lambda}$ from the

point of application of the concentrated force, the effect of the soil stiffness on the deflection can be neglected. Therefore, a beam of length $l > 2\frac{3}{2}\frac{\pi}{\lambda}$ with a concentrated load applied at midspan exhibits approximately the same deflection curve as an infinitely long beam under the action of a concentrated load of the same intensity.

Example 5.9 (A shear beam of infinite length on an elastic foundation)



Consider a shear beam of infinite length on an elastic foundation subjected to a concentrated force. We make use of symmetry and study only half of the beam with half of the load. The governing equation is (5.3) which is analogous to the equation for the axial deformation problem that we have studied in Section 1.5. Hence

$$v = \frac{F}{GA_s \alpha} e^{-\alpha x} \quad \text{and} \quad V = -F e^{-\alpha x}. \quad (5.25)$$

In deriving these expressions, we have considered the following boundary conditions: $v = 0$ for $x \rightarrow \infty$ and $V = -F$ at $x = 0$. These expressions are valid for $x > 0$. Because of symmetry conditions, the displacement function is an even function (it is the same on both sides of the vertical axis) which implies $v(x) = v(-x)$. On the other hand, the shear, being the derivative of the displacement field according to $V = GA_s \frac{dv}{dx}$ is an odd function which implies that $V(x) = -V(-x)$.

5.3 Classification of beams according to stiffness

This section is based on [1, Section 17]. The quantity λl characterizes the relative stiffness of a beam on an elastic foundation. This quantity determines the magnitude of the curvature of the elastic line and defines the rate at which the effect of a loading force dies out in the form of a damped wave along the length of the beam. According to these λl values we may classify beams in three groups:

- I Short beams ($\lambda l < \pi/4$): we can neglect the bending deformation of the beam as it is small compared with the deformation produced in the foundation. Hence, beams with $\lambda l < \pi/4$ can be considered rigid;
- II Beams of medium length ($\pi/4 < \lambda l < \pi$): we need to do accurate computation of

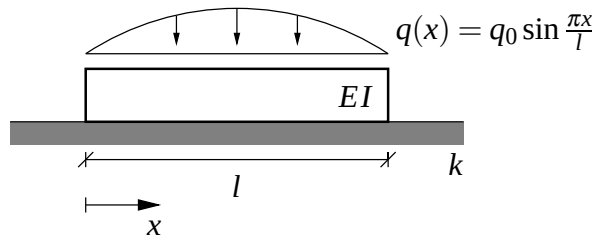
the beam. These beams are such that a force acting at one end has a finite and not negligible effect at the other end;

- III Long beams ($\lambda l > \pi$): These beams are such that a force acting at one end has a negligible effect at the other end. This means that λl is so large that we can take in all the formulas $A_{\lambda l} = B_{\lambda l} = C_{\lambda l} = D_{\lambda l} = 0$.

The classification is made from a practical point of view since it offers the possibility of using simplifications by neglecting certain quantities in particular instances. Of course, these limits depend on the accuracy required in the computations.

Exercises

- 5.1 Determine the deflection and bending moment at $x = l/2$. Discuss the role of λl .



5.2

References

- [1] M. Hetényi. *Beams on elastic foundation*. The University of Michigan Press, eight edition, 1967.
- [2] Y. H. Wang, L. G. Tham, and Y. K. Cheung. Beams and plates on elastic foundations: A review. *Progress in Structural Engineering and Materials*, 7(4):174–182, 2005.

Chapter 6

Transverse deflection of cables

Flexible cables are used in suspension bridges, transmission lines, lifts and in many other structures. In the design of these structures it is necessary to know the relation between cable sag, tension and span – cable sag is defined as the maximum vertical displacement of the cable. We shall determine these quantities by examining the cable as a body in equilibrium. In the analysis of flexible cables we assume that any resistance offered to bending is negligible. This implies that the force in the cable is always in the direction of the cable. In these problems we are concerned with the stiffness or flexibility of cables rather than with their strength. In the following derivations, the mechanical model is defined by the cable in its loaded, or deformed, configuration. The cable is assumed to be a very flexible string able to resist tensile forces only. The cable assumes a configuration which is known as the funicular curve of the load applied to the cable – a funicular curve is a curve in which the bending moment at any point is theoretically zero for a given transverse load.

This chapter is based on [1, 2, 3].

6.1 Kinematic relation

Consider a differential element of cable as shown in Figure 6.1. The primary unknown is the transverse deflection $y(x)$. With simple geometrical consideration we derive $dy = \tan \alpha dx$ from which $\frac{dy}{dx} = \tan \alpha$. Note that we do not approximate $\tan \alpha$ with α since the effect of loads on the overall geometry of cables cannot be neglected. Therefore, the superposition principle does not hold.

6.2 Constitutive relation

Unlike the previous cases, the kinematic parameter is a geometrical quantity which can be related to a force by considering the decomposition of the cable tension T into its vertical and horizontal components as shown in Figure 6.1:

$$V = H \tan \alpha. \quad (6.1)$$

Note that H does not depend on the coordinate x since by horizontal equilibrium $dH = 0$ in the absence of horizontally applied loads.

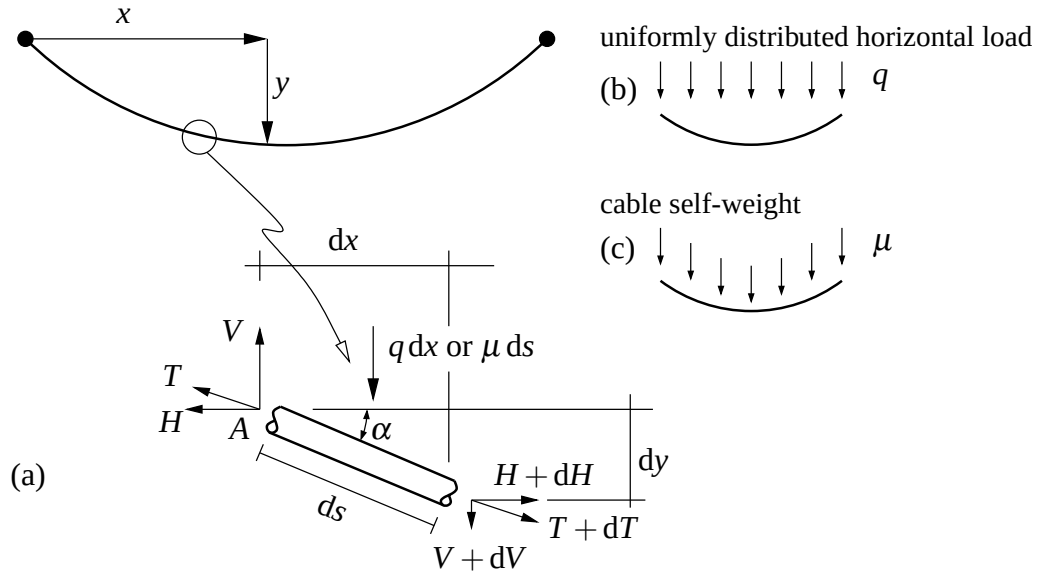


Figure 6.1

6.3 Governing equation

Depending upon the loading condition, the cable can be described by a parabolic curve or a hyperbolic cosine curve. When a load of intensity q is uniformly distributed along the horizontal projection of the cable, like in a suspension bridge, the cable deforms according to a parabolic curve (parabolic cable). When a cable sags under the action of its own weight, under the influence of gravity, its shape can be described by a hyperbolic cosine curve (catenary cable). In a catenary the vertical load on the chain is uniform with respect to the arc length.

6.3.1 The parabolic cable

Consider Figure 6.1. In a parabolic cable, the equilibrium in the vertical direction yields

$$\frac{dV}{dx} = -q, \quad (6.2)$$

while the rotational equilibrium around point A, neglecting second order terms of the type $(dx)^2$, gives

$$V = H \frac{dy}{dx}. \quad (6.3)$$

We may then express the relation between the deflection y and the applied load q through

$$-H \frac{d^2y}{dx^2} = q. \quad (6.4)$$

Successive integrations of (6.4) yield

$$y = -\frac{q}{H} \frac{x^2}{2} + C_1 x + C_2, \quad (6.5)$$

where the integration constants can be determined by considering $y(0) = 0$ and $y(l) = 0$, where l is the cable span ($C_2 = 0$, $C_1 = q_0 l / 2H$). Hence

$$y = \frac{q_0}{2H} x(l - x) \quad (6.6)$$

and

$$V = q_0 \left(\frac{l}{2} - x \right). \quad (6.7)$$

Note that the deflection y is a function of H .

In this case the deflection at the mid-point of the cable is the cable sag f and it is equal to

$$f = \frac{ql^2}{8H}. \quad (6.8)$$

The expression of the deflection at midspan holds also in the more general case when f is measured from the middle of the line joining the ends of the cable (see Figure 6.10).

As a side remark, the equilibrium equation (6.3) has been derived considering a deformed configuration and is therefore a geometrically non-linear equation. Although this differential equation is similar to the previous differential equations related to axial extension and shear deformation, it describes a different equilibrium state.

6.3.2 The catenary cable

The catenary is the curve described by a uniform, perfectly flexible chain hanging under the influence of gravity. Its equation was obtained by Leibniz, Huygens and Johann Bernoulli in 1691 who responded to a challenge put out by Jacob Bernoulli to find the equation of the "chain curve". Galileo (1564–1642) claimed that the curve of a chain under gravity would be a parabola (this was disproved by Jungius in 1669). Nonetheless, the shape of suspension bridge chains or cables, tied to the bridge deck at uniform intervals, is that of a parabola. As a side remark, it "is interesting to note that when suspension bridges are constructed, the suspension cables initially sag as the catenary function, before being tied to the deck below, and then gradually assume a parabolic curve as additional connecting cables are tied to connect the main suspension cables with the bridge deck below" [4] (see Figure 6.2).

Equation (6.4) is not valid for a cable hanging under the influence of gravity as it was derived considering a uniformly distributed horizontal load q . A simple patch to (6.4) consists in replacing $q(x)$ by the cable weight. This means that the resultant $q dx$ must be equal to μds where μ is the weight per unit length of the cable. Given that the infinitesimal length ds of the cable is equal to

$$ds = \sqrt{dy^2 + dx^2}, \quad (6.9)$$



Figure 6.2 Hercilio Luz Bridge (City of Florianopolis, state of Santa Catarina (SC), Brazil; picture taken by Sérgio Schmiegelow Cesarious [4]).



Figure 6.3 “The Capilano Suspension Bridge is a simple suspension bridge crossing the Capilano River in the District of North Vancouver, British Columbia, Canada. The current bridge is 136 meters long and 70 meters above the river. The current bridge was built in 1956. The cables are encased in 11.8 tonnes of concrete at either end” [5].

we obtain

$$\frac{d^2y}{dx^2} = -\frac{\mu}{H} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \quad (6.10)$$

This is the differential equation of the catenary curve assumed by the cable. Integration with the boundary conditions $y(0) = 0$ and $y(l) = 0$ yields [1, equation (1.7)]

$$y = \frac{H}{\mu} \left(\cosh \frac{\mu l}{2H} - \cosh \left(\frac{\mu}{H} \left(\frac{l}{2} - x \right) \right) \right). \quad (6.11)$$

The catenary is a hyperbolic cosine curve, and its slope varies as the hyperbolic sine.

A friendlier version of the catenary expression can be derived by placing the origin of the coordinate system at the point in which the curve has a horizontal tangent and considering the vertical axis pointing upwards. With the boundary conditions $\frac{dy}{dx} = y = 0$ at $x = 0$, the expression of the catenary becomes

$$y = \frac{H}{\mu} \left(\cosh \frac{\mu x}{H} - 1 \right). \quad (6.12)$$

It is worth noting that the lowest term of a Taylor series expansion of the above catenary curve yields

$$y = \frac{\mu x^2}{2H} \quad (6.13)$$

which can also be obtained by (6.4) through simple derivations setting $\mu = q$.

For small sag-to-span ratios the geometry of a catenary and a parabola are practically the same. If the ratio of sag to span is 1:8 or less, a uniform cable hanging under its own weight between two supports at the same level can be accurately described by (6.4) with the substitution $q = \mu$. This approximation is equivalent to ignoring the term $(dy/dx)^2$ in comparison to unity in (6.10). The parabolic assumption for flat profile cables is accurate enough even with ratios sag to span up to 1:5.

6.4 The horizontal component of the cable tension

In the previous derivations we have assumed that the horizontal component H of the cable tension T is known in advance. In this section we seek the relation between H and the applied load. We shall show that this relation is non linear.

Consider the cable shown in Figure 6.4. The left-hand side end is fixed while the right-hand side can move horizontally and is the point of application of the horizontal force H . The cable is under the action of a distributed load q which is expressed as a function of the cable tension T at $x = l$ through $q = \lambda T(l)/l$, where $\lambda > 0$ is a load factor.

horizontal components to give

By using Phytagoras' theorem we obtain the relation $H^2 = T^2(x) - V^2(x)$. Given that the distributed load is expressed as a function of $T(l)$, we can express $V(x)$ at $x = l$ and factor the common term $T(l)$. To this end, armed with the expression

$$V(x) = -qx + \frac{1}{2}ql$$

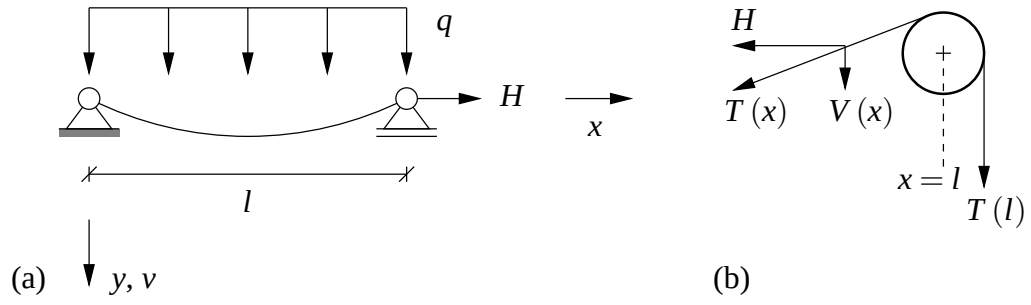


Figure 6.4

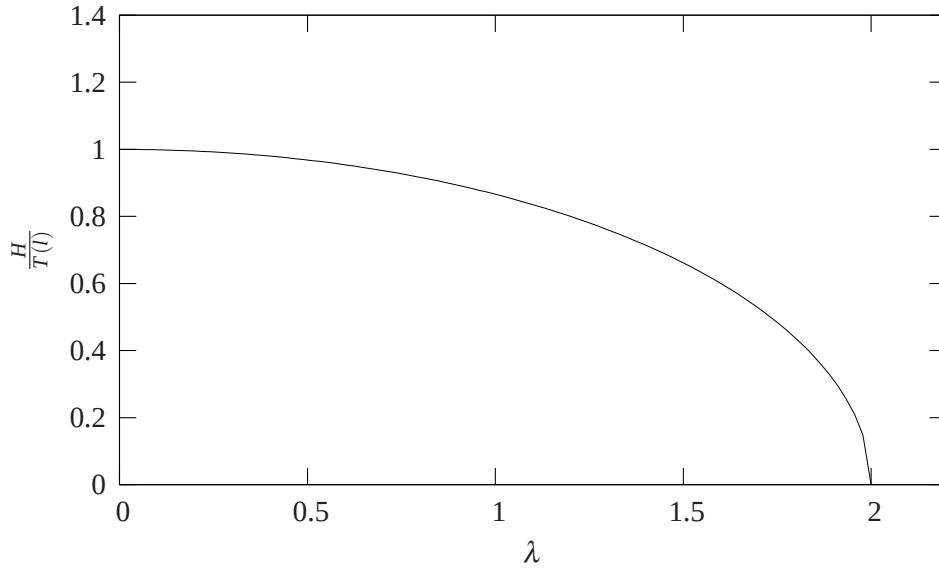


Figure 6.5

of the vertical component V of the cable tension, we determine its value at $x = l$:

$$V(l) = -\frac{1}{2}ql = -\frac{1}{2}\lambda T(l).$$

By making use of the expression $H^2 = T^2(l) - V^2(l)$ we can express the horizontal component of the cable tension as

$$H = T(l) \sqrt{1 - \frac{1}{4}\lambda^2}.$$

The principle of superposition of the horizontal component of the cable tension is not valid since the relation between H and the applied load, expressed through the load factor λ , is not linear as shown in Figure 6.5.

6.5 The relationship between the length of the cable and its sag

Consider a uniform cable of span l and length L hanging under the action of a uniformly distributed load q between two supports at the same level. The uniformly distributed load represents the cable weight – with this assumption we have already approximated the catenary shape of the cable by a parabola. Under these circumstances, we have

$$f = y\left(\frac{l}{2}\right) = \frac{ql^2}{8H}. \quad (6.14)$$

We now seek the relationship between the length of the cable and its sag f . Consider an infinitesimal slice dx of the cable as shown in Figure 6.1. The corresponding cable length ds can be approximated by

$$ds = \sqrt{dy^2 + dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx. \quad (6.15)$$

Since the cable length L is the sum from 0 to L of all the infinitesimal slices ds , we have

$$L = \sum \lim_{ds \rightarrow 0} ds = \int_0^L ds. \quad (6.16)$$

By replacing ds with the expression derived above and changing the range $[0, L]$ related to L with the corresponding x range $[0, l]$, we obtain

$$L = \int_0^l \sqrt{1 + \frac{dy^2}{dx}} dx \quad (6.17)$$

which can be approximated by making use of the binomial theorem as

$$\begin{aligned} L &= \int_0^l \left(1 + \frac{dy^2}{dx}\right)^{1/2} dx = \int_0^l \left(1 + \frac{1}{2} \frac{dy^2}{dx} - \frac{1}{8} \frac{dy^4}{dx} + \dots\right) dx \\ &= l + \frac{1}{2} \int_0^l \frac{dy^2}{dx} dx - \frac{1}{8} \int_0^l \frac{dy^4}{dx} dx + \dots \end{aligned} \quad (6.18)$$

The above formula is valid only for $|\frac{dy}{dx}| < 1$. We can now approximate the difference Δ between cable length L and cable span l by

$$\Delta = L - l \approx \frac{1}{2} \int_0^l \frac{dy^2}{dx} dx, \quad (6.19)$$

where we have kept only the first terms of the expansion. If we now assume that the cable deflection can be represented by the parabolic curve (6.6), we have

$$\Delta = \frac{8}{3} \frac{f^2}{l} = \frac{q^2 l^3}{24H^2}, \quad (6.20)$$

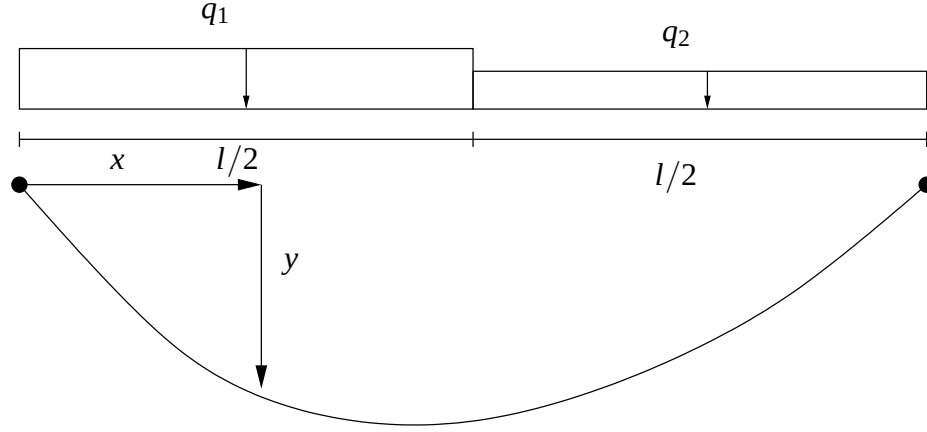


Figure 6.6

from which we can express the cable length as $L = l + \Delta$.

The approximation (6.19) is valid in most of the cases of practical interest. The correction to Δ is indeed very small for a sag-to-span ratio $f/l = 0.1$:

$$\Delta = \frac{8}{3} \frac{f^2}{l} \left(1 - \frac{12}{5} \frac{f^2}{l^2} + \dots \right) = \frac{8}{3} \frac{1}{10} f (1 - 0.024 + \dots). \quad (6.21)$$

6.5.1 On the principle of superposition for a cable under non-uniform load

The peculiarity of cables stems from their extreme flexibility resulting in big structural changes which are very much influenced by how the load is applied. Consider for instance the system in Figure 6.6. The differential equation (6.4) cannot be used due to the discontinuity at the midspan. We can however solve the differential equation in each of the two domains and “join” the two contributions by means of some compatibility conditions at the midspan. The problem is completely specified by considering two boundary conditions at the two ends (deflection equal to zero) plus a displacement compatibility and an equilibrium condition at the midspan – the latter matching condition implies that the tension, i.e. the vertical forces in the cable, must be the same. By using these four conditions we derive the expression of the deflection in the two parts and, by (6.19) and $\Delta = L - l$, given L and l , the horizontal force

$$H^2 = \frac{l^3}{384\Delta} (5q_1^2 + 6q_1q_2 + 5q_2^2). \quad (6.22)$$

Evidently, the principle of superposition cannot be applied.

Let us now take a different look at the non-uniform load in this example. It is evident that we can consider that non-uniform load as the sum of a uniform and a contra-symmetric load as shown in Figure 6.7. By means of p_0 and q_0 we redefine the applied load as $q_1 = q_0 + p_0$

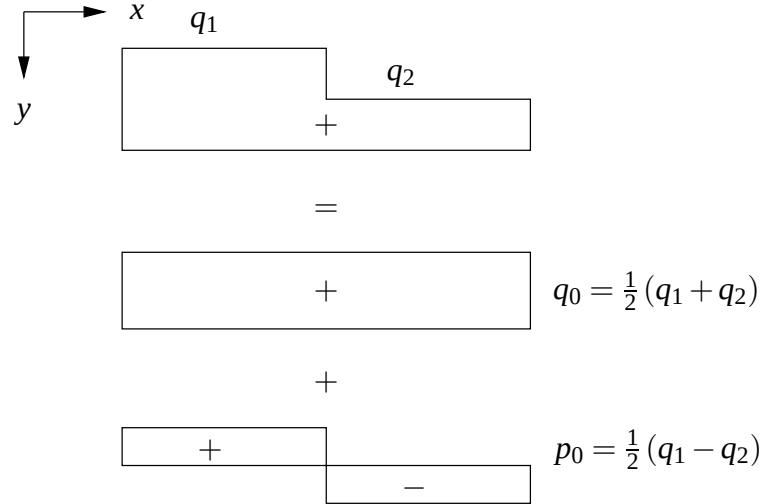


Figure 6.7

and $q_2 = q_0 - p_0$. Hence, the cable force H in (6.22) can be expressed as

$$H^2 = H_0^2 \left(1 + \frac{1}{4} \left(\frac{p_0}{q_0} \right)^2 \right), \quad (6.23)$$

where

$$H_0^2 = \frac{q_0^2 l^3}{24\Delta} \quad (6.24)$$

is the horizontal cable force when the contra-symmetric load is zero ($p_0 = 0$).

By an appropriate choice of the ratio p_0/q_0 we are allowed to say that H is linearly proportional, in an approximate way of course, to the applied load. Indeed by selecting $p_0 = 1/4q_0$ we have that

$$H^2 = H_0^2 \left(1 + \frac{1}{64} \right) = H_0^2 (1 + 0.0156), \quad (6.25)$$

which means that p_0 increases the square of the horizontal component of the cable tension by 1.56% (and H by 0.78%). Under these circumstances, the system acts close to linear and the principle of superposition holds (in an approximate way).

6.5.2 The horizontal deflection of cables

Let us have a better look at what happens to the cable depicted in Figure 6.6. Since there is more load on the left-hand side, the shape of the cable will be different from that of a parabola. The loading condition will also shift the middle point of the cable to the left and every point of the cable will suffer a horizontal and a vertical displacement $u(x)$ and $v(x)$.

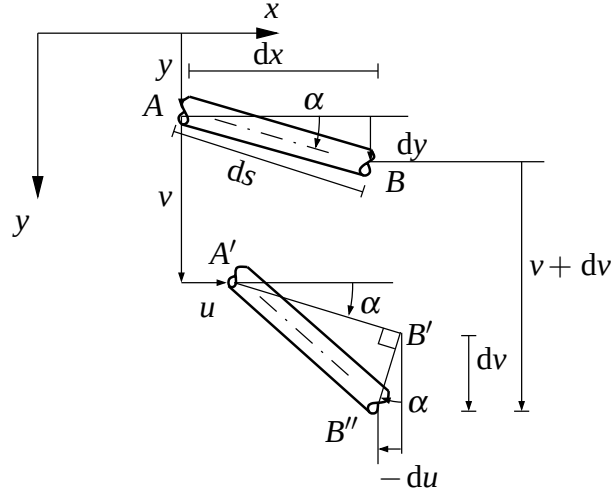


Figure 6.8

We can derive a relationship between u and v by considering a differential element of length ds as shown in Figure 6.8.

To set the scene, consider the element ds in its initial configuration due to the uniform load q_0 and in its deformed consideration after the application of the load p_0 (the load is now $q_0 + p_0$). The deformed configuration can be identified by a vertical displacement v , a horizontal displacement u and a rotation α about A' to $A'B'$ which takes place starting from the segment in the position $A'B'$ – note that $A'B'$ is inclined as AB .

Since we consider only small displacements, $v \ll y$ and the arc $B'B''$ can be replaced by a perpendicular line to $A'B'$. This implies that point B , which is now B' , undergoes an extra vertical displacement dv and an extra horizontal displacement $-du$. By eliminating $\tan \alpha$ from the expression of du obtained considering the triangle with the segment $B'B''$ as hypotenuse, $-du = dv \tan \alpha$, and from dy derived considering the initial configuration of the cable segment AB , $dy = dx \tan \alpha$, we have

$$du = -dv \tan \alpha = -dv \frac{dy}{dx} \quad (6.26)$$

which can be rewritten as

$$\frac{du}{dx} = -\frac{dv}{dx} \frac{dy}{dx}. \quad (6.27)$$

The horizontal displacement can now be found by integration:

$$u = \int du = - \int \frac{dv}{dx} \frac{dy}{dx} dx + C_1. \quad (6.28)$$

Hence, given the deflections due to the uniform load q_0 (i.e. y) and to the contra-symmetric load p_0 (i.e. v) it is possible to express the horizontal displacement of the cable. The integration constant can be found with the condition $u(0) = 0$.

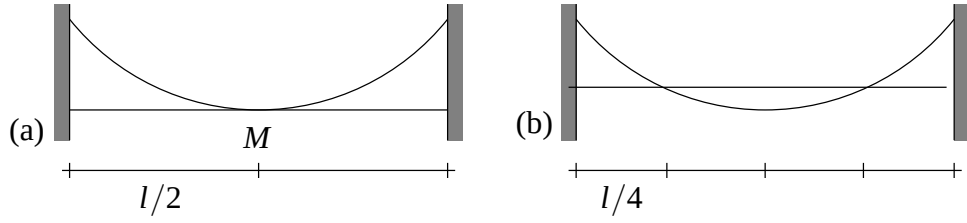


Figure 6.9

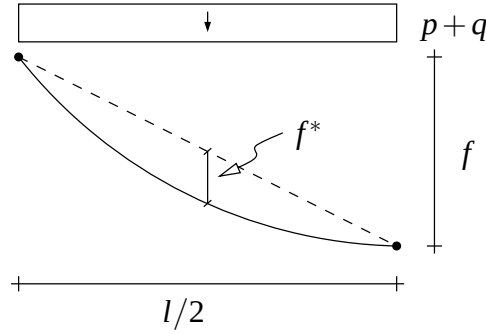


Figure 6.10

6.5.3 Cable stiffening

How do we stiffen a cable as to reduce its horizontal displacement? One simple way is to add an inextensible tension wire like depicted in Figure 6.9.

Consider the case depicted in Figure 6.9(a) under the action of a uniform load q and a contra-symmetric load p . Obviously, the deflection at midspan is due to q only (the contra-symmetric load p does not produce any deflection at midspan due to symmetry). Further, due to the tension wire, the horizontal deflection is zero. Hence, as a good approximation of the original system we consider the cable depicted in Figure 6.10 and term f^* the deflection at midspan. Worth noting is that the deflection line corresponding to the initial load q is not influenced by the additional load p (it is still a parabola!). However, the force H has changed and its value follows from

$$H + \Delta H = \frac{(p+q) \left(\frac{l}{2}\right)^2}{8f^*} = \frac{(q+p)l^2}{8f} \quad (6.29)$$

where we considered the reduced system with H the original horizontal cable force, ΔH the extra contribution coming from the stiffening cable, and $f^* = 1/4f$ where f is the deflection at midspan of the whole system ($f = ql^2/8H$). The expression $f^* = 1/4f$ derives from $f = pl^2/8H$ if we consider half cable span $l^* = 1/2l$; alternatively this expression can be easily derived as $f^* = \tilde{v}(l/4) - f/2$ with $\tilde{v}$ the deflection of the parabolic cable found

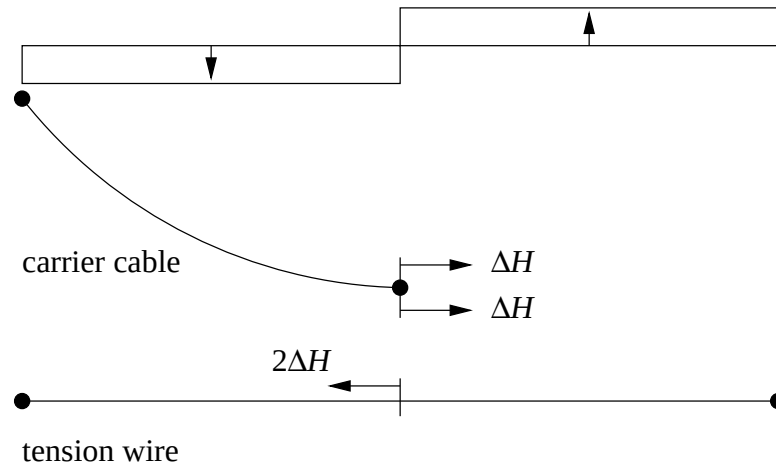


Figure 6.11

solving the differential equation of the cable with the boundary conditions $\tilde{v}(0) = 0$ and $\tilde{v}(l/2) = f$.

After simple manipulations, the increase in cable force reads as

$$\Delta H = \frac{pl^2}{8f}. \quad (6.30)$$

This force is provided by the tension wire at the point M .

Since the load in the right part of the cable is in the upward direction, the cable force in that point has decreased with an equal amount ΔH . So, also this part of the cable causes at point M a force to the left on the tension wire. The sum of both forces ($2\Delta H$) is carried by the right hand part of the tension wire (see Figure 6.11).

Exercises

6.1

6.2

References

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Chapter 7

Combined systems

Combined systems are structural systems in which different components contribute to the global load-carrying capacity. In these systems, the applied load is redistributed to the various components as a function of the corresponding stiffness. Suspension bridges are amongst the most common combined systems.

7.1 Spring systems as prototypes of combined systems

Springs are flexible elastic devices used to store and release energy. Springs can be combined in series, parallel and in parallel/series assemblages. Any spring or spring system can be characterized by its spring constant k which can be determined by Hooke's law $F = ku$, where F is the applied force and u the resulting displacement (see Figure 7.1).

7.1.1 Hooke's law

Hooke's law of elasticity states that the extension of an elastic spring is linearly proportional to the applied force through a constant of proportionality called the spring constant. This law is valid up to the elastic limit after which springs enter the plastic regime and suffer irrecoverable plastic deformation.

The law is named after the 17th century physicist Robert Hooke who described it by using stretched springs as shown in Figure 7.2. This law holds also for compression springs under physical constraints to prevent buckling. In 1676, Robert Hooke announced the law



a portrait claimed by historian Lisa Jardine to be of Robert Hooke

Robert Hooke

★ Freshwater, Isle of Wight, England, July 18, 1635

† London, England, March 3, 1703

English polymath who played an important role in the scientific revolution, through both experimental and theoretical work.

In 1660, he discovered Hooke's law of elasticity, which describes the linear variation of tension with extension in an elastic spring.

Hooke coined the biological term *cell*.

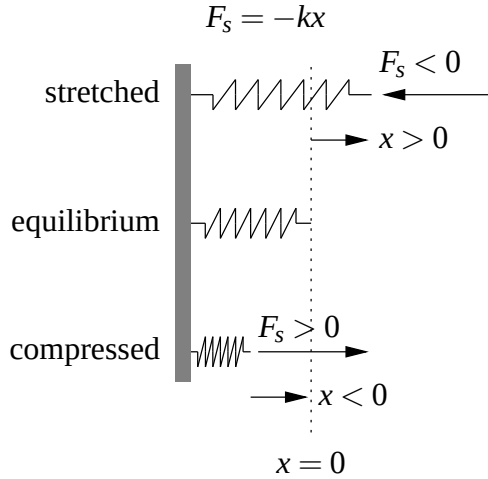


Figure 7.1 Hooke's law: the negative sign in F_s shows that any movement on the spring will be resisted by an equal but opposite force. The spring force is a restoring force and acts in the opposite direction from the direction in which the system is displaced. The origin has to be placed at the position where the spring system would be in equilibrium for the equation $F_s = -kx$ to be valid (in this position the net force on the object to which the spring is attached is equal to zero. If not, then $F_s = -k(x - x_0)$ where x_0 is equilibrium position relative to the origin.

that bears his name by using the anagram *ceiinossttuv*. In 1678 he explained it as *ut tensio sic vis*, Latin for *the stretch is proportional to the force*, which is comprehensible only to the initiates. Fortunately, in his “Lectures on Natural Philosophy” (1807), Thomas Young demystified Hooke's law by expressing it as the equation $F = ku$. Young, however, failed in reproducing the terseness of Hooke's original enunciate [1]:

... the modulus of elasticity of any substance is a column of the same substance, capable of producing a pressure on its base which is to the weight causing a certain degree of compression as the length of the substance is to the diminution of its length ...

7.1.2 Springs in parallel and in series

Parallel and series systems of springs are the two ways to arrange a set of springs. These systems are shown in Figures 7.3 and 7.6.

The equivalent spring constant for the parallel system in Figure 7.3 is derived by expressing the translational equilibrium in the horizontal direction for the free body diagram shown in Figure 7.3(b). This results in the equation

$$F = F_1 + F_2 = k_1 u_1 + k_2 u_2 = (k_1 + k_2) u, \quad (7.1)$$

where we have used Hooke's law to express the relation between force and displacement in the springs. We have also assumed that both springs undergo the same deformation $u (= u_1 = u_2)$, so that only one degree of freedom suffices to describe the system. From the above equation we define the equivalent spring constant as $k_e = k_1 + k_2$. This result can be generalized for a system of N springs in parallel:

$$k_e = \sum_{i=1}^N k_i. \quad (7.2)$$

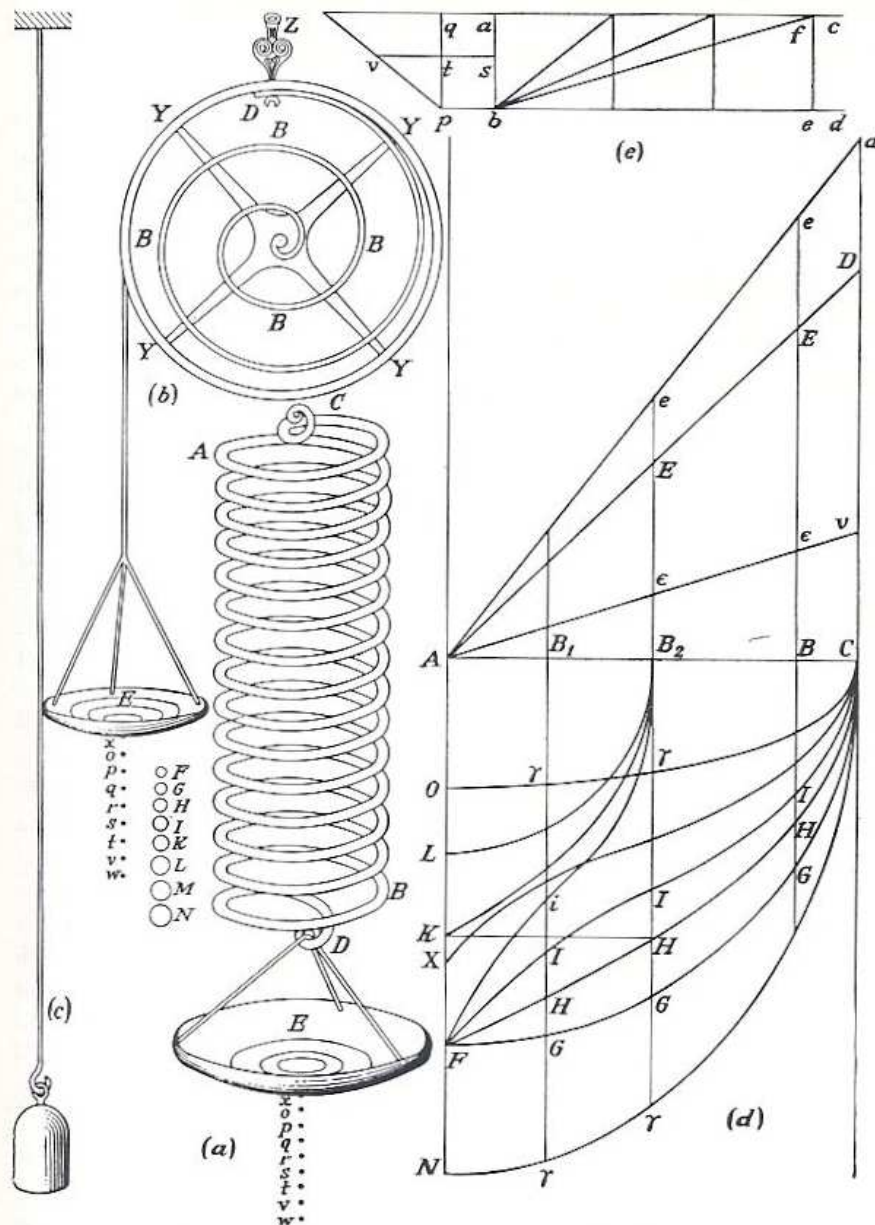


Figure 7.2 Plate to Hooke’s “De Potentia Restitutiva, or, of Spring: Explaining the Power of Springing Bodies”, Sixth Cutler Lecture, Printed for John Martyn, printer to the Royal Society, at the Bell in St. Pauls Churchyard, London: (a) wire helical spring stretched to points o, p, q, r, s, t, v, w , by weights F, G, H, I, K, L, M, N ; (b) watch spring similarly stretched by weights put in pan; (c) the “Springing of a string of brass wire 36 ft long; (d) diagram of velocities of springs; (e) diagram of law of ascent and descent of heavy bodies.

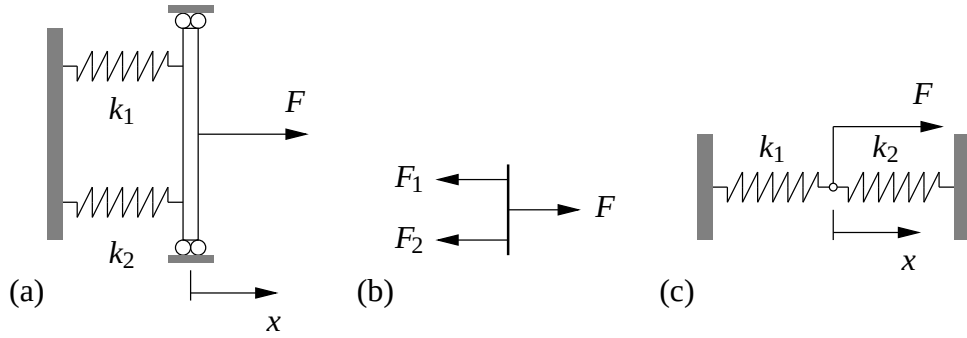


Figure 7.3 Springs in parallel.

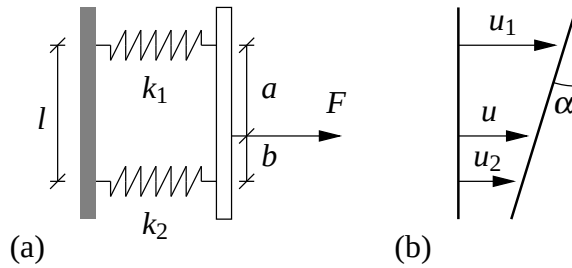


Figure 7.4 A parallel combination of springs (unsymmetric case).

In the above derivations we have assumed that the springs are constrained as to undergo the same deformation. This was achieved by preventing the vertical bar to undergo any rotation.

By releasing the rotation constraint, the system shown in Figure 7.4 has two degrees of freedom, a translation and a rotation. The extensional equivalent spring constant in the direction of the horizontal applied force F can be found by expressing the horizontal displacement u of the point of application of the load as a function of the two spring stiffnesses. This is easily accomplished by noting that $u_1 - u = a \tan \alpha$ and $u - u_2 = b \tan \alpha$ (these equations have been obtained by considering similar triangles in Figure 7.4). By eliminating $\tan \alpha$ we obtain

$$u = u_1 \frac{b}{l} + u_2 \frac{a}{l}. \quad (7.3)$$

The force F_i in each spring can be expressed as a function of the applied force F by imposing the rotational equilibrium at the two points where the springs are attached to the vertical bar. From $F_1 l = F b$ and $F_2 l = F a$, and by using Hooke's law for the two springs, we obtain

$$u = F \left(\frac{b^2}{k_1 l^2} + \frac{a^2}{k_2 l^2} \right) = \frac{F}{k_e}, \quad (7.4)$$

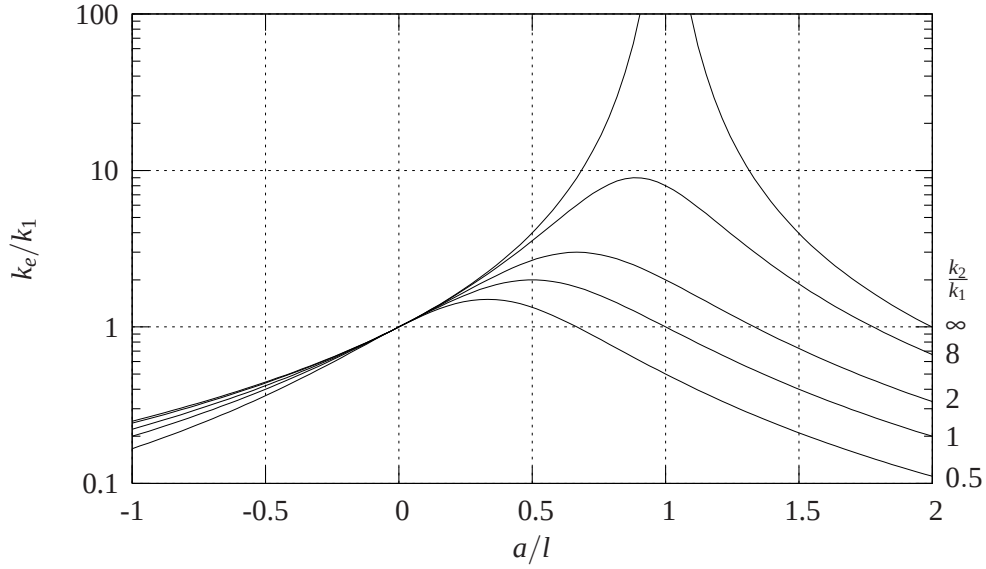


Figure 7.5 Equivalent spring constant for an unconstrained parallel combination of springs for various ratios k_2/k_1 .

from which the equivalent spring constant for unconstrained parallel springs can be expressed as

$$k_e = \frac{l^2 k_1 k_2}{a^2 k_1 + b^2 k_2}. \quad (7.5)$$

This results is valid also when the force F is not applied between the two springs.

Following Folkerts [2], some insight into (7.5) can be gained by considering its adimensional form,

$$\frac{k_e}{k_1} = \frac{k_2/k_1}{(a/l)^2 + (1 - a/l)^2 k_2/k_1}, \quad (7.6)$$

for various ratios k_2/k_1 . Figure 7.5 shows that the equivalent spring constant for a system of two unconstrained parallel springs depends on the location of the applied force.

We now consider some limiting cases of (7.5).

(a) $a = 0$ or $b = 0$: The force is applied in line with one spring and $k_e = k_1$ or $k_e = k_2$, respectively.

(b) $a \rightarrow \infty$ or $b \rightarrow \infty$: $k_e \rightarrow 0$ which implies that the system becomes less stiff as the lever arm becomes longer.

(c) $k_1 \rightarrow \infty$ or $k_2 \rightarrow \infty$: One of the two springs behaves like a rigid constraint and $k_e \rightarrow l^2 k_2/a^2$ or $k_e \rightarrow l^2 k_1/b^2$, respectively. This implies that as the distance between the applied force and the rigid constraint increases the spring constant rapidly drops.

Quoting Folkerts [2]:

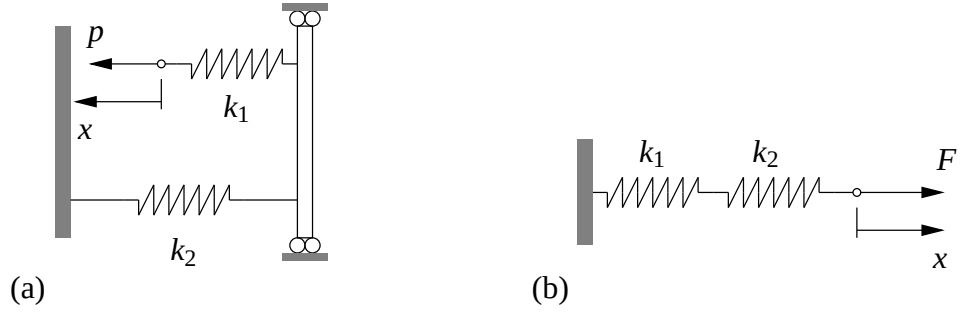


Figure 7.6 Springs in series.

When two springs are not constrained to compress the same amount, a situation common in physics problems and in real life, then the textbook result of $k_e = k_1 + k_2$ is often a poor approximation to the correct value.

Suppose now that we are interested in finding where to apply the load F such that the bar does not rotate after the application of the load. The condition to enforce is that $u_1 = u_2$ or, equivalently, $F_1/k_1 = F_2/k_2$. By using the expression previously derived for F_1 and F_2 , the constraint on the displacement yields the relation $ak_1 = bk_2$.

When springs are in series, like in the system depicted in Figure 7.6, the force in each spring equals the applied force: $F = F_1 = F_2$. The total deformation, on the other hand, is the sum of the deformations of the single components and is given by

$$u = u_1 + u_2 = \frac{F_1}{k_1} + \frac{F_2}{k_2} = F \left(\frac{1}{k_1} + \frac{1}{k_2} \right). \quad (7.7)$$

Hence the equivalent spring constant is

$$k_e = \frac{F}{u} = \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}}. \quad (7.8)$$

In general, for a system of N springs in series, we have that

$$\frac{1}{k_e} = \sum_{i=1}^N \frac{1}{k_i}. \quad (7.9)$$

By using the previous concepts is possible to derive the equivalent stiffness of combined system, like the one shown in Figure 7.7. The equivalent stiffness of this system is equal to

$$k_e = \frac{1}{\frac{1}{k_1+k_2+k_1} + \frac{1}{k_3+k_3}}. \quad (7.10)$$

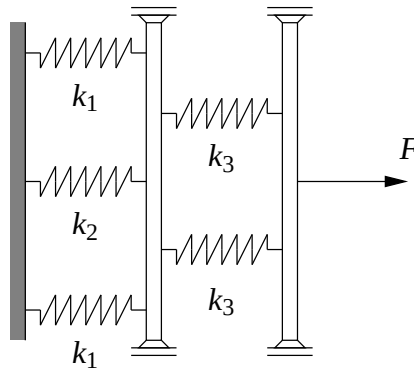


Figure 7.7 A parallel-series combination of springs.

7.2 Beam-cable systems: Deflection of stiffened suspension bridges

A stiffened suspension bridge is defined as a bridge with a stiff horizontal deck “where the deck load is carried vertically (or mostly vertically) by suspenders up to cables that transfer the loads by nearly horizontal forces to towers and then over them to side spans ending in heavy concrete anchorages” [3]. This system is also called a two-hinged truss suspension bridge [4].

In these structural systems, the transverse deflection is reduced by the introduction of a stiffening beam. A schematic is depicted in Figure 7.9 and consists of a single span cable stiffened by a simply supported beam of constant cross section. With reference to Figure 7.8, this could be considered as a good approximation of the part of the bridge between the two towers.

7.2.1 Key assumptions

The assumptions in the analysis of two-hinged truss suspension bridges are as follow [4], [5, Section 11.3]:

- the dead load of the structure, uniformly distributed along the span, is entirely transmitted to the cable which takes a parabolic form;
- the initial dead load is carried by the cable with no bending in the beam;
- the stiffening beam has constant moment of inertia EI (this is not essential, just a simplification);
- there is a continuous shear of vertical hangers connecting the cable to the stiffening girder;
- the hangers are vertical in all deflected configurations of the structure i.e. the inclination of the hangers may be neglected under live loads;
- axial elastic deformation of towers and hangers are neglected.

7.2.2 Governing equation and its solution

The photo of the Clifton suspension bridge in Figure 7.8 highlights one of the main constituents of a suspension bridge, the suspenders. Ironically, suspenders, or hangers, are not represented in schematics of suspension bridges. Their “contribution” is however key to the derivation of the differential equation.

Consider the schematic of a suspension bridge shown in Figure 7.9. Here we have considered a suspension bridge as a system composed by a bending beam with flexural stiffness EI and a cable with no bending stiffness ($EI = 0$) and negligible extension ($EA = \infty$) connected by means of rigid hanger cables, not shown in the figure, with no extension under the action of a distributed load. The space between hangers is considered to be small compared to the bridge span so that the hangers can be considered as continuously distributed along the span. A live load produces deflection of both cable and bending beam.

The inextensibility of the hanger cables is the most important characteristic of the system as it allows us to relate the deflection v_b of the beam and that v_c of the cable through the equation

$$v_b(x) = v_c(x) (= v(x)). \quad (7.11)$$

One of the implications of this assumption is that load redistribution can be easily defined by assigning a portion q_c of the total load q to the cable and the remainder of the load ($q_b = q - q_c$) to the beam. By assuming the same transverse displacement v we have defined a parallel system and separated the two components: we have a cable under the action of



Figure 7.8 Clifton suspension bridge, Bristol, England. The Clifton Suspension Bridge was designed as the longest single-span road bridge in the world (span: 214 m). It spans the Avon Gorge and links Clifton in Bristol to Leigh Woods in North Somerset. The bridge was designed by Isambard Kingdom Brunel; its construction begun in 1831 and was completed in 1864 (photo courtesy of Erik Stensland, Morning Light Photography, <http://morninglight.us>).

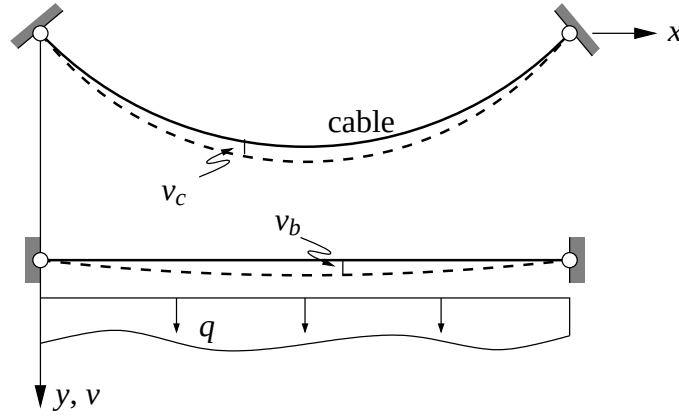


Figure 7.9

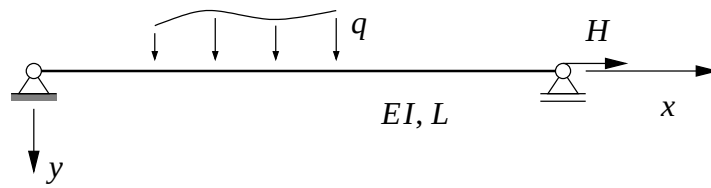


Figure 7.10

load q_c and a beam under the action of load $q_b = q - q_c$. Both components are described by the same transverse deflection v . We can therefore write the governing equations for cable and beam separately as

$$\text{cable: } H \frac{d^2 v}{dx^2} = -q_c, \quad \text{beam: } EI \frac{d^4 v}{dx^4} = q_b = q - q_c. \quad (7.12)$$

A single differential equation in terms of the transverse deflection v can be finally obtained by eliminating the cable load q_c :

$$EI \frac{d^4 v}{dx^4} - H \frac{d^2 v}{dx^2} = q. \quad (7.13)$$

Equation (7.13) describes the behavior of the combined, parallel cable-beam system. The same equation can be derived by considering the vertical equilibrium for both cable and beam of a differential element dx .

As a side remark, it is worth noticing that the transverse deflection of a simply supported beam under the combined action of an axial tensile load and a distributed load is also governed by (7.13). Further, a prestressed concrete girder can be regarded as a parallel cable-beam system. The beam shown in Figure 7.10 can be related to the suspension bridge by noticing that the differential equation (7.13) remains valid for beam-cable systems with hangers of zero length – indeed, we did not consider the length of the hangers in the derivation of (7.13).

The solution of (7.13) comprises the solution of the homogeneous equation and the particular solution that accounts for the inhomogeneous term $q(x)$. The solution of the homogeneous differential equation

$$EI \frac{d^4 v}{dx^4} - H \frac{d^2 v}{dx^2} = 0 \quad \text{or} \quad \frac{d^4 v}{dx^4} - \alpha^2 \frac{d^2 v}{dx^2} = 0 \quad (7.14)$$

is

$$v(x) = C_1 e^{\alpha x} + C_2 e^{-\alpha x} + C_3 + C_4 x, \quad (7.15)$$

with $\alpha^2 = H/EI$ ($[L]^{-2}$).

To determine the particular solution related to the distributed load q , we consider a uniform distributed load $q(x) = q_0$. In order to proceed, we have to determine the lowest order in the derivative of the differential equation which, in this case, is two. Hence, we consider $v = Cx^2$ as particular solution of (7.13) and, by replacing the particular solution in (7.13), we find that

$$C = -\frac{q_0}{2H}. \quad (7.16)$$

The particular solution is then

$$v(x) = -\frac{q_0}{2H} x^2 \quad (7.17)$$

with which we can define the general solution to our problem:

$$v(x) = C_1 e^{\alpha x} + C_2 e^{-\alpha x} + C_3 + C_4 x - \frac{q_0}{2H} x^2. \quad (7.18)$$

The four integration constants can be found by using the following boundary conditions:

$$\text{at } x = 0 : \quad v(0) = 0 \quad \text{and} \quad M(0) = 0, \quad (7.19a)$$

$$\text{at } x = L : \quad v(L) = 0 \quad \text{and} \quad M(L) = 0 \quad (7.19b)$$

which yield

$$C_1 = \frac{1 - e^{-\alpha l}}{e^{\alpha l} - e^{-\alpha l}} \frac{q_0}{\alpha^2 H}, \quad C_2 = \frac{e^{\alpha l} - 1}{e^{\alpha l} - e^{-\alpha l}} \frac{q_0}{\alpha^2 H}, \quad C_3 = -\frac{q_0}{\alpha^2 H}, \quad C_4 = \frac{q_0 l}{2H}. \quad (7.20)$$

With these integration constants we can express the transverse displacement v .

7.2.3 An approximate solution valid for long span bridges

The integration constants (7.20) can be simplified in long span bridges as the quantity αl is a number of considerable magnitude – Timoshenko and Young [5, Section 11.4] report two examples for which $\alpha l = 9.52$ and $\alpha l = 35$. By eliminating $e^{-\alpha l}$, the deflection at midspan is equal to

$$v\left(\frac{l}{2}\right) \approx \frac{q_0 l^2}{8H} \left(1 - \frac{8}{\alpha^2 l^2}\right) = \frac{q_0 l^2}{8H} \left(1 - \frac{8EI}{Hl^2}\right) \quad (7.21)$$

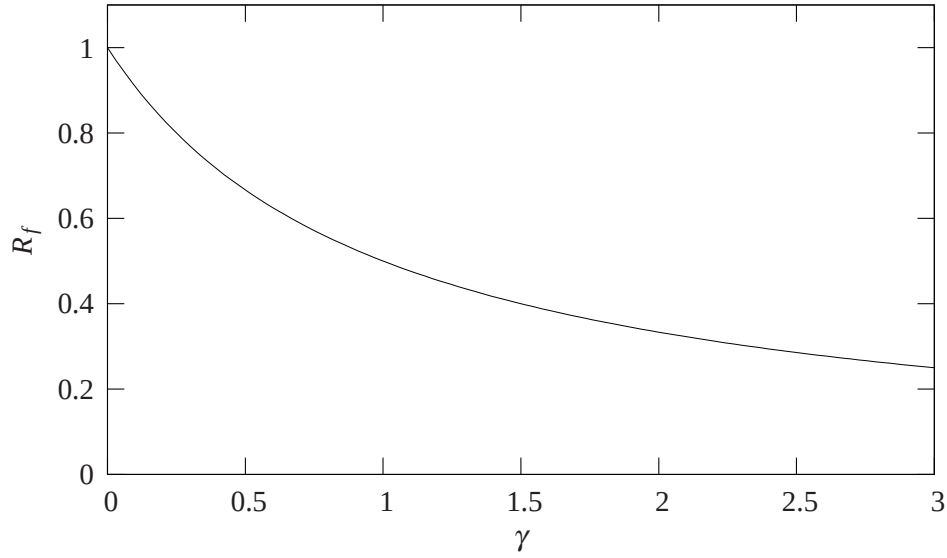


Figure 7.11

which can be compared to the cable sag

$$f = \frac{q_0 l^2}{8H} \quad (7.22)$$

of an unstiffened suspension bridge. It is thus evident that in a stiffened suspension bridge the ratio EI/Hl^2 is responsible for the reduction of the stresses in the beam. A direct consequence of this finding is the possibility of using beams with smaller cross sections.

7.2.4 Sinusoidal load function

The effect of the axial load can be better appreciated if we assume a load function of the type

$$q(x) = q_0 \sin \pi \frac{x}{l}. \quad (7.23)$$

The solution of the governing equation is then given by

$$v(x) = \frac{1}{1 + \frac{H}{EI} \left(\frac{l}{\pi}\right)^2} \frac{q_0}{EI} \left(\frac{l}{\pi}\right)^4 \sin \frac{\pi x}{l}, \quad (7.24)$$

while the bending moment

$$M(x) = -EI \frac{d^2 v}{dx^2} = \frac{1}{1 + \frac{H}{EI} \left(\frac{l}{\pi}\right)^2} q_0 \left(\frac{l}{\pi}\right)^2 \sin \frac{\pi x}{l}. \quad (7.25)$$

Both expressions contains the reduction factor

$$R_f = \frac{1}{1 + \frac{H}{EI} \left(\frac{l}{\pi}\right)^2}. \quad (7.26)$$

It is worth noticing that by letting $H = 0$ we recover the solution of a bending beam without axial force.

The reduction factor can be made a function of the Eulerian buckling load of a beam $P_E = \pi^2 \frac{EI}{l^2}$:

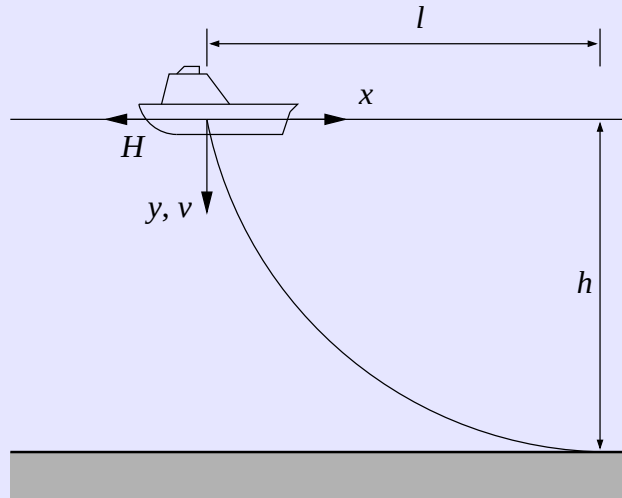
$$R_f = \frac{1}{1 + \frac{H}{EI} \left(\frac{l}{\pi}\right)^2} = \frac{1}{1 + \frac{H}{P_E}} = \frac{1}{1 + \gamma} \quad (7.27)$$

with the parameter γ equal to the ratio of the cable force to the Eulerian buckling force. The influence of the reduction factor on $v(x)$ and $M(x)$ can be appreciated by analyzing the curve shown in Figure 7.11.

Example 7.1 (Seabed pipe-laying)

The problem of seabed pipe-laying can be analyzed by using (7.14). Consider the figure below where a vessel is positioning a cable on the bottom of the sea. In placing the cable, the vessel exerts a horizontal tension H in the cable.

The problem consists in defining the point at which the cable is in perfect contact with the sea floor given the force H , the weight of the cable q , the stiffness EI and the depth of the water h .



We take as origin of the coordinate system the boat and consider that the cable is subjected to its own weight q which we imagine as the applied load q acting on the suspension bridge. The general solution of (7.14) is

$$v = C_1 e^{\alpha x} + C_2 e^{-\alpha x} + C_3 + C_4 x - \frac{q}{2H} x^2, \quad (7.28)$$

where the last term comes into play because we have a distributed load. This equation can be solved considering four boundary conditions. The first two are specified at $x = 0$: $v = 0$ and $M = 0$. The remaining boundary conditions are defined by determining the point at which the cable is in perfect contact with the sea floor. We place this point at $x = l$. Hence, at $x = l$ we have $\frac{dv}{dx} = 0$ as the slope of the cable is zero – the cable is horizontal – and $M = 0$. With these boundary conditions we find

$$C_1 = \frac{q}{\alpha^2 H} e^{-\alpha l}, \quad C_2 = \frac{q}{\alpha^2 H}, \quad C_3 = -\frac{q}{\alpha H}, \quad C_4 = \frac{ql}{H} - \frac{q}{\alpha H}, \quad (7.29)$$

which yield the expression of the deflection

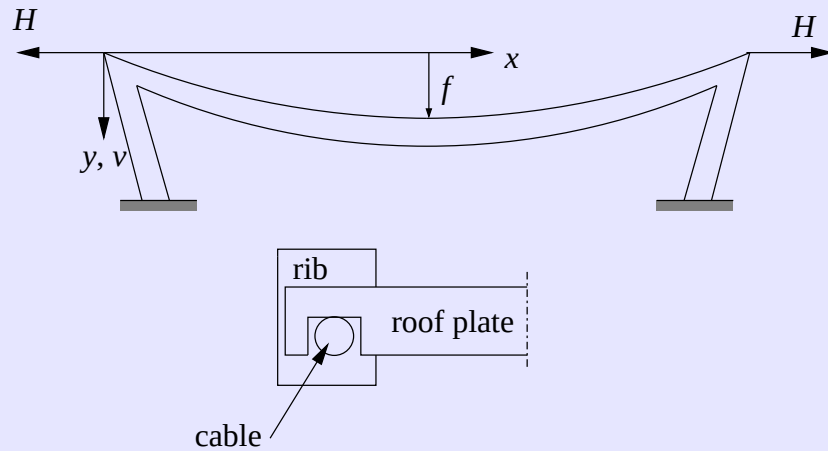
$$v = \frac{q}{\alpha^2 H} \left(e^{-\alpha(l-x)} + e^{-\alpha x} - 1 \right) + \left(\frac{ql}{H} - \frac{q}{\alpha H} \right) x - \frac{1}{2} \frac{q}{H} x^2. \quad (7.30)$$

At $x = l$ we find

$$v(l) = -\frac{ql}{\alpha H} + \frac{1}{2} \frac{q}{H} l^2 \quad (7.31)$$

from which, using $v(l) = h$ we can find the length l .

Example 7.2 (Parabolic roof structures)



The deflection of roof structures can be reduced by adding flexural ribs. Let us take a look at a simple model of a roof structure like the parabolic roof structure depicted in the figure above. In the construction of such a structure, the first step is to hang cables between the two vertical beams. Next, roof plates are added on the cables and then concrete ribs are casted. The total weight q of the cables, plating and ribs is carried by the cables and the governing

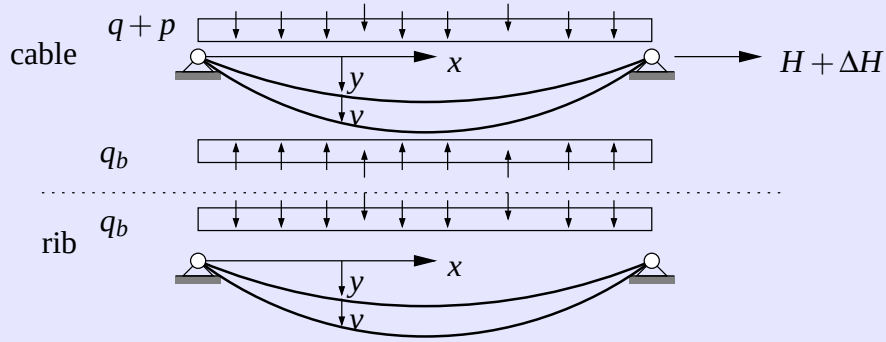
equation for this system is

$$-H \frac{d^2 y}{dx^2} = q. \quad (7.32)$$

Since the load is distributed along the curve, the shape should be a catenary, but we can approximate it by a parabola, Therefore, we may say that the force in the roof structure is

$$H = \frac{qL^2}{8f}. \quad (7.33)$$

Well, now the roof is pretty solid, concrete has hardened and we are ready to have a party on it. In doing so, we load the roof with some extra load p which causes an extra deflection v . Part of the load will be carried by the cable and part by the concrete ribs. Note that the concrete ribs have used the cable to define their shape. Of course, extra deflection means extra force ΔH in the structure which is now $H + \Delta H$. We indicate by q_b the force taken by the ribs and therefore the load taken by the cable is $q_c = q + p - q_b$ where $q + p$ is the applied load. The implicit assumption is that since cable and ribs are glued together they will undergo the same deformation. We have a parallel system. Therefore we can draw the contributions of the two systems as shown in the figure below.



The governing equations are now

$$-(H + \Delta H) \frac{d^2(y + v)}{dx^2} = q + p - q_b, \quad \text{for the cable} \quad (7.34)$$

and

$$EI \frac{d^4 v}{dx^4} = q_b \quad \text{for the ribs.} \quad (7.35)$$

Here we have used v since the ribs have been created on the shape given by y . Eliminating q_b we have

$$EI \frac{d^4 v}{dx^4} - H \frac{d^2 y}{dx^2} - \Delta H \frac{d^2 y}{dx^2} - (H + \Delta H) \frac{d^2 v}{dx^2} = q + p. \quad (7.36)$$

Since $-H \frac{d^2 y}{dx^2} = q$ we have

$$EI \frac{d^4 v}{dx^4} - (H + \Delta H) \frac{d^2 v}{dx^2} = -\frac{\Delta H}{H} q + p. \quad (7.37)$$

If ΔH is very small compared to H we may write

$$EI \frac{d^4 v}{dx^4} - H \frac{d^2 v}{dx^2} = p, \quad (7.38)$$

which is the same equation we have seen for the suspension bridge.

The ratio $\Delta H/H$ is $\ll 1$ if we have a contra-symmetric load with small ratio p/q as we have seen in Chapter 6.

7.3 Shear beam-bending beam systems

A common class of buildings is characterized by the presence of some walls made of continuous slabs of concrete. The behavior of this kind of system is in between that of a shear beam and a bending beam. Generally speaking, when a building is much wider than it is tall we can consider the bending to be approximately zero and we call this system a shear beam – this is due to the fact that the total stiffness against shear is given by the shear modulus times the cross sectional area; hence, a wide building is dominated by shear and not by bending. On the other hand, when the height of the building is large compared to the width, the building deforms primarily by bending. Note however that if we assume that the building has a square cross section of width w , the building becomes very stiff against flexure as w becomes large.

7.3.1 Basic assumptions

Shear beams are an important component in many buildings. They are usually rigidly connected to beams with flexural stiffness like shown in Figure 7.12 and form a parallel system.

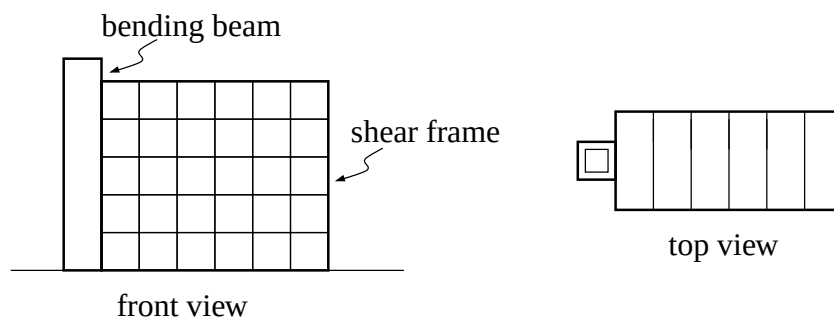


Figure 7.12

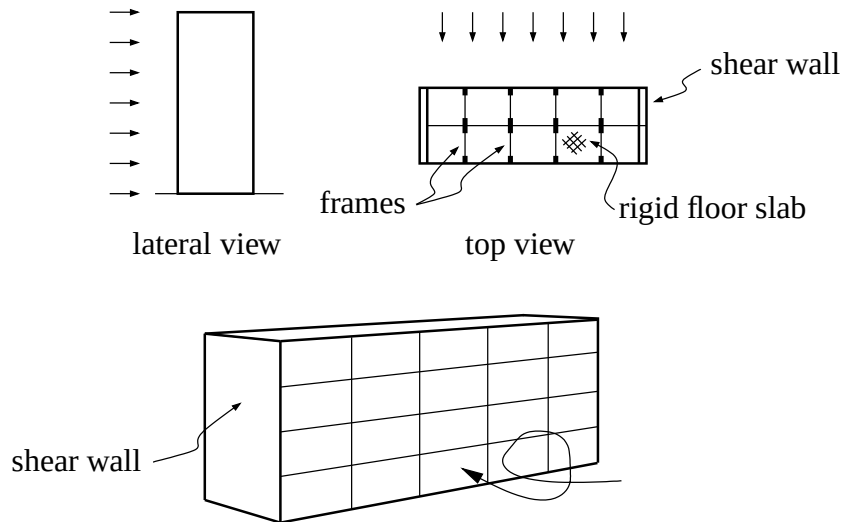


Figure 7.13

If this system is loaded with a horizontal system of forces, the frame and the beam will experience the same horizontal deflection.

Often, special shear walls are employed to sustain horizontal forces as shown in Figure 7.13. In this case, the building is made of a number of slender high frames placed next to each other and is enclosed by flexible walls. A horizontal load is partly carried by the frames and partly transmitted to the shear walls via the floor slabs. If these floor slabs are rigid in their plane, the frames and walls experience equal displacements. This is also a combined system.

To investigate the behavior of this kind of combined system, we consider the structure shown in Figure 7.14a where a shear beam is connected with rigid members to a bending beam, or shear wall. Both beams are clamped to the foundation. In this case, the shear beam is subjected to a horizontal distributed load q . As illustrated in Figure 7.14b the deformation of the two systems, taken separately, is obviously different. However, when connected, they are constrained to deform in a common deflected shape.

7.3.2 Governing equations and solution

Since the links are rigid, the displacements of frame and shear wall can be considered the same: $v_s = v_b$. As a consequence, the bending beam carries a portion q_b of the total load q and the shear beam will carry the remaining part $q_s = q - q_b$. Similar to the case of the suspension bridge, we have two separate structural systems, a shear wall under the action of q_b and a frame under the action of $q_s = q - q_b$. The two systems are linked through the

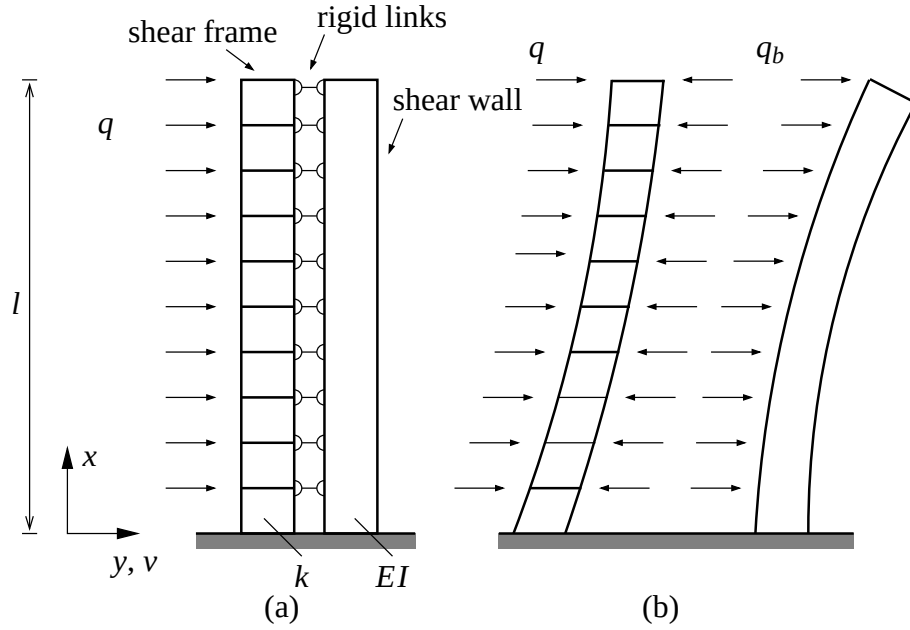


Figure 7.14

displacement v . We can therefore write the following system of governing equations:

$$\begin{aligned} \text{frame:} \quad & -k \frac{d^2 v}{dx^2} = q_s = q - q_b \\ \text{beam:} \quad & EI \frac{d^4 v}{dx^4} = q_b \end{aligned} \quad (7.39)$$

in which the only unknown is q_b . Elimination of q_b yields

$$EI \frac{d^4 v}{dx^4} - k \frac{d^2 v}{dx^2} = q. \quad (7.40)$$

This differential equation is similar to the one governing the deflection of the suspension bridge (cf (7.13)). The boundary conditions are however different and, with reference to Figure 7.14a, they are:

$$\begin{aligned} \text{at } x = 0: \quad & v = 0 \quad \text{and} \quad \frac{dv}{dx} = 0; \\ \text{at } x = l: \quad & M = -EI \frac{d^2 v}{dx^2} = 0 \quad \text{and} \quad V = V_s + V_b = k \frac{dv}{dx} - EI \frac{d^3 v}{dx^3} = 0. \end{aligned}$$

Using these boundary conditions we find the expression of the transverse deflection and the bending moment:

$$v = \frac{ql^2}{GA} \left(-\frac{1 + \alpha l \sinh \alpha l}{(\alpha l)^2 \cosh \alpha l} + \frac{\cosh \alpha x + \alpha l \sinh \alpha (l - x)}{(\alpha l)^2 \cosh \alpha l} + \frac{x}{l} - \frac{x^2}{2l^2} \right), \quad (7.41)$$

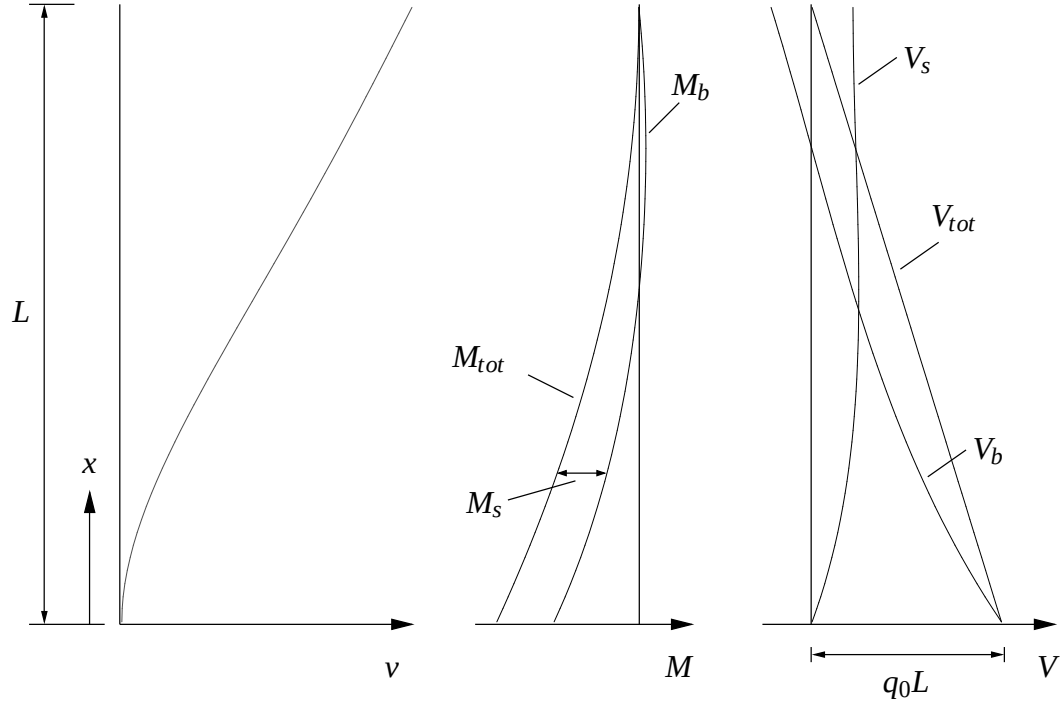


Figure 7.15

$$M_b = -EI \frac{d^2 v}{dx^2} = -\frac{EI}{GA} q \left(-\frac{\cosh \alpha l + \alpha l \sinh \alpha (l-x)}{\cosh \alpha l} + 1 \right), \quad (7.42)$$

with $\alpha^2 = GA/EI$. The expression of the shear force in the the frame and the shear wall can be derived by using

$$V_s = k \frac{dv}{dx} \quad \text{and} \quad V_b = \frac{dM}{dx}. \quad (7.43)$$

A plot of these quantities is reported in Figure 7.15 for $\alpha l = 2$.

Note that the internal shear forces V_s and V_b at $x = l$ are equal and opposite in sign. This implies that there is a concentrated shear force acting at that point in each of the two systems.

As a measure of the effectiveness of this combined system, we can compare the deflection v at $x = l$ with that of a pure shear beam (v_s) and of a bending beam (v_b). By setting $\alpha l = 2$ we obtain:

$$v = 0.2015 \frac{ql^2}{GA} = 0.0504 \frac{ql^4}{EI}, \quad v_s = 0.5 \frac{ql}{GA}, \quad v_b = 0.125 \frac{ql^4}{EI}. \quad (7.44)$$

Clearly, the combination shear wall-shear frame works pretty well as the deflection at the top is significantly reduced.

Example 7.3 (Solving the shear wall-shear frame system with Maple)

The differential equation (7.40) can be solved with the aid of a general-purpose computer algebra system. An example of a Maple solution is reported below (courtesy of Hans Welleman). M and V are the total moment and shear force of the system from equilibrium.

```
> restart;
> alpha:=sqrt(k/EI);
> w:=C1+C2*x+C3*exp(-alpha*x)+C4*exp(alpha*x)-(qo*x^2)/(2*k);
      w:=C1+C2x+C3e(-√EI x)+C4e(√EI x)- $\frac{qox^2}{2k}$ 
> phi:=-diff(w,x): Va:=k*diff(w,x): kappa:=diff(phi,x):
> Mb:=EI*kappa: Vb:=diff(Mb,x):
> x:=0: eq1:=w=0: eq2:=phi=0:
> x:=L: eq3:=Mb=0: eq4:=Va+Vb=0:
> sol:=solve({eq1,eq2,eq3,eq4},{C1,C2,C3,C4}): assign(sol);
> x:=x':
> M:=(-1/2)*qo*(x-L)^2: V:=diff(M,x):
> L:=10: EI:=10000: k:=400: evalf(alpha*L): qo:=5;
> plot(w,x=0..L,y=0..(1/(8*EI))*qo*L^4,title="displacement v",labels=["x
[m]","v [m]"]);
> plot([Va,Vb,V],x=0..L,y=(-1/2)*qo*L..qo*L,title="Shear
force",labels=["x [m]","V [kN]"],legend=["Va","Vb","V total"]);
> plot([Mb,M],x=0..L,y=(-1/2)*qo*L^2..(1/2)*qo*L^2,title="Bending
moment",legend=["Mb","M total"],labels=["x [m]","M [kNm]"]);
```

Exercises**7.1****7.2****References**

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Chapter 8

Fundamentals of matrix structural analysis: The matrix displacement method

The matrix displacement method is a numerical procedure to determine displacement and stress fields of a structural system under the action of applied loads. The matrix displacement method can be considered as a form of the finite element method.

There is a plethora of documentation on all aspects of matrix structural analysis. An excellent starting point is the historical outline by Felippa [1] and references therein. For those interested in a complete reference, the book by Przemieniecki [2] is a must. This chapter is based on [2, 3].

8.1 Introduction

Given a physical system, in the matrix displacement method we seek a relation, set in a matrix form of the type

$$ka = f, \quad (8.1)$$

between applied forces f and unknown displacements a through a stiffness matrix k . Hence the definition “matrix displacement method”. In structural systems, the array a contains nodal unknowns such as axial extensions, transverse deflections and rotations while the array f collects the corresponding axial forces, shearing forces and bending moments. k is the stiffness of the complete system, also known as global stiffness matrix, and is given by

$$k = \sum_{e=1}^{N_{el}} A^e k^e, \quad (8.2)$$

where k^e is the element stiffness matrix. A indicates the assembly operator which scatters the degrees of freedom from the element to the structure. The assembly operator can be replaced by the summation operator Σ provided that the element stiffnesses are fully expanded to the global matrix.

A typical analysis involves the following steps:

1. idealization of the physical system;
2. discretization of the mathematical model;
3. application of the matrix displacement method procedure;

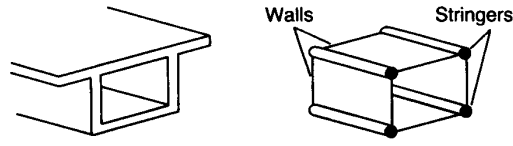


Figure 8.1 Stringer and Wall model of a beam (from Model Code 90)

4. solution of the equations;
5. evaluation of stresses;
6. interpretation of numerical results.

The idealization of the physical system (step 1), a structure in our case, consists in its reduction to simpler units, like if the structure were a Lego assembly – an example is reported in Figure 8.1. For each of these units, or finite elements, we seek the stiffness properties in terms of force-displacement relations (step 2). Through assembly (step 3), we define the force stiffness relation of the global structure which, once solved (step 4), yields the sought displacements. Given the displacement field, we can derive the strain and stress fields (step 5). The final step, usually the most difficult, consists in understanding the outcome of the numerical analysis (step 6).

8.2 The force-displacement relationship

There are many ways to derive the force-displacement relationship of individual elements: the unit-displacement theorem, Castigliano's theorem, the solution of the differential equations for the element displacements, variational methods. These methods are detailed in Cook et al. [3], Buchanan [4], Przemieniecki [2] and Hughes [5]. Here we shall use the solution of the differential equation. The stiffness matrices derived with this method, as well with that of the methods above are exact within the limit of validity of the differential equation in the sense that no further approximations will be used to derive the stiffness matrix. This method is however limited to simple elements and is restricted to situations in which actual displacements are assumed to be small (i.e. we can approximate $\cos \theta$ by 1 and $\sin \theta$ by 0) and the material of the structural member obeys linear elasticity.

The result of this operation can be condensed into

$$ka = f \quad \text{or} \quad k_{i,j}a_j = f_i, \quad (8.3)$$

where $k_{i,j}$ is the reaction (axial force, bending moment, shearing force) at the degree of freedom i due to a unit deformation (translation, rotation, deflection) at the degree of freedom j . In other words, a column of k lists nodal loads that must be applied to nodal degrees of freedom in order to create the deformation state associated with unit value of the corresponding element degrees of freedom while all other element degrees of freedom are zero.

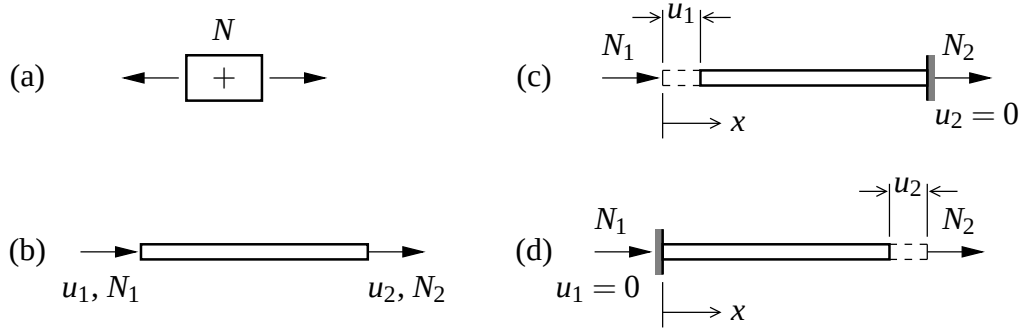


Figure 8.2

Next, we determine the stiffness matrices of uniform and weightless bar and beam elements taking into account the definition of the stiffness coefficients $k_{i,j}$ just given.

8.3 Axial deformation

We consider a bar finite element with one axial degree of freedom per node. Positive axial displacements and forces are depicted in Figure 8.2(b) while Figure 8.2(a) reports the convention for positive axial internal forces. The differential equation governing the axial displacement u in the uniform bar shown in Figure 8.2(c) is

$$N(x) = EA \frac{du}{dx}. \quad (8.4)$$

This differential equation is integrated directly to obtain

$$u(x) = \frac{N(x)}{EA}x + C_1, \quad (8.5)$$

where C_1 is an integration constant. Making use of the boundary condition at $x = 0$ ($u = u_1$) we derive $C_1 = u_1$. The second boundary condition at $x = l$ ($u = u_2 = 0$) yields $N_2 = -u_1EA/L$. From the equilibrium equation $N_1 = -N_2$ we obtain $N_1 = u_1EA/L$. Hence

$$k_{1,1} = \frac{N_1}{u_1} = \frac{EA}{L} \quad (8.6)$$

and

$$k_{4,1} = \frac{N_2}{u_1} = -\frac{EA}{L}. \quad (8.7)$$

The remaining terms of the stiffness matrix can be derived by symmetry or by considering the bar shown in Figure 8.2(d). Following the above procedure we obtain $k_{4,4} = k_{1,1}$ and $k_{1,4} = k_{4,1}$.

Note that we have used the ordering in (8.35) for the coefficients $k_{i,j}$.

Quantity	Definition
curvature	$\kappa = \frac{d^2v}{dx^2}$
bending moment	$M = EI\kappa$
shear force	$V = \frac{dM}{dx}$
distributed load	$q = \frac{dV}{dx}$

Table 8.1 Summary of governing equation for beam with sign convention of Figure 8.3.

8.4 Shear and bending deformation

The beam element has four degrees of freedom: a displacement perpendicular to the axis of the beam and a rotation at each end. Nodal forces corresponds to nodal displacements while nodal moments corresponds to nodal rotations. Sign conventions are shown in Figure 8.3 while the corresponding governing equation are summarized in Table 8.1. Note that nodal moments M_i are positive in the direction of nodal rotations θ_i .

When bending and shearing strains are present, the transverse deflection of the beam is given by the sum of the bending and shearing contributions

$$v = v_b + v_s. \quad (8.8)$$

The bending deflection is governed by the differential equation

$$EI \frac{d^2v_b}{dx^2} = V_1x - M_1, \quad (8.9)$$

where the right-hand side is defined as the moment acting at distance x from left-hand end of the beam. The shear deflection v_s is such that

$$\frac{dv_s}{dx} = -\frac{V_1}{GA_s}, \quad (8.10)$$

where A_s represents the beam cross sectional area effective in shear. By integrating (8.9) twice, replacing $v_b = v - v_s$ in this equation and considering v_s from the integration of (8.10)

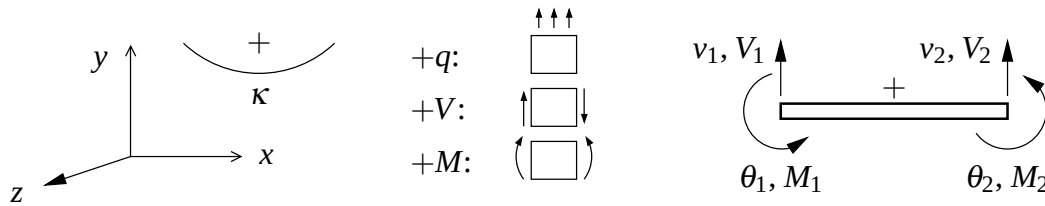


Figure 8.3 Conventions for a beam element: (a) positive curvature κ , (b) forces on an infinitesimal beam element, (c) degrees of freedom v_1 , v_2 and θ_1 , θ_2 with associated nodal forces V_1 , V_2 and moments M_1 , M_2 .

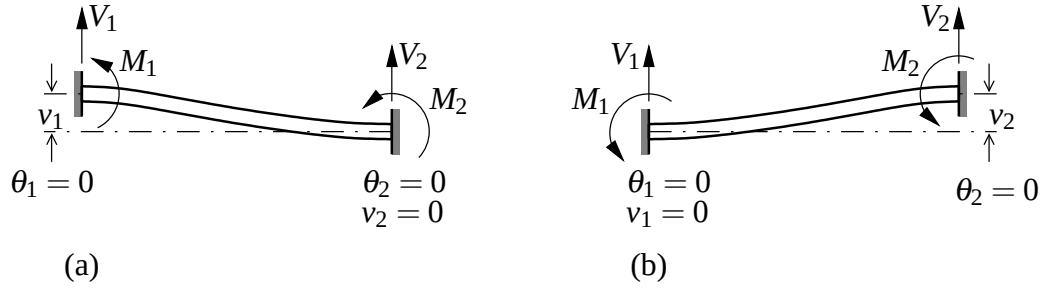


Figure 8.4 Deformation states associated with the activation of v_1 and v_2 .

we obtain

$$EIv = \frac{1}{6}V_1x^3 - \frac{1}{2}M_1x^2 + \left(C_1 - \frac{EIV_1}{GA_s}\right)x + C_2 \quad (8.11)$$

with C_1 and C_2 integration constants.

8.4.1 Shear

The boundary conditions related to Figure 8.4(a) for (8.11) yield [6, Section 7.12]

$$\frac{dv}{dx} = \frac{dv_s}{dx} = -\frac{V_1}{GA_s} \quad \text{at} \quad x = 0, \quad x = L \quad (8.12)$$

and

$$v = 0 \quad \text{at} \quad x = L, \quad (8.13)$$

from which

$$C_1 = 0 \quad (8.14)$$

and

$$C_2 = \frac{1}{12} \frac{LV_1}{GA_s} (12EI + GA_sL^2), \quad (8.15)$$

where we have considered that $M = 0$ at $x = L/2$ and $M_1 = V_1L/2$. With these integration constants, (8.11) becomes

$$EIv = \frac{V_1x^3}{6} - \frac{M_1x^2}{2} - \frac{V_1\Phi L^2}{12}x + \frac{1}{12}L^3V_1(1 + \Phi) \quad (8.16)$$

where

$$\Phi = \frac{12EI}{GA_sL^2} \quad (8.17)$$

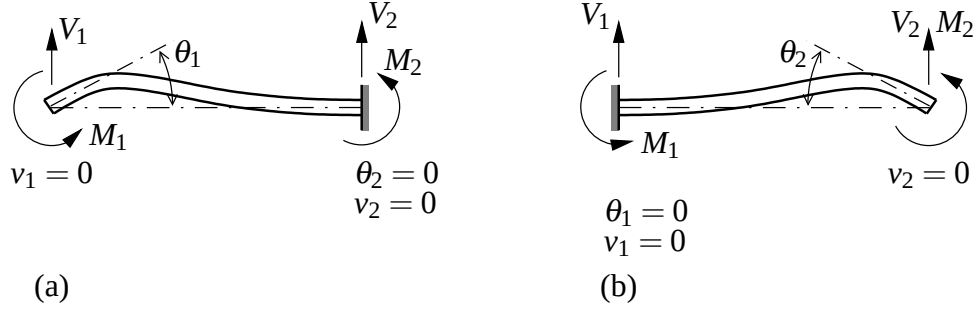


Figure 8.5 Deformation state associated with the activation of θ_1 and θ_2 .

is a dimensionless coefficient.

By using equilibrium we can define the remaining forces acting on the beam as $V_2 = -V_1$ and $M_2 = -M_1 + V_1L$.

We are now ready to define the coefficients of the stiffness matrix. According to the definition of $k_{i,j}$ and with the ordering in (8.35), letting $v = v_1$ at $x = 0$, we have

$$v_1 = (1 + \Phi) \frac{L^3 V_1}{12EI} \quad (8.18)$$

from which

$$k_{2,2} = \frac{V_1}{v_1} = \frac{12EI}{(1 + \Phi)L^3}, \quad (8.19)$$

$$k_{3,2} = \frac{M_1}{v_1} = \frac{V_1 L}{2v_1} = \frac{6EI}{(1 + \Phi)L^2}, \quad (8.20)$$

$$k_{5,2} = \frac{V_2}{v_1} = -\frac{12EI}{(1 + \Phi)L^3}, \quad (8.21)$$

and

$$k_{6,2} = \frac{-M_1 + V_1 L}{v_1} = \frac{6EI}{(1 + \Phi)L^2}. \quad (8.22)$$

The remaining coefficients in the second column are zero.

If the left-hand end of the beam is built-in as in Figure 8.4(b) we can, by symmetry considerations or by using the differential equation, derive the remaining coefficients:

$$k_{5,5} = k_{2,5} = \frac{12EI}{(1 + \Phi)L^3}, \quad k_{6,5} = k_{3,5} = -k_{6,2} = -\frac{6EI}{(1 + \Phi)L^2}. \quad (8.23)$$

8.4.2 Bending

In order to define the stiffness coefficients associated with the rotations θ_1 and θ_2 we subject the beam to bending moments and the associated shear forces as shown in Figure 8.5. Equation (8.11) is still valid but the boundary conditions need to be adapted to the new loading

condition. With reference to Figure 8.5(a) we have

$$v = 0 \quad \text{at} \quad x = 0, \quad x = L \quad (8.24)$$

and

$$\frac{dv}{dx} = \frac{dv_s}{dx} = -\frac{V_1}{GA_s} \quad \text{at} \quad x = L. \quad (8.25)$$

With $v = 0$ at $x = 0$ we obtain $C_2 = 0$, while $v = 0$ at $x = L$ yields

$$\frac{V_1 L^3}{6} - \frac{M_1 L^2}{2} + \left(C_1 - \frac{EI}{GA} V_1 \right) L = 0. \quad (8.26)$$

The boundary condition (8.25) gives $C_1 = M_1 L - \frac{V_1 L^2}{2}$. By replacing C_1 in (8.26) we obtain the following relation between V_1 and M_1 :

$$V_1 = \frac{6M_1}{(4 + \Phi)L}. \quad (8.27)$$

The equations derived before are still valid and we report them below:

$$V_2 = -V_1, \quad M_2 = -M_1 + V_1 L. \quad (8.28)$$

We are now ready to derive the coefficients $k_{i,j}$. At $x = 0$ we have

$$\frac{dv_b}{dx} = \frac{dv}{dx} - \frac{dv_s}{dx} = \theta_1 \quad (8.29)$$

which implies

$$\theta_1 = \frac{M_1 (1 + \Phi) L}{EI (4 + \Phi)} \quad (8.30)$$

from which

$$k_{3,3} = \frac{M_1}{\theta_1} = \frac{(4 + \Phi) EI}{(1 + \Phi) L}, \quad (8.31)$$

$$k_{5,3} = \frac{V_2}{\theta_1} = -\frac{6EI}{(1 + \Phi) L^2} \quad (= -k_{2,3} = -\frac{V_1}{\theta_1}), \quad (8.32)$$

and

$$k_{6,3} = \frac{M_2}{\theta_1} = \frac{-M_1 + V_1 L}{\theta_1} = \frac{(2 - \Phi) EI}{(1 + \Phi) L}. \quad (8.33)$$

If the deflection of the left-hand end of the beam is zero as shown in Figure 8.5(b) it is then evident from symmetry that

$$k_{6,6} = k_{3,3} = \frac{(4 + \Phi) EI}{(1 + \Phi) L}. \quad (8.34)$$

The remaining coefficients can be derived in a similar manner.

8.5 Putting it all together: the plane frame element

Assembling the terms corresponding to axial, shear and flexural deformation, we arrive at the stiffness matrix of a two dimensional frame element:

$$\begin{bmatrix} N_1 \\ V_1 \\ M_1 \\ N_2 \\ V_2 \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3(1+\Phi)} & \frac{6EI}{L^2(1+\Phi)} & 0 & -\frac{12EI}{L^3(1+\Phi)} & \frac{6EI}{L^2(1+\Phi)} \\ 0 & \frac{6EI}{L^2(1+\Phi)} & \frac{(4+\Phi)EI}{L(1+\Phi)} & 0 & -\frac{6EI}{L^2(1+\Phi)} & \frac{(2-\Phi)EI}{L(1+\Phi)} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3(1+\Phi)} & -\frac{6EI}{L^2(1+\Phi)} & 0 & \frac{12EI}{L^3(1+\Phi)} & -\frac{6EI}{L^2(1+\Phi)} \\ 0 & \frac{6EI}{L^2(1+\Phi)} & \frac{(2-\Phi)EI}{L(1+\Phi)} & 0 & -\frac{6EI}{L^2(1+\Phi)} & \frac{(4+\Phi)EI}{L(1+\Phi)} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{bmatrix}. \quad (8.35)$$

8.6 Reduction to particular cases

8.6.1 Truss element

The truss element is the simplest structural elements. It has only one degree of freedom associated with each node and is obtained by considering only the degrees of freedom corresponding to axial deformation in (8.35):

$$\begin{bmatrix} N_1 \\ N_2 \end{bmatrix} = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}. \quad (8.36)$$

8.6.2 Beam element

There are two major beam theory, the Timoshenko beam and the Euler-Bernoulli beam. Both assume no axial deformation.

Timoshenko beam

The Timoshenko beam accounts for transverse shear deformation and is obtained by neglecting axial deformation in (8.35):

$$\begin{bmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{12EI}{L^3(1+\Phi)} & \frac{6EI}{L^2(1+\Phi)} & -\frac{12EI}{L^3(1+\Phi)} & \frac{6EI}{L^2(1+\Phi)} \\ \frac{6EI}{L^2(1+\Phi)} & \frac{(4+\Phi)EI}{L(1+\Phi)} & -\frac{6EI}{L^2(1+\Phi)} & \frac{(2-\Phi)EI}{L(1+\Phi)} \\ -\frac{12EI}{L^3(1+\Phi)} & -\frac{6EI}{L^2(1+\Phi)} & \frac{12EI}{L^3(1+\Phi)} & -\frac{6EI}{L^2(1+\Phi)} \\ \frac{6EI}{L^2(1+\Phi)} & \frac{(2-\Phi)EI}{L(1+\Phi)} & -\frac{6EI}{L^2(1+\Phi)} & \frac{(4+\Phi)EI}{L(1+\Phi)} \end{bmatrix} \begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}. \quad (8.37)$$

Euler-Bernoulli beam

The Euler-Bernoulli is the classical formulation for beams and is obtained by neglecting the effects of shear deformation in (8.37). This is accomplished by setting the shear deformation parameter Φ to zero:

$$\begin{bmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{12EI}{L^3} & \frac{6EI}{L^2} & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{4EI}{L} & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{2EI}{L} & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \begin{bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{bmatrix}. \quad (8.38)$$

8.6.3 Properties of the element stiffness matrix

- The stiffness matrices of the frame element (8.35), the truss element (8.36), the Timoshenko beam element (8.37), and the Euler-Bernoulli beam element (8.38) are all singular and symmetric.
- Singularity of the stiffness matrix is caused by the linear relations introduced by the equilibrium equations. In the bar element, the first row is equal to the second row but for the minus sign. In the beam element we encounter a similar situation for rows 1 and 3 plus the relation L times row 1 minus row 2 equals row 4. Physically, it means that the element can undergo rigid body motion due to insufficient constraints.
- The stiffness matrix is symmetric and this can be justified with the reciprocal theorem of Maxwell-Betti.
- When displacement are directly proportional to the applied loads the stiffness matrix is always symmetric.
- A stiffness terms $k_{i,j}$ is zero unless at least one element is attached to both degree of freedom i and j .
- Diagonal terms are non-negative. If a diagonal term is zero, then deformation with no strain energy is possible (this is the case of a mechanism).

8.7 The assembly procedure

As briefly discussed in the introduction, in the matrix displacement method we seek a relation of the type

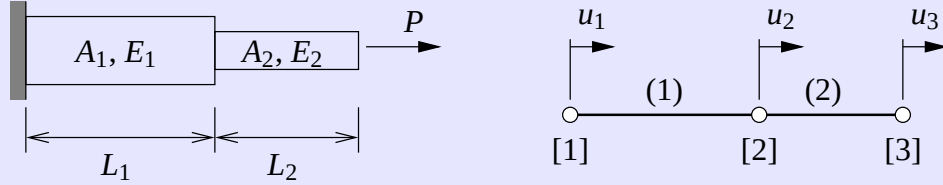
$$ka = f \quad (8.39)$$

valid at the structural level in which

$$k = \sum_{e=1}^{N_{el}} k^e, \quad (8.40)$$

where k^e is the element stiffness matrix. The operator A denotes a procedure used to construct the stiffness matrix of the structure from the stiffness matrices of the N_{el} elements in which the structure has been discretized. The assembly procedure is best described with the aid of an example.

Example 8.1



Consider the axially loaded bar shown above. The unknown degrees of freedom are indicated by u_1 , u_2 and u_3 . In order to define the 3×3 system of equations related to the bar, we first have to compute the element stiffness matrices. In this case they are given by

$$k^{(1)} = \frac{E_1 A_1}{L_1} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \text{ and } k^{(2)} = \frac{E_2 A_2}{L_2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad (8.41)$$

which, for the sake of the next derivations, can be conveniently written as

$$k^{(1)} = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} k_{1,1}^{(1)} & k_{1,2}^{(1)} \\ k_{2,1}^{(1)} & k_{2,2}^{(1)} \end{bmatrix} \end{matrix}, \text{ and } k^{(2)} = \begin{matrix} & \begin{matrix} 2 & 3 \end{matrix} \\ \begin{matrix} 2 \\ 3 \end{matrix} & \begin{bmatrix} k_{1,1}^{(2)} & k_{1,2}^{(2)} \\ k_{2,1}^{(2)} & k_{2,2}^{(2)} \end{bmatrix} \end{matrix}, \quad (8.42)$$

where the number outside the matrix indicates the global degree of freedom number related to each of the elements. A simple way to visualize the construction of the global stiffness matrix is to proceed as follows by making use of the definition of the stiffness matrix k ($k_{i,j}$ is the reaction at degree of freedom i due to a unit deformation at degree of freedom j). With this definition we can populate the stiffness matrix for the whole bar by separately considering the contributions of the two bars. Consider for instance the second bar. The two degrees of freedom are numbered 2 and 3. Hence

$$k_{el. 2} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & k_{1,1}^{(2)} & k_{1,2}^{(2)} \\ 0 & k_{2,1}^{(2)} & k_{2,2}^{(2)} \end{bmatrix} \end{matrix}. \quad (8.43)$$

Analogously, for element 1

$$k_{el. 1} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} k_{1,1}^{(1)} & k_{1,2}^{(1)} & 0 \\ k_{2,1}^{(1)} & k_{2,2}^{(1)} & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}. \quad (8.44)$$

Adding the two contributions, we arrive at the stiffness matrix for the whole bar:

$$k = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} k_{1,1}^{(1)} & k_{1,2}^{(1)} & 0 \\ k_{2,1}^{(1)} & k_{2,2}^{(1)} + k_{1,1}^{(2)} & k_{1,2}^{(2)} \\ 0 & k_{2,1}^{(2)} & k_{2,2}^{(2)} \end{bmatrix} \end{matrix}. \quad (8.45)$$

The applied load must be assembled as well. It is applied in correspondence of the third degree of freedom. Hence

$$f = \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \begin{bmatrix} R \\ 0 \\ P \end{bmatrix}. \quad (8.46)$$

Note the reaction force R in the RHS. This is an additional unknown in this problem.

What is left to do is to solve the system $ka = f$ for the displacement vector $a = [u_1, u_2, u_3]$. Note that the matrix k is singular (its inverse does not exist!). This means that the bar is not constrained and can undergo rigid body motion—with reference to the differential equation related to this problem, it is like if we were trying to solve it without considering boundary conditions. To constrain the bar we have to specify the value of a sufficient number of degrees of freedom. In this simple one-dimensional example the application of the displacement boundary condition related to the node 1 suffices. Practically, this is done by striking out the row and the column related to that degree of freedom. This operation yields the 2×2 system

$$\begin{matrix} & \begin{matrix} 2 & 3 \end{matrix} \\ \begin{matrix} 2 \\ 3 \end{matrix} & \begin{bmatrix} k_{2,2}^{(1)} + k_{1,1}^{(2)} & k_{1,2}^{(2)} \\ k_{2,1}^{(2)} & k_{2,2}^{(2)} \end{bmatrix} \end{matrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{matrix} 2 \\ 3 \end{matrix} \begin{bmatrix} 0 \\ P \end{bmatrix} \quad (8.47)$$

which can be easily solved. The row-striking technique just described can be applied with homogeneous constraints ($u_1 = 0$). The recipe for the general case of non-homogeneous constraints like e.g. $u_3 = \bar{u}$ is discussed in Section 8.10.

8.7.1 The matrix assembly procedure in a finite element code

A widely used procedure consists in defining a connectivity array $\mathbf{ld}$ for each element ($\mathbf{ld}$ is acronym for "local degrees of freedom"). In this array local degrees of freedom are linked to global degrees of freedom. In this example $\mathbf{ld}^{(1)} = [1, 2]$ and $\mathbf{ld}^{(2)} = [2, 3]$, where the array index i defines the local element degree of freedom and $\mathbf{ld}(i)$ the global degree of freedom number. These arrays can be used to populate the global stiffness matrix by using Algorithm 1. An example is reported in the code listed on page 91.

Input : ld , number of elements, number of element degrees of freedom (dof),
element stiffness matrix $k^{(e)}$

Output: global stiffness matrix k

```

1 zero global stiffness matrix  $k$ 
2 for each element  $e$  do
3   for  $i = 1, dof$  do
4     for  $j = 1, dof$  do
5        $k_{ld(i),ld(j)} = k_{ld(i),ld(j)} + k_{i,j}^{(e)}$ 
6     end
7   end
8 end

```

Algorithm 1: Assembly procedure.

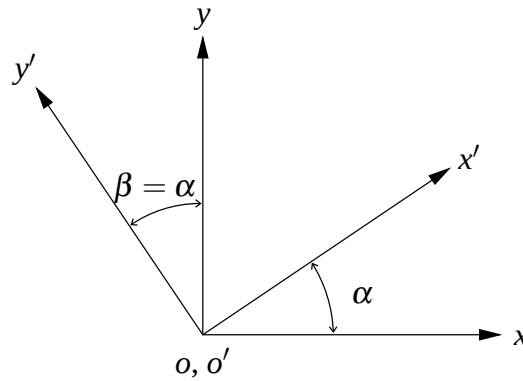


Figure 8.6

8.8 Transformations

We have computed element arrays such as stiffness matrices and force vectors in local coordinates with the aim of ease computations of these quantities. However, beams and bars are arbitrarily placed and this requires a way to express element arrays in a generic system knowing their expression in the local coordinate systems. This can be accomplished by making use of some simple manipulations.

8.8.1 Transformation of vectors

This section is based upon [3, Section 7.2]. Consider a vector V which can be thought of as a force vector, a displacement vector etc. The components of V in the reference system xoy are (V_x, V_y) . The same vector can be defined in a system $x'o'y'$ rotated of an angle α with respect to the reference system. We call the system $x'o'y'$ the local coordinate system. The components of V in the local system are $(V_{x'}, V_{y'})$.

Our goal is to express $(V_{x'}, V_{y'})$ in terms of (V_x, V_y) and of the cosines of the angles between the two systems. These cosines are called direction cosines.

The direction cosines of the x' axis are indicated by $(l_{x'}, m_{x'})$ and those of the y' axis by $(l_{y'}, m_{y'})$. We know, from analytical geometry that we can express $V_{x'}$ as the sum of the projection of V_x and V_y along the x' axis as $V_{x'} = l_{x'}V_x + m_{x'}V_y$. The component $V_{y'}$ can be expressed in a similar way. In matrix format these two expressions reads

$$\begin{pmatrix} V_{x'} \\ V_{y'} \end{pmatrix} = \begin{bmatrix} l_{x'} & m_{x'} \\ l_{y'} & m_{y'} \end{bmatrix} \begin{pmatrix} V_x \\ V_y \end{pmatrix} = R \begin{pmatrix} V_x \\ V_y \end{pmatrix}. \quad (8.48)$$

The matrix R is orthogonal ($R^{-1} = R^T$) and corresponds to pure rotation since $\det R = 1$.

With reference to Figure 8.6 we have that $l_{x'} = \cos \alpha$, $m_{x'} = \cos(\pi/2 - \alpha) = \sin \alpha$, $l_{y'} = \cos(\pi/2 + \beta) = \cos(\pi/2 + \alpha) = -\sin \alpha$, and $m_{y'} = \cos \beta = \cos \alpha$. We can therefore express the rotation matrix as

$$R = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}. \quad (8.49)$$

8.8.2 Transformation of element arrays

The stiffness matrix of the bar element has been expressed as a function of the two degrees of freedom u_1 and u_2 in the local system (see (8.36)). It is however convenient, in view of its expression in the global xoy coordinates system, to express it as a function of four degrees of freedom, two in the axial direction, u'_1 and u'_2 , and two normal to the bar axis, v'_1 and v'_2 . A similar transformation can be done on the local force array. Hence, starting from the stiffness matrix

$$\begin{matrix} u'_1 & u'_2 \\ u'_1 & \begin{bmatrix} k_{1,1} & k_{1,2} \\ k_{2,1} & k_{2,2} \end{bmatrix} \\ u'_2 & \end{matrix}, \quad (8.50)$$

the augmented stiffness matrix reads

$$\begin{matrix} u'_1 & v'_1 & u'_2 & v'_2 \\ u'_1 & \begin{bmatrix} k_{1,1} & 0 & k_{1,2} & 0 \\ 0 & 0 & 0 & 0 \\ k_{2,1} & 0 & k_{2,2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ v'_1 & \\ u'_2 & \\ v'_2 & \end{matrix}. \quad (8.51)$$

Next, the array $a' = [u'_1, v'_1, u'_2, v'_2]^T$ is expressed as a function of the array $a = [u_1, v_1, u_2, v_2]^T$ in the global coordinate system through the use of the transformation matrix

$$T_{4 \times 4} = \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix}. \quad (8.52)$$

Hence, $a' = Ta$, $f' = Tf$, and $k'a' = f'$ can be written as $T^T k' T a = f$ which yields the stiffness matrix of a bar element inclined of an angle α in the xoy coordinate system:

$$k = T^T k' T = \frac{EA}{L} \begin{bmatrix} k_0 & -k_0 \\ -k_0 & k_0 \end{bmatrix}, \quad (8.53)$$

where

$$k_0 = \begin{bmatrix} c^2 & cs \\ cs & s^2 \end{bmatrix} \quad (8.54)$$

with $c = \cos \alpha$ and $s = \sin \alpha$. From a practical point of view, it is convenient to retain the structure of the stiffness matrix of the truss in the original 2×2 format and use a different transformation matrix. By using the 2×4 transformation matrix

$$T = \begin{bmatrix} l_{x'} & m_{x'} & 0 & 0 \\ 0 & 0 & l_{y'} & m_{y'} \end{bmatrix} \quad (8.55)$$

and the 2×2 element stiffness matrix (see (8.36)), we arrive at the same expression of the 4×4 stiffness matrix in the xoy coordinate system. For a frame element with the degrees of freedom ordered as in (8.35), the transformation matrix is [2, page 82]

$$T_{6 \times 6} = \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix} \quad (8.56)$$

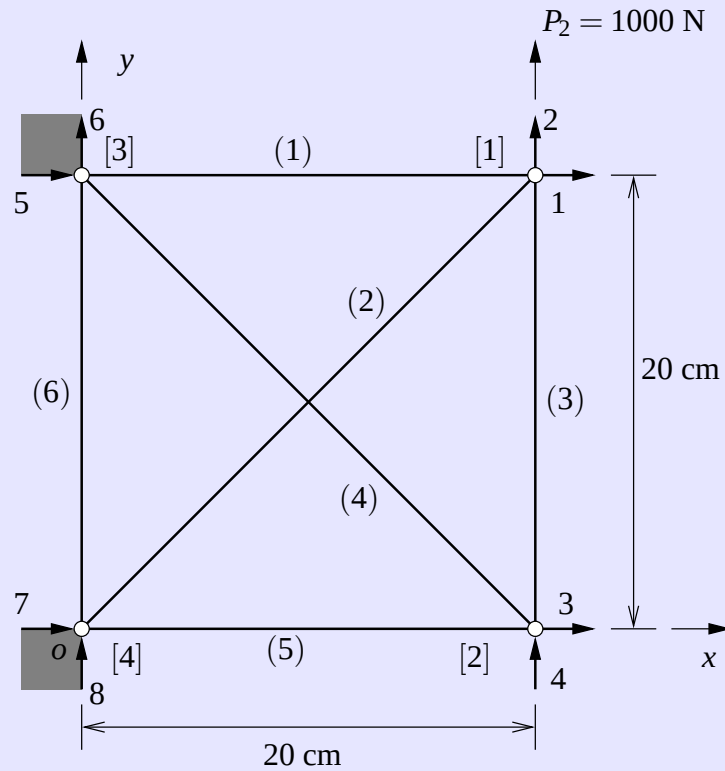
with

$$R = \begin{bmatrix} l_{x'} & m_{x'} & 0 \\ l_{y'} & m_{y'} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (8.57)$$

The next example shows how to apply the transformation procedure described above.

Example 8.2

This example is a slight modification of the one found in the book by Przemieniecki [2, Section 6.9]. Consider the statically undetermined pin-jointed truss shown in the next figure. The structure is loaded by a vertical force at node 1. The numbering for elements, nodes and degrees of freedom is shown in the figure together with the dimension of the truss, the intensity of the load and the boundary conditions (nodes 3 and 4 are constrained in both directions). Other data are listed in the next table where the $p - q$ direction for each member is identified by the node numbers, and the direction cosines are defined in the xoy coordinate system by means of the node coordinates. The Young's modulus of the material is $10 \times 10^6 \text{ N/mm}^2$.



Member number	Cross-sectional area	Length	Location	Direction cosines	
i	A^i [mm ²]	L^i [mm]	$p-q$	l_{pq}	m_{pq}
1	1.0	20	3,1	1	0
2	$\sqrt{2}/2$	$20\sqrt{2}$	4,1	$\sqrt{2}/2$	$\sqrt{2}/2$
3	1.0	20	1,2	0	-1.0
4	$\sqrt{2}/2$	$20\sqrt{2}$	3,2	$\sqrt{2}/2$	$-\sqrt{2}/2$
5	1.0	20	4,2	1.0	0
6	1.0	20	3,4	0	-1.0

The stiffness matrix for a pin-jointed truss in the xoy coordinate system is defined through $k = T^T k' T$ where k' is the stiffness matrix (8.36) defined in a local coordinate system and T is the transformation matrix (8.55). Hence

$$k^{(1)} = \begin{matrix} & \begin{matrix} 5 & 6 & 1 & 2 \end{matrix} \\ \begin{matrix} 5 \\ 6 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} 0.5 \times 10^6 \frac{\text{N}}{\text{mm}}, \quad (8.58)$$

$$k^{(2)} = \begin{matrix} & \begin{matrix} 7 & 8 & 1 & 2 \end{matrix} \\ \begin{matrix} 7 \\ 8 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \end{matrix} 0.125 \times 10^6 \frac{\text{N}}{\text{mm}}, \quad (8.59)$$

$$k^{(3)} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{matrix} 0.5 \times 10^6 \frac{\text{N}}{\text{mm}}, \quad (8.60)$$

$$k^{(4)} = \begin{matrix} & \begin{matrix} 5 & 6 & 3 & 4 \end{matrix} \\ \begin{matrix} 5 \\ 6 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \end{matrix} 0.125 \times 10^6 \frac{\text{N}}{\text{mm}}. \quad (8.61)$$

Noting that $k^{(5)} = k^{(1)}$ and $k^{(6)} = k^{(3)}$, the element matrices are then assembled into the global system stiffness matrix of the structure. This leads to

$$k = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{matrix} & \begin{bmatrix} 5 & & & & & & & \\ 1 & 5 & & & & & & \\ 0 & 0 & 5 & & & & & \\ 0 & -4 & -1 & 5 & & & & \\ -4 & 0 & -1 & 1 & 5 & & & \\ 0 & 0 & 1 & -1 & -1 & 5 & & \\ -1 & -1 & -4 & 0 & 0 & 0 & 5 & \\ -1 & -1 & 0 & 0 & 0 & -4 & 1 & 5 \end{bmatrix} \end{matrix} 0.125 \times 10^6 \frac{\text{N}}{\text{mm}}. \quad (8.62)$$

Similarly, assembly of the global force array yields

$$f = \begin{matrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{matrix} & \begin{bmatrix} 0 \\ 1000 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{matrix} \text{N}. \quad (8.63)$$

Removing row and columns of constrained degrees of freedom (5 to 8) we may now write the equilibrium equations as

$$ka = f \quad (8.64)$$

or

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 5 & 1 & 0 & 0 \\ 1 & 5 & 0 & -4 \\ 0 & 0 & 5 & -1 \\ 0 & -4 & -1 & 5 \end{bmatrix} \end{matrix} 0.125 \times 10^6 \frac{\text{N}}{\text{mm}} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 \\ 10^3 \\ 0 \\ 0 \end{bmatrix} \text{N} \quad (8.65)$$

from which

$$a = k^{-1}f = \frac{2}{11} \times 10^{-3} \begin{bmatrix} -6 \\ 30 \\ 5 \\ 25 \end{bmatrix} \text{mm}. \quad (8.66)$$

Once the displacements are known, we can derive the strain and stress fields in the truss.

8.9 A minimal Matlab/Octave 2D finite element truss code

The whole procedure described in Example 8.2 can be made automatic. A minimal Matlab/Octave 2D finite element code is listed below and can be downloaded from the course web page. The result of Example 8.2 is shown in Figure 8.7 with displacements magnified 400 times.

```
% truss.m
%
% a minimal Matlab/Octave 2D finite element truss code
%
% coded by Angelo Simone, TU Delft, December 2007
%
%
% --- clear memory
%
clear all

%
% --- give truss properties: nodal coordinates,
% connectivity table, degrees of freedom/node,
% axial stiffness
%
coord=[20 20; 20 0; 0 20; 0 0];
conn=[3 1; 4 1; 1 2; 3 2; 4 2; 3 4];
dofNode=2;
E=10*10^6;
A(1)=1;
A(2)=sqrt(2)/2;
A(3)=1;
A(4)=sqrt(2)/2;
A(5)=1;
A(6)=1;
```

```

AE=A*E;
nNodes=size(coord,1);
nElements=size(conn,1);

%
% — allocate arrays in ka=f
%
nDofs=dofNode*nNodes; % total number of dofs
k=zeros(nDofs,nDofs); % stiffness matrix
a=zeros(nDofs,1); % solution array
f=zeros(nDofs,1); % rhs

%
% — define boundary conditions:
% constrained dofs and applied load
%
constrainedDofs=[5 6 7 8];
f(2)=1000;

%
% — assemble stiffness matrix
%
for e=1:nElements

    % element connectivity table
    eleConn=conn(e,:);

    % element coordinates and length
    x1=coord(eleConn(1),1);
    x2=coord(eleConn(2),1);
    y1=coord(eleConn(1),2);
    y2=coord(eleConn(2),2);
    len=sqrt((x2-x1)*(x2-x1)+(y2-y1)*(y2-y1));

    % c and s are cosine and sine of the
    % angle between local and global axes
    c=(x2-x1)/len;
    s=(y2-y1)/len;

    % element stiffness matrix (local system)
    ke_loc=AE(e)/len*[1 0 -1 0;
                     0 0 0 0;
                     -1 0 1 0;
                     0 0 0 0];

    % rotation and transformation matrix
    R=[c s;
       -s c];

    Zero=[0 0;
          0 0];

    T=[R Zero;
       Zero R];

    % element stiffness matrix (global system)
    ke=T'*ke_loc*T

    % ... faster alternative
    % ke=AE(e)/len*[c*c c*s -c*c -c*s;
    %               c*s s*s -c*s -s*s;
    %               -c*c -c*s c*c c*s;

```

```

%           -c*s  -s*s  c*s  s*s]

% compute system dofs associated with each element
% (node 1 has dofs 1 and 2, node 2 has dofs 3 and 4...)
index(1)=dofNode*eleConn(1)-1;
index(2)=dofNode*eleConn(1);
index(3)=dofNode*eleConn(2)-1;
index(4)=dofNode*eleConn(2);

% assemble ke into K
edof = length(index);
for i=1:edof
    ii=index(i);
    for j=1:edof
        jj=index(j);
        k(ii,jj)=k(ii,jj)+ke(i,j);
    end
end

end

%
% — apply boundary conditions by zeroing out
%   rows and columns and putting ones on the diagonal
%
k(constrainedDofs,:)=0;
k(:,constrainedDofs)=0;
k(constrainedDofs,constrainedDofs)=eye(length(constrainedDofs));

%
% — solve ka=f for a
%
a=inv(k)*f

%
% — plot results
%
mag=400; % scale factor for plot
clf
hold on
for e=1:nElements
    x=coord(conn(e,:),1);
    y=coord(conn(e,:),2);
    u=a(2*conn(e,:)-1);
    v=a(2*conn(e,:));
    title('Deformed plot')
    axis equal
    plot(x,y,'r—o')
    plot(x+mag*u,y+mag*v,'k-o')
end

```

8.10 Constraints: application of prescribed displacements

Degrees of freedom can be given a value different than zero by following the procedure described in this section. To this end, consider the relation

$$ka = f, \quad (8.67)$$

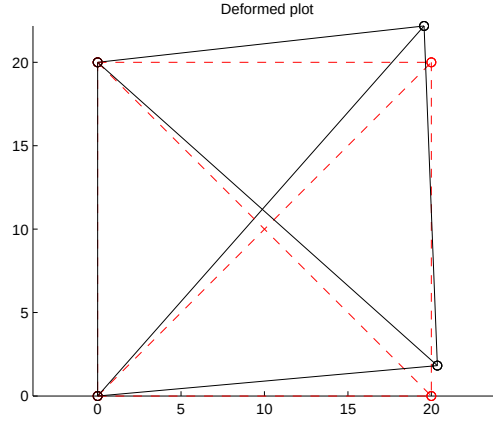


Figure 8.7 Truss after the application of the load (displacements are magnified 400 times).

and partition the a array in free and prescribed components:

$$a = \begin{bmatrix} a_f \\ a_p \end{bmatrix}. \quad (8.68)$$

Consequently, the relation (8.67) can now be written as

$$\begin{bmatrix} k_{ff} & k_{fp} \\ k_{pf} & k_{pp} \end{bmatrix} \begin{bmatrix} a_f \\ a_p \end{bmatrix} = \begin{bmatrix} f_f \\ f_p \end{bmatrix} \quad (8.69)$$

which can be re-written as

$$k_{ff}a_f + k_{fp}a_p = f_f \quad (8.70a)$$

$$k_{pf}a_p + k_{pp}a_p = f_p. \quad (8.70b)$$

In the above system of equations, known quantities are a_p and f_f (prescribed degrees of freedom and applied forces at free nodes) while unknown quantities are a_f and f_p (unknown degrees of freedom at free nodes and forces at constrained nodes, i.e. the reaction forces).

Considering equation (8.70a), it is possible to solve for a_f :

$$k_{ff}a_f = f_f - k_{fp}a_p, \quad (8.71)$$

where the effect of the prescribed degrees of freedom is expressed as extra forces applied to the free degrees of freedom. Equation (8.70b) can be used to compute the reaction forces once the unknown vector a is known.

Splitting the relation (8.67) into two systems of equations and using only the first corresponds to cancelling the rows related to the prescribed degrees of freedom—this is indicated as step 1 in the example below; moving the terms related to a_p to the RHS allows to solve for a_f —step 2 in the example below.

Example 8.3

Let us reconsider Example 8.1 discussed in Section 8.7. We now constrain the third degree of freedom to be equal to $\bar{u}$. With this constraint,

$$a = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ u_2 \\ \bar{u} \end{bmatrix}, \quad (8.72)$$

while the RHS

$$f = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} R_1 \\ 0 \\ R_3 \end{bmatrix}, \quad (8.73)$$

where R_i are reaction forces. There are three unknowns in this problem: u_2 , R_1 , and R_3 and the stiffness matrix (8.45) is still valid.

To solve this problem we proceed as follows:

step 1 remove the equations related to known displacements (equations 1 and 3) in $ka = f$:

$$_2 \begin{bmatrix} \overset{1}{k_{2,1}^{(1)}} & \overset{2}{k_{2,2}^{(1)} + k_{1,1}^{(2)}} & \overset{3}{k_{1,2}^{(2)}} \end{bmatrix} \begin{bmatrix} 0 \\ u_2 \\ \bar{u} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix} \quad (8.74)$$

step 2 move the columns related to known displacements (columns 1 and 3) to the RHS multiplying them by the corresponding displacement values (0 and $\bar{u}$):

$$\begin{bmatrix} k_{2,2}^{(1)} + k_{1,1}^{(2)} \end{bmatrix} \begin{bmatrix} u_2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix} - \begin{bmatrix} k_{2,1}^{(1)} \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix} - \begin{bmatrix} k_{1,2}^{(2)} \end{bmatrix} \begin{bmatrix} \bar{u} \end{bmatrix} \quad (8.75)$$

step 3 solve the system for the unknowns (u_2 in this case):

$$u_2 = -\frac{k_{1,2}^{(2)}}{k_{2,2}^{(1)} + k_{1,1}^{(2)}} \bar{u} \quad (8.76)$$

The last step usually implies the inversion of the matrix k_{ff} .

8.11 Equivalent concentrated forces

To take into account distribute loads, we need to derive equivalent nodal forces. This is necessary as we work with elements that accept only nodal contributions to the RHS. The energy approach allows the computation of concentrated forces equivalent, in terms of work, to the corresponding distributed force.

We consider any one of the beam (or bar) dofs a_i to be of unit intensity and compute the

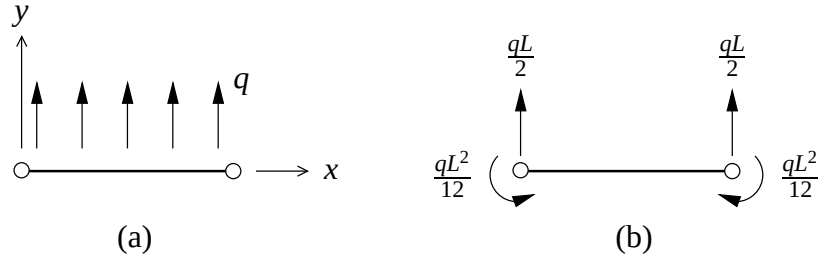


Figure 8.8

work done by the distributed load q against the deformation due to a_i ; we then equate this work, which is related to the distributed load, to the work done by the unit displacement a_i against an equivalent concentrated force P_{eq} .

Consider a Euler-Bernoulli beam of length L with a distributed load q as shown in Figure 8.8(a). The work-equivalent load related to the distributed load q is computed with the relation

$$\int_L v_D(x) q dx = P_{eq}^{(D)} D = P_{eq}^{(D)}, \quad (8.77)$$

where the last term has been obtained by considering a unit value of the degree of freedom D and v_D is the displacement in the beam due to D . When applied to all the dofs a_i , this procedure yields a vector of concentrated forces P_{eq} equivalent to the distributed load q .

Example 8.4

Let us compute the nodal equivalent force in the direction of v_1 due to the distributed load q for the beam shown in Figure 8.8(a). To compute the integral we need the expression of the deflection due to $v_1 = 1$ while all the remaining dofs are set equal to zero. This is easily done by considering the elementary beam theory developed in the previous chapters and results in the first expression reported in table below.

$v_1 = 1$	$v = 1 - 3\frac{x^2}{L^2} + 2\frac{x^3}{L^3}$	
$\theta_1 = 1$	$v = x - 2\frac{x^2}{L} + \frac{x^3}{L^2}$	
$v_2 = 1$	$v = 3\frac{x^2}{L^2} - 2\frac{x^3}{L^3}$	
$\theta_2 = 1$	$v = -\frac{x^2}{L} + \frac{x^3}{L^2}$	

Euler-Bernoulli beam deflection due to unit values of the transverse and rotational dofs.

With this expression at hand, the nodal equivalent load is computed from

$$\int_L v_D q dx = \int_L v_{v_1} q dx = \int_L \left(1 - 3\frac{x^2}{L^2} + 2\frac{x^3}{L^3}\right) q dx = \frac{1}{2}qL = P_{eq}^{(v_1)}. \quad (8.78)$$

Considering all the remaining contributions we have

$$P_{eq} = qL \begin{bmatrix} 1/2 \\ L/12 \\ 1/2 \\ -L/12 \end{bmatrix}. \quad (8.79)$$

P_{eq} is a vector of work-equivalent nodal loads. They are equivalent to support reactions for a uniform beam fixed at both ends and uniformly loaded with a load of intensity $-q$ (see Figure 8.8(b)). These nodal forces and moments are statically equivalent to the distributed force: they have the same resultant force and moment about an arbitrary chosen point as does the original distributed loading.

The procedure described above can be used to compute nodal forces equivalent to concentrated forces applied between two nodes. In this case, the integral must be replaced by the product of the deflection evaluated at the point of application of the force and the force itself.

Exercises

8.1 Compute the stiffness matrix of a bar of embedded in a medium. The bar has length L and axial stiffness EA . The action of the medium on the bar can be replaced by a set of continuously distributed elastic forces of the type $p = ku$ where u is the axial displacement and k is the elastic stiffness of the medium. What happens to the stiffness matrix when $k \rightarrow 0$ and $k \rightarrow \infty$?

8.2 Demonstrate that the vector of work-equivalent nodal loads obtained employing a Timoshenko beam element is identical to that obtained with a Euler-Bernoulli beam when a beam is loaded by a uniformly distributed transverse load. What happens with an arbitrarily distributed load?

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